

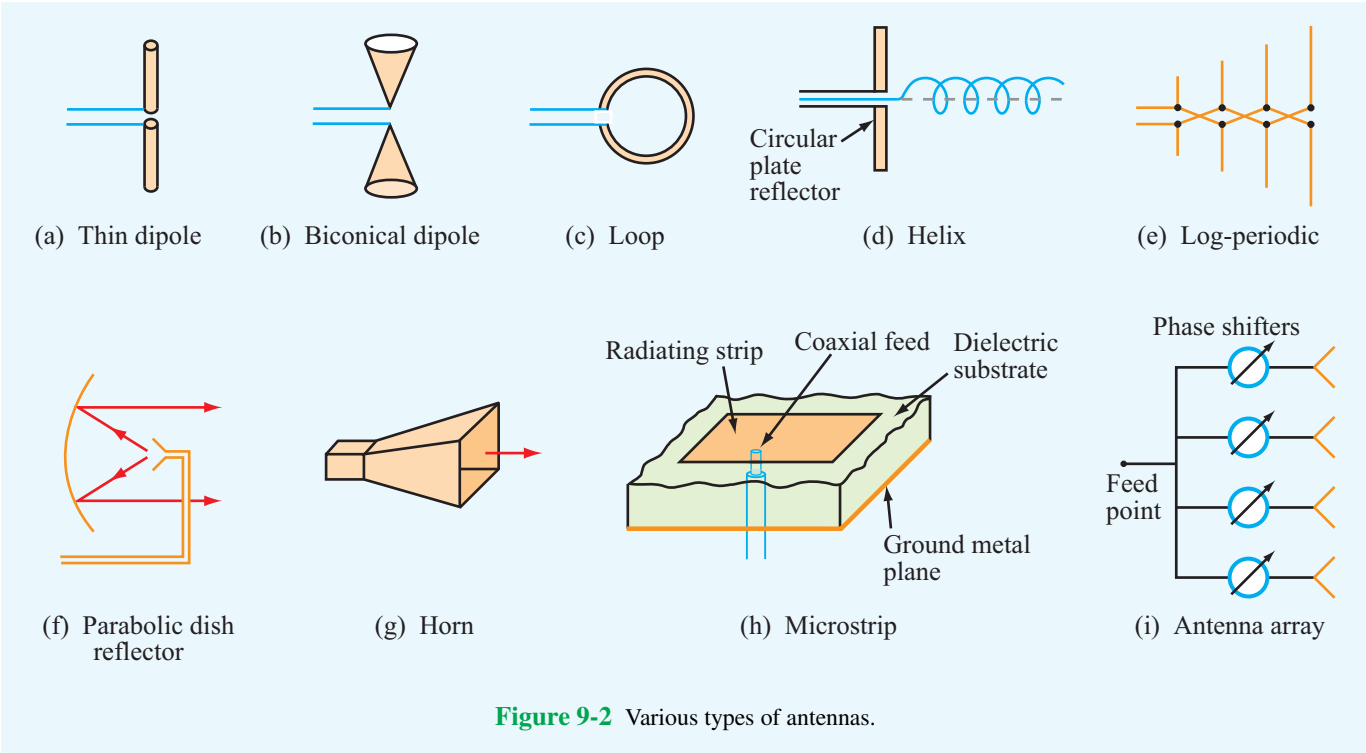


Figure 9-2 Various types of antennas.

Radiation Sources

Radiation sources fall into two categories: currents and aperture fields. The dipole and loop antennas (Fig. 9-2(a) and (c)) are examples of current sources; the time-varying currents flowing in the conducting wires give rise to the radiated electromagnetic fields. A horn antenna (Fig. 9-2(g)) is an example of the second group because the electric and magnetic fields across the horn's aperture serve as the sources of the radiated fields. The aperture fields are themselves induced by time-varying currents on the surfaces of the horn's walls. Therefore, all radiation ultimately is due to time-varying currents. The choice of currents or apertures as the sources is merely a computational convenience arising from the structure of the antenna. We will examine the radiation processes associated with both types of sources.

Far-Field Region

The wave radiated by a point source is spherical in nature, with the wavefront expanding outward at a rate equal to the phase velocity  $u_p$  (or the velocity of light  $c$  if the medium is free space). If  $R$ , which is the distance between the transmitting antenna and the receiving antenna, is sufficiently large enough for the wavefront across the receiving aperture to be considered planar (Fig. 9-3), then the receiving aperture is said to be

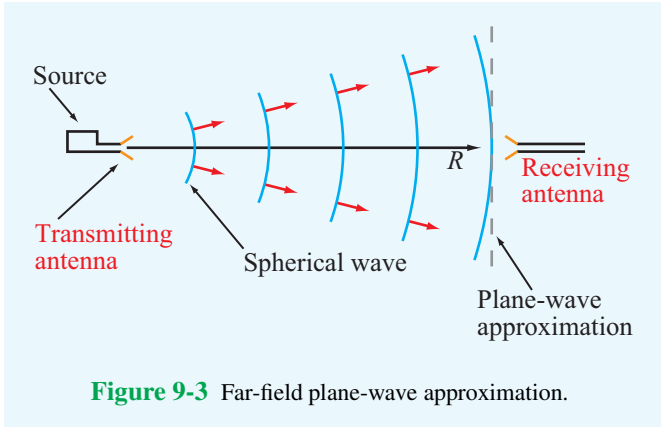


Figure 9-3 Far-field plane-wave approximation.

in the *far-field* (or *far-zone*) region of the transmitting point source. This region is of particular significance because—for most applications—the location of the observation point is indeed in the far-field region of the antenna. The far-field plane-wave approximation allows the use of certain mathematical approximations that simplify the computation of the radiated field and, conversely, provide convenient techniques for synthesizing the appropriate antenna structure that would give rise to the desired far-field antenna pattern.

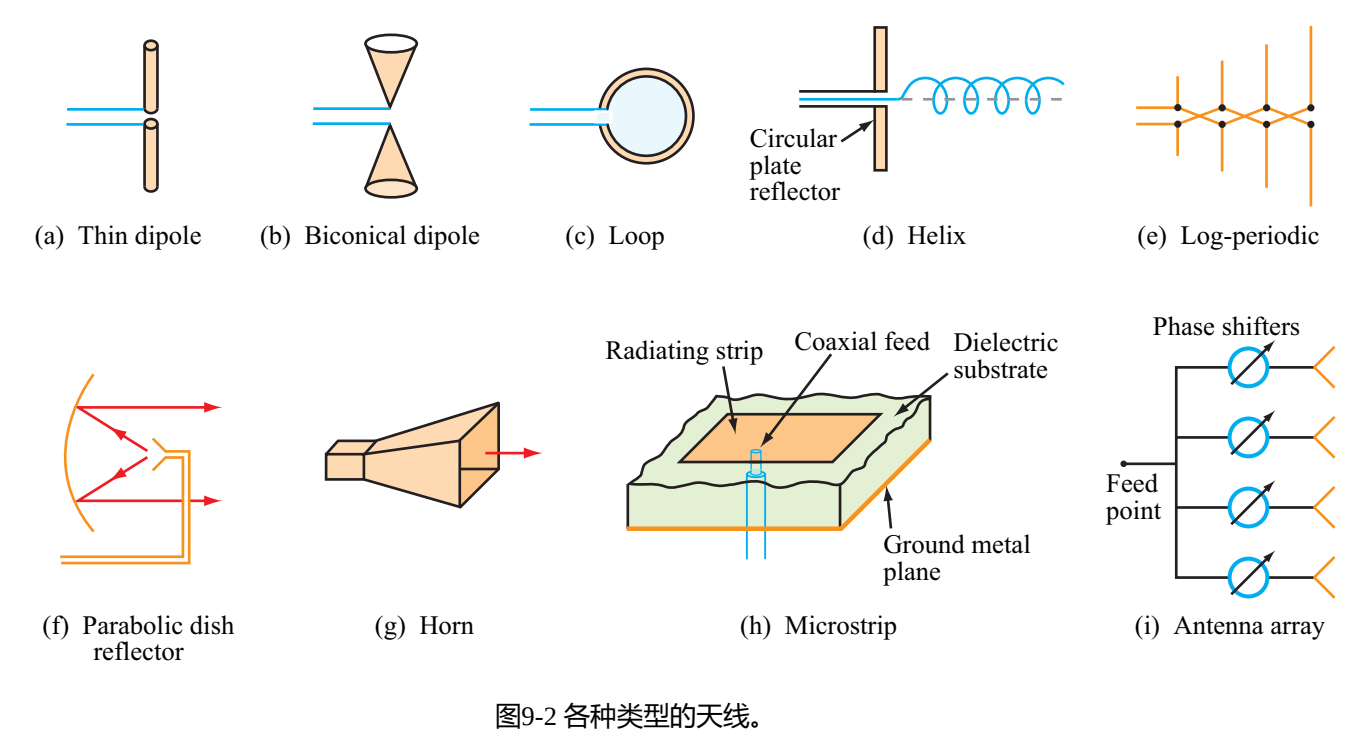


图9-2 各种类型的天线。

辐射源

辐射源分为两类：电流和孔径场。偶极子天线和环形天线（图9-2 (a) 和 (c)）是电流源的示例；导线中流动的时变电流会产生辐射电磁场。喇叭天线（图 9-2(g)）是第二组的一个例子，因为喇叭孔径上的电场和磁场充当辐射场的来源。孔径场本身是由喇叭壁表面上的时变电流引起的。因此，所有辐射最终都是由于时变电流造成的。选择电流或孔径作为源只是出于天线结构的计算方便。我们将检查与这两种类型的源相关的辐射过程。

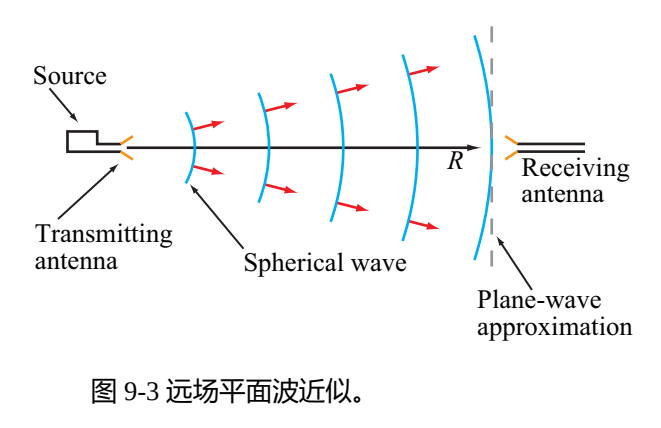


图 9-3 远场平面波近似。

远场区

点源辐射的波本质上是球形的，波前以等于相速度  $u_p$ （或光速  $c$ （如果介质是自由空间））的速率向外扩展。如果  $R$ （发射天线和接收天线之间的距离）足够大，足以使穿过接收孔径的波前被视为平面（图 9-3），则接收孔径称为

在发射点源的远场（或远区）区域。该区域具有特别重要的意义，因为对于大多数应用而言，观察点的位置确实位于天线的远场区域。远场平面波近似允许使用某些数学近似来简化辐射场的计算，相反，提供了合成适当天线结构的便捷技术，从而产生所需的远场天线方向图。

### Antenna Arrays

When multiple antennas operate together, the combination is called an **antenna array** (Fig. 9-2(i)), and the array as a whole behaves as if it were a single antenna. By controlling the magnitude and phase of the signal feeding each antenna, it is possible to *shape the radiation pattern* of the array and to *electronically steer the direction of the beam electronically*. These topics are treated in Sections 9-9 to 9-11.

## 9-1 The Hertzian Dipole

By regarding a linear antenna as consisting of a large number of infinitesimally short conducting elements where each is so short that current may be considered uniform over its length, the field of the entire antenna may be obtained by integrating the fields from all these differential antennas with the proper magnitudes and phases taken into account. We shall first examine the radiation properties of such a differential antenna, known as a **Hertzian dipole**, and in Section 9-3, we will extend the results to compute the fields radiated by a half-wave dipole, which is commonly used as a standard antenna for many applications.

► A **Hertzian dipole** is a thin, linear conductor whose length  $l$  is very short compared with the wavelength  $\lambda$ ;  $l$  should not exceed  $\lambda/50$ . ◀

In this case, the wire is oriented along the  $z$  direction in Fig. 9-4 and carries a sinusoidally varying current given by

$$i(t) = I_0 \cos \omega t = \Re[I_0 e^{j\omega t}] \quad (\text{A}), \quad (9.1)$$

where  $I_0$  is the current amplitude. From Eq. (9.1), the phasor current  $\tilde{I} = I_0$ . Even though the current has to go to zero at the two ends of the dipole, we shall treat it as constant across its entire length.

The customary approach for finding the electric and magnetic fields at a point  $Q$  in space (Fig. 9-4) due to radiation by a current source is through the retarded vector potential  $\tilde{\mathbf{A}}$ . From Eq. (6.84), the phasor retarded vector potential  $\tilde{\mathbf{A}}(\mathbf{R})$  at a distance vector  $\mathbf{R}$  from a volume  $\mathcal{V}'$  containing a phasor current distribution  $\tilde{\mathbf{J}}$  is given by

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu_0}{4\pi} \int_{\mathcal{V}'} \frac{\tilde{\mathbf{J}} e^{-jkR'}}{R'} d\mathcal{V}', \quad (9.2)$$

where  $\mu_0$  is the magnetic permeability of free space (because the observation point is in air) and  $k = \omega/c = 2\pi/\lambda$  is the wavenumber. For the dipole, the current density is simply

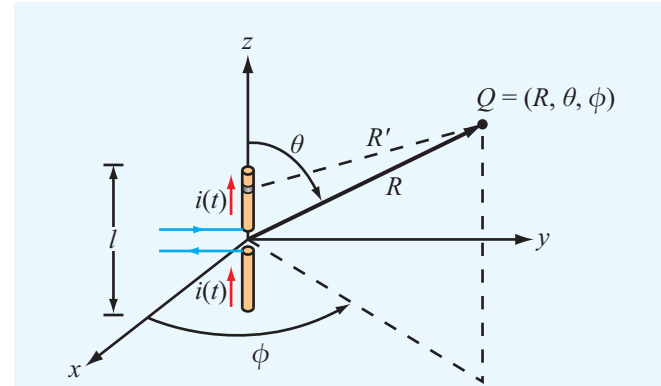


Figure 9-4 Short dipole placed at the origin of a spherical coordinate system.

$\tilde{\mathbf{J}} = \hat{\mathbf{z}}(I_0/s)$ , where  $s$  is the cross-sectional area of the dipole wire. Also,  $d\mathcal{V}' = s dz$ , and the limits of integration are from  $z = -l/2$  to  $z = l/2$ . In Fig. 9-4, the distance  $R'$  between the observation point and a given point along the dipole is not the same as the distance to its center,  $R$ , but because we are dealing with a very short dipole, we can set  $R' \approx R$ . Hence,

$$\tilde{\mathbf{A}} = \frac{\mu_0}{4\pi} \frac{e^{-jkR}}{R} \int_{-l/2}^{l/2} \hat{\mathbf{z}} I_0 dz = \hat{\mathbf{z}} \frac{\mu_0}{4\pi} I_0 l \left( \frac{e^{-jkR}}{R} \right), \quad (9.3)$$

► The function  $(e^{-jkR}/R)$  is called the **spherical propagation factor**. It accounts for the  $1/R$  decay of the magnitude with distance as well as the phase change represented by  $e^{-jkR}$ . ◀

The direction of  $\tilde{\mathbf{A}}$  is the same as that of the current ( $z$  direction).

Because our objective is to characterize the directional character of the radiated power at a fixed distance  $R$  from the antenna, antenna pattern plots are presented in a spherical coordinate system (Fig. 9-5). Its variables,  $R$ ,  $\theta$ , and  $\phi$ , are called the **range**, **zenith angle**, and **azimuth angle**, respectively. To that end, we need to write  $\tilde{\mathbf{A}}$  in terms of its spherical coordinate components, which is realized with the help of Eq. (3.65c) by expressing  $\hat{\mathbf{z}}$  in terms of spherical coordinates:

$$\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta. \quad (9.4)$$

Upon substituting Eq. (9.4) into Eq. (9.3), we obtain

$$\tilde{\mathbf{A}} = (\hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta) \frac{\mu_0 I_0 l}{4\pi} \left( \frac{e^{-jkR}}{R} \right) = \hat{\mathbf{R}} \tilde{A}_R + \hat{\boldsymbol{\theta}} \tilde{A}_\theta + \hat{\boldsymbol{\phi}} \tilde{A}_\phi, \quad (9.5)$$

### 天线阵列

当多个天线一起工作时，这种组合称为天线阵列（图9-2(i)），整个阵列的行为就像单个天线一样。通过控制馈送到每个天线的信号的幅度和相位，可以塑造阵列的辐射方向图并以电子方式控制波束的方向。这些主题将在第 9-9 至 9-11 节中讨论。

## 9-1 赫兹偶极子

通过将线性天线视为由大量无限短的导电元件组成，其中每个元件都很短，以至于可以认为电流在其长度上是均匀的，则可以通过对所有这些差分天线的场进行积分并考虑适当的幅度和相位来获得整个天线的场。我们将首先检查这种差分天线（称为赫兹偶极子）的辐射特性，在第 9-3 节中，我们将扩展结果以计算半波偶极子辐射的场，半波偶极子通常用作许多应用的标准天线。

► 赫兹偶极子是一种薄的线性导体，其长度  $l$  与波长  $\lambda$  相比非常短； $l$  不应超过  $\lambda/50$ 。 ◀

在这种情况下，导线沿图 9-4 中的  $z$  方向定向，并承载由下式给出的正弦变化电流：

$$i(t) = I_0 \cos \omega t = \Re[I_0 e^{j\omega t}] \quad (\text{A}), \quad (9.1)$$

其中  $I_0$  是电流幅度。从方程 (9.1) 相量电流  $\tilde{I} = I_0$ 。尽管偶极子两端的电流必须为零，但我们仍将其视为整个长度上的常数。

寻找空间  $Q$  点（图 9-4）由于电流源辐射而产生的电场和磁场的常规方法是通过延迟矢量势  $\tilde{\mathbf{A}}$ 。(6.84)，距包含相量电流分布  $\tilde{\mathbf{J}}$  的体积  $\mathcal{V}'$  距离矢量  $\mathbf{R}$  处的相量延迟矢量势  $\tilde{\mathbf{A}}(\mathbf{R})$  由下式给出

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu_0}{4\pi} \int_{\mathcal{V}'} \frac{\tilde{\mathbf{J}} e^{-jkR'}}{R'} d\mathcal{V}', \quad (9.2)$$

其中  $\mu_0$  是自由空间的磁导率（因为观察点在空气中）， $k = \omega/c = 2\pi/\lambda$  是波数。对于偶极子，电流密度很简单

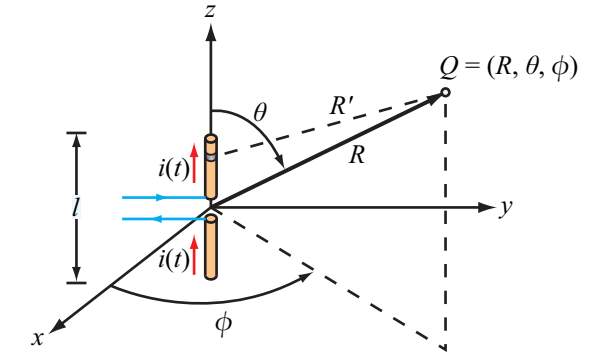


图 9-4 放置在球坐标系原点的短偶极子。

$\tilde{\mathbf{J}} = \hat{\mathbf{z}}(I_0/s)$ ，其中  $s$  是偶极子的横截面积。另外， $d\mathcal{V}' = s dz$ ，积分的极限是从  $z = -l/2$  到  $z = l/2$ 。在图 9-4 中，观察点与偶极子上给定点之间的距离  $R'$  与到其中心  $R$  的距离不同，但因为我们处理的是非常短的偶极子，所以可以设置  $R' \approx R$ 。因此，

$$\tilde{\mathbf{A}} = \frac{\mu_0}{4\pi} \frac{e^{-jkR}}{R} \int_{-l/2}^{l/2} \hat{\mathbf{z}} I_0 dz = \hat{\mathbf{z}} \frac{\mu_0}{4\pi} I_0 l \left( \frac{e^{-jkR}}{R} \right), \quad (9.3)$$

► 函数  $(e^{-jkR}/R)$  称为球面传播因子。它解释了  $1/R$  幅度随距离的衰减以及由  $e^{-jkR}$  表示的相位变化。 ◀

$\tilde{\mathbf{A}}$  的方向与电流方向（ $z$  方向）相同。

因为我们的目标是表征距天线固定距离  $R$  处的辐射功率的方向特性，所以天线方向图以球坐标系呈现（图 9-5）。它的变量  $R$ 、 $\theta$  和  $\phi$  分别称为距离、天顶角和方位角。为此，我们需要将  $\tilde{\mathbf{A}}$  写成其球坐标分量，这可以借助方程 1 来实现。(3.65c) 用球坐标表示  $\hat{\mathbf{z}}$ ：

$$\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta. \quad (9.4)$$

代入方程式后。(9.4) 代入等式。(9.3)，我们得到

$$\tilde{\mathbf{A}} = (\hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta) \frac{\mu_0 I_0 l}{4\pi} \left( \frac{e^{-jkR}}{R} \right) = \hat{\mathbf{R}} \tilde{A}_R + \hat{\boldsymbol{\theta}} \tilde{A}_\theta + \hat{\boldsymbol{\phi}} \tilde{A}_\phi, \quad (9.5)$$

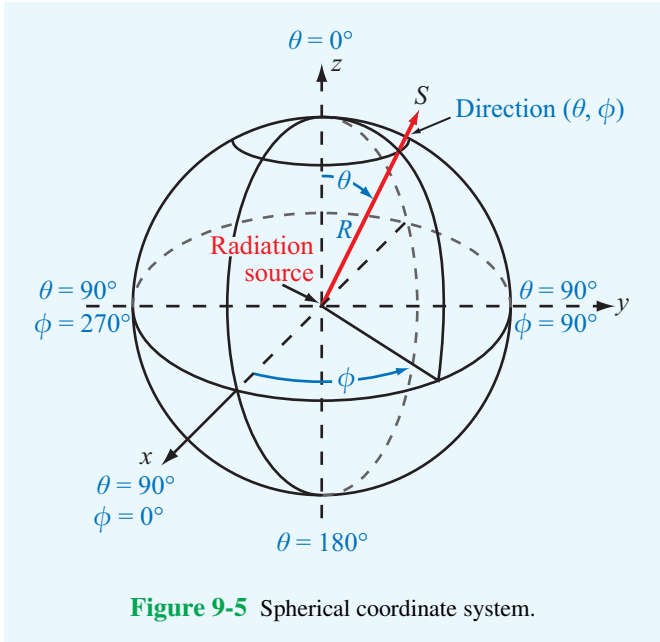


Figure 9-5 Spherical coordinate system.

where  $\eta_0 = \sqrt{\mu_0/\epsilon_0} \approx 120\pi$  ( $\Omega$ ) is the intrinsic impedance of free space. The remaining components ( $\tilde{H}_R$ ,  $\tilde{H}_\theta$ , and  $\tilde{E}_\phi$ ) are everywhere zero. **Figure 9-6** depicts the electric field lines of the wave radiated by the short dipole.

### 9-1.1 Far-Field Approximation

As was stated earlier, in most antenna applications, we are primarily interested in the radiation pattern of the antenna at great distances from the source. For the electric dipole, this corresponds to distances  $R$  so that  $R \gg \lambda$  or, equivalently,  $kR = 2\pi R/\lambda \gg 1$ . This condition allows us to neglect the terms varying as  $1/(kR)^2$  and  $1/(kR)^3$  in Eqs. (9.8a) to (9.8c) in favor of the terms varying as  $1/kR$ , which yields the far-field expressions

$$\tilde{E}_\theta = \frac{jI_0 l k \eta_0}{4\pi} \left( \frac{e^{-jkR}}{R} \right) \sin \theta \quad (\text{V/m}), \quad (9.9a)$$

$$\tilde{H}_\phi = \frac{\tilde{E}_\theta}{\eta_0} \quad (\text{A/m}), \quad (9.9b)$$

and  $\tilde{E}_R$  is negligible. At the observation point  $Q$  (**Fig. 9-4**), the wave now appears similar to a uniform plane wave with its electric and magnetic fields in phase, related by the intrinsic impedance of the medium  $\eta_0$ , and their directions orthogonal to each other and to the direction of propagation ( $\hat{\mathbf{R}}$ ). Both fields are proportional to  $\sin \theta$  and independent of  $\phi$  (which is expected from symmetry considerations).

### 9-1.2 Power Density

Given  $\tilde{\mathbf{E}}$  and  $\tilde{\mathbf{H}}$ , the **time-average Poynting vector** of the radiated wave, which is also called the **power density**, can be obtained by applying Eq. (7.100); that is,

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \Re \left( \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right) \quad (\text{W/m}^2). \quad (9.10)$$

For the short dipole, use of Eqs. (9.9a) and (9.9b) yields

$$\mathbf{S}_{\text{av}} = \hat{\mathbf{R}} S(R, \theta), \quad (9.11)$$

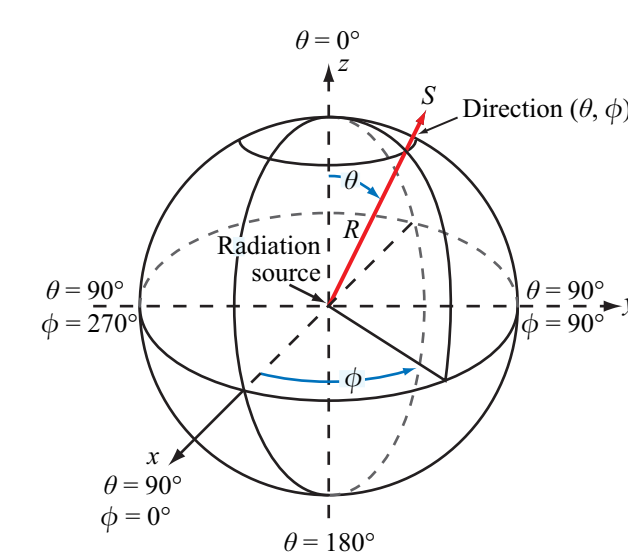


图 9-5 球坐标系。

其中  $\eta_0 = \sqrt{\mu_0/\epsilon_0} \approx 120\pi$  ( $\Omega$ ) 是自由空间的固有阻抗。其余分量 ( $\tilde{H}_R$ ,  $\tilde{H}_\theta$  和  $\tilde{E}_\phi$ ) 处处为零。图 9-6 描绘了短偶极子辐射的电场线。

### 9-1.1 远场近似

如前所述，在大多数天线应用中，我们主要对距源较远的天线的辐射方向图感兴趣。对于电偶极子，这对应于距离  $R$ ，使得  $R \gg \lambda$  或等效地  $kR = 2\pi R/\lambda \gg 1$ 。这个条件允许我们忽略等式中  $1/(kR)^2$  和  $1/(kR)^3$  变化的项。(9.8a) 到 (9.8c) 支持项变化为  $1/kR$ ，从而产生远场表达式

$$\tilde{E}_\theta = \frac{jI_0 l k \eta_0}{4\pi} \left( \frac{e^{-jkR}}{R} \right) \sin \theta \quad (\text{V/m}), \quad (9.9a)$$

$$\tilde{H}_\phi = \frac{\tilde{E}_\theta}{\eta_0} \quad (\text{A/m}), \quad (9.9b)$$

并且  $\tilde{E}_R$  可以忽略不计。在观察点  $Q$  (图 9-4)，波现在看起来类似于均匀平面波，其电场和磁场同相，与介质的固有阻抗  $\eta_0$  相关，并且它们的方向彼此正交并与传播方向 ( $\hat{\mathbf{R}}$ ) 正交。两个场都与  $\sin \theta$  成正比，并且与  $\phi$  (无关，这是出于对称性考虑而预期的)。

### 9-1.2 功率密度

给定  $\tilde{\mathbf{E}}$  和  $\tilde{\mathbf{H}}$ ，辐射波的时均坡印廷矢量，也称为功率密度，可以通过应用式 (1) 获得。(7.100); 那是，

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \Re \left( \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right) \quad (\text{W/m}^2). \quad (9.10)$$

对于短偶极子，使用方程。(9.9a) 和 (9.9b) 产量

$$\mathbf{S}_{\text{av}} = \hat{\mathbf{R}} S(R, \theta), \quad (9.11)$$

with

$$\tilde{A}_R = \frac{\mu_0 I_0 l}{4\pi} \cos \theta \left( \frac{e^{-jkR}}{R} \right), \quad (9.6a)$$

$$\tilde{A}_\theta = -\frac{\mu_0 I_0 l}{4\pi} \sin \theta \left( \frac{e^{-jkR}}{R} \right), \quad (9.6b)$$

$$\tilde{A}_\phi = 0.$$

With the spherical components of  $\tilde{\mathbf{A}}$  known, the next step is straightforward; we simply apply the free-space relationships given by Eqs. (6.85) and (6.86) as

$$\tilde{\mathbf{H}} = \frac{1}{\mu_0} \nabla \times \tilde{\mathbf{A}} \quad (9.7a)$$

and

$$\tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon_0} \nabla \times \tilde{\mathbf{H}} \quad (9.7b)$$

to obtain the expressions

$$\tilde{H}_\phi = \frac{I_0 l k^2}{4\pi} e^{-jkR} \left[ \frac{j}{kR} + \frac{1}{(kR)^2} \right] \sin \theta, \quad (9.8a)$$

$$\tilde{E}_R = \frac{2I_0 l k^2}{4\pi} \eta_0 e^{-jkR} \left[ \frac{1}{(kR)^2} - \frac{j}{(kR)^3} \right] \cos \theta, \quad (9.8b)$$

$$\tilde{E}_\theta = \frac{I_0 l k^2}{4\pi} \eta_0 e^{-jkR} \left[ \frac{j}{kR} + \frac{1}{(kR)^2} - \frac{j}{(kR)^3} \right] \sin \theta, \quad (9.8c)$$

和

$$\tilde{A}_R = \frac{\mu_0 I_0 l}{4\pi} \cos \theta \left( \frac{e^{-jkR}}{R} \right), \quad (9.6a)$$

$$\tilde{A}_\theta = -\frac{\mu_0 I_0 l}{4\pi} \sin \theta \left( \frac{e^{-jkR}}{R} \right), \quad (9.6b)$$

$$\tilde{A}_\phi = 0.$$

已知  $\tilde{\mathbf{A}}$  的球形分量后，下一步就很简单了；我们简单地应用方程给出的自由空间关系。(6.85) 和 (6.86) 为

$$\tilde{\mathbf{H}} = \frac{1}{\mu_0} \nabla \times \tilde{\mathbf{A}} \quad (9.7a)$$

和

$$\tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon_0} \nabla \times \tilde{\mathbf{H}} \quad (9.7b)$$

获得表达式

$$\tilde{H}_\phi = \frac{I_0 l k^2}{4\pi} e^{-jkR} \left[ \frac{j}{kR} + \frac{1}{(kR)^2} \right] \sin \theta, \quad (9.8a)$$

$$\tilde{E}_R = \frac{2I_0 l k^2}{4\pi} \eta_0 e^{-jkR} \left[ \frac{1}{(kR)^2} - \frac{j}{(kR)^3} \right] \cos \theta, \quad (9.8b)$$

$$\tilde{E}_\theta = \frac{I_0 l k^2}{4\pi} \eta_0 e^{-jkR} \left[ \frac{j}{kR} + \frac{1}{(kR)^2} - \frac{j}{(kR)^3} \right] \sin \theta, \quad (9.8c)$$

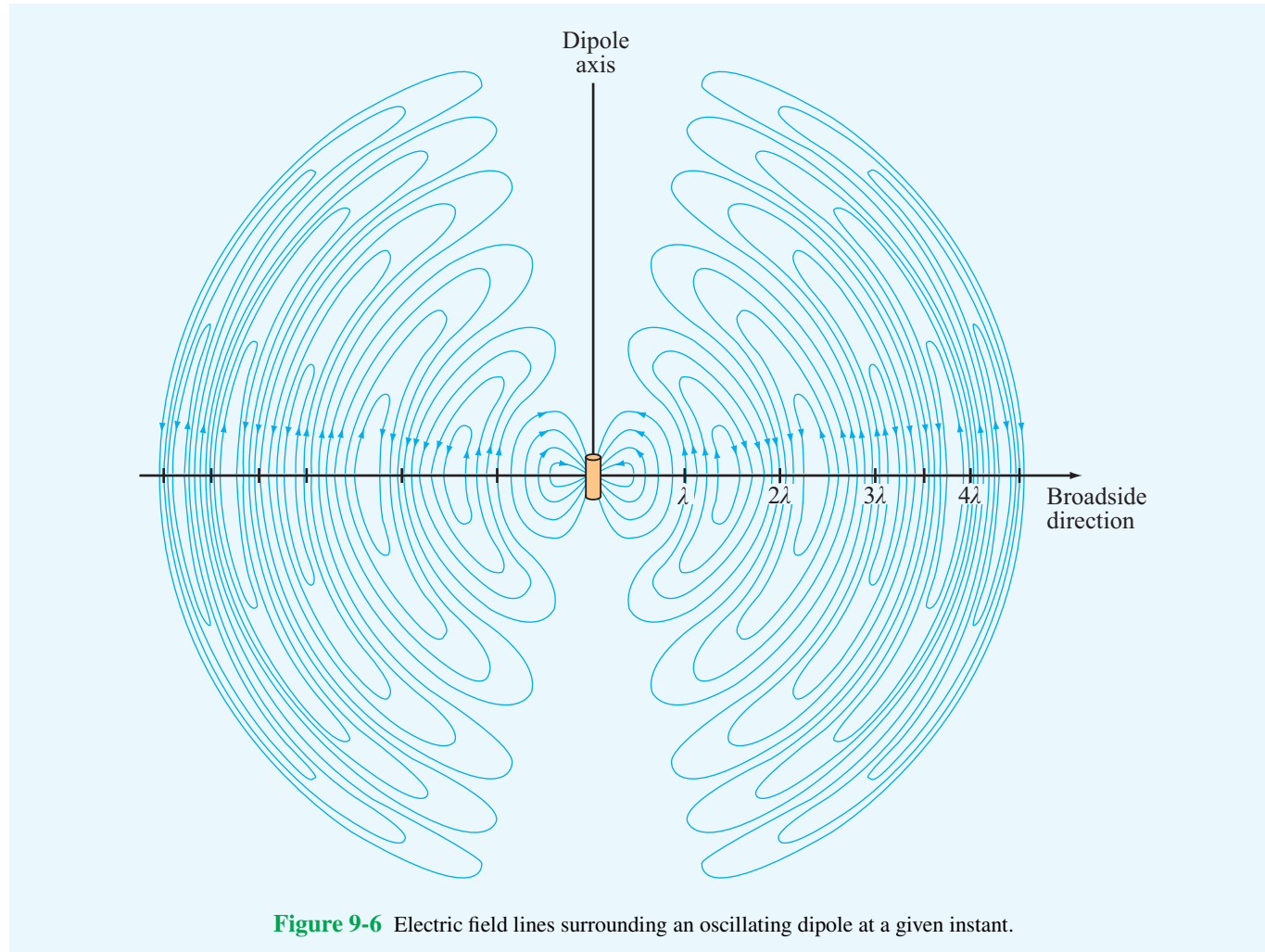


Figure 9-6 Electric field lines surrounding an oscillating dipole at a given instant.

with

$$S(R, \theta) = \left( \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta = S_0 \sin^2 \theta \quad (\text{W/m}^2). \quad (9.12)$$

The directional pattern of any antenna is described in terms of the **normalized radiation intensity**  $F(\theta, \phi)$  and is defined as the ratio of the power density  $S(R, \theta, \phi)$  at a specified range  $R$  to  $S_{\max}$ , which is the maximum value of  $S(R, \theta, \phi)$  at the same range. Thus,

$$F(\theta, \phi) = \frac{S(R, \theta, \phi)}{S_{\max}}. \quad (\text{dimensionless}) \quad (9.13)$$

For the Hertzian dipole, the  $\sin^2 \theta$  dependence in Eq. (9.12)

indicates that the radiation is maximum in the broadside direction ( $\theta = \pi/2$ ), corresponding to the azimuth plane, and is given by

$$S_{\max} = S_0 = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} = \frac{15\pi I_0^2}{R^2} \left( \frac{l}{\lambda} \right)^2 \quad (\text{W/m}^2), \quad (9.14)$$

where use was made of the relations  $k = 2\pi/\lambda$  and  $\eta_0 \approx 120\pi$ . We observe that  $S_{\max}$  is directly proportional to  $I_0^2$  and  $l^2$  (with  $l$  measured in wavelengths) and that it decreases with distance as  $1/R^2$ .

From the definition of the normalized radiation intensity given by Eq. (9.13), it follows that

$$F(\theta, \phi) = F(\theta) = \sin^2 \theta. \quad (9.15)$$

Plots of  $F(\theta)$  are shown in Fig. 9-7 in both the elevation plane (the  $\theta$  plane) and the azimuth plane ( $\phi$  plane).

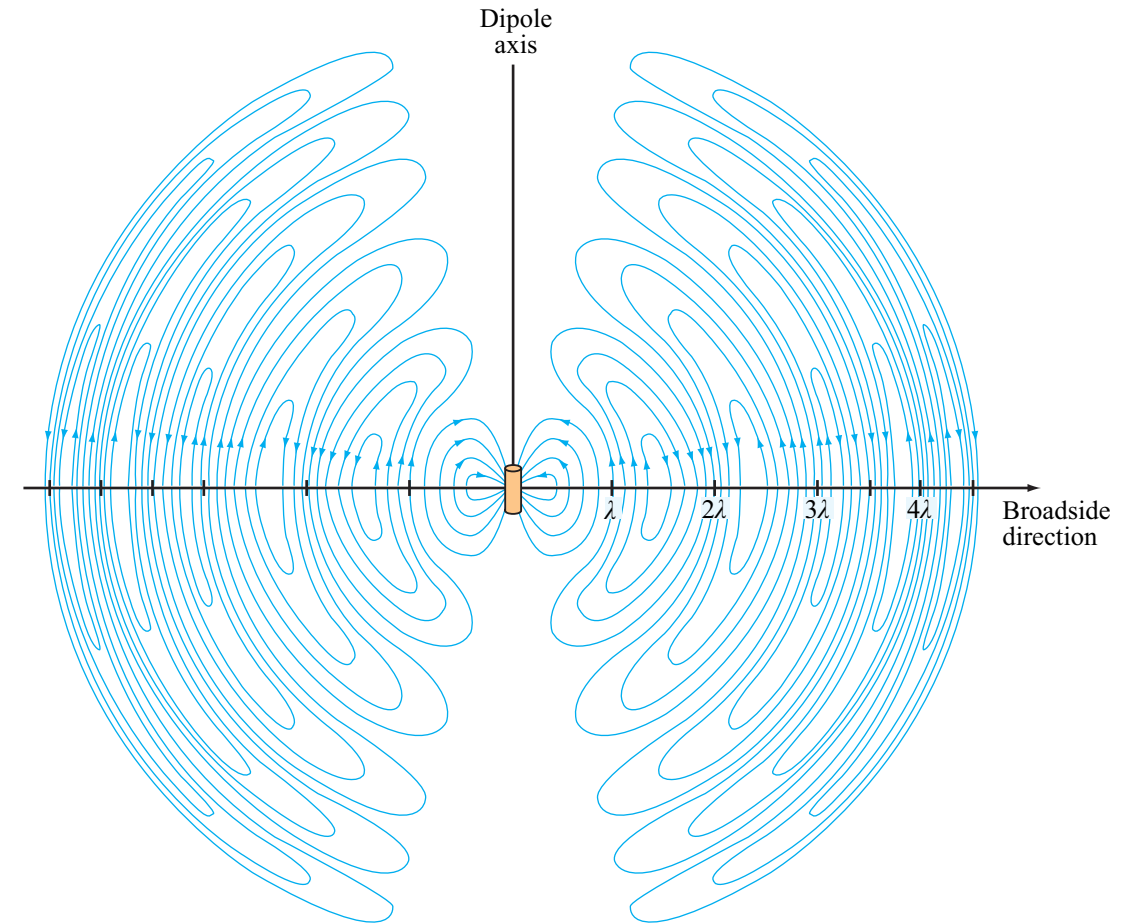


图 9-6 在给定时刻围绕振荡偶极子的电场线。

和

$$S(R, \theta) = \left( \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta = S_0 \sin^2 \theta \quad (\text{W/m}^2). \quad (9.12)$$

任何天线的方向图都用归一化辐射强度  $F(\theta, \phi)$  来描述，并定义为在指定范围  $R$  到  $S_{\max}$  上功率密度  $S(R, \theta, \phi)$  的比率，它是同一范围内  $S(R, \theta, \phi)$  的最大值。因此，

$$F(\theta, \phi) = \frac{S(R, \theta, \phi)}{S_{\max}}. \quad (\text{dimensionless}) \quad (9.13)$$

对于赫兹偶极子，方程中的  $\sin^2 \theta$  依赖性：(9.12)

表示辐射在对应于方位平面的侧边方向 ( $\theta = \pi/2$ ) 上最大，由下式给出

$$S_{\max} = S_0 = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} = \frac{15\pi I_0^2}{R^2} \left( \frac{l}{\lambda} \right)^2 \quad (\text{W/m}^2), \quad (9.14)$$

其中使用关系  $k = 2\pi/\lambda$  和  $\eta_0 \approx 120\pi$ 。我们观察到  $S_{\max}$  与  $I_0^2$  和  $l^2$  (成正比，其中  $l$  以波长) 测量，并且它随着距离的增加而减小，即  $1/R^2$ 。

根据方程给出的归一化辐射强度的定义：(9.13)，由此可知

$$F(\theta, \phi) = F(\theta) = \sin^2 \theta. \quad (9.15)$$

$F(\theta)$  的图如图 9-7 所示，在仰角平面 ( $\theta$  平面) 和方位角平面 ( $\phi$  平面) 中。

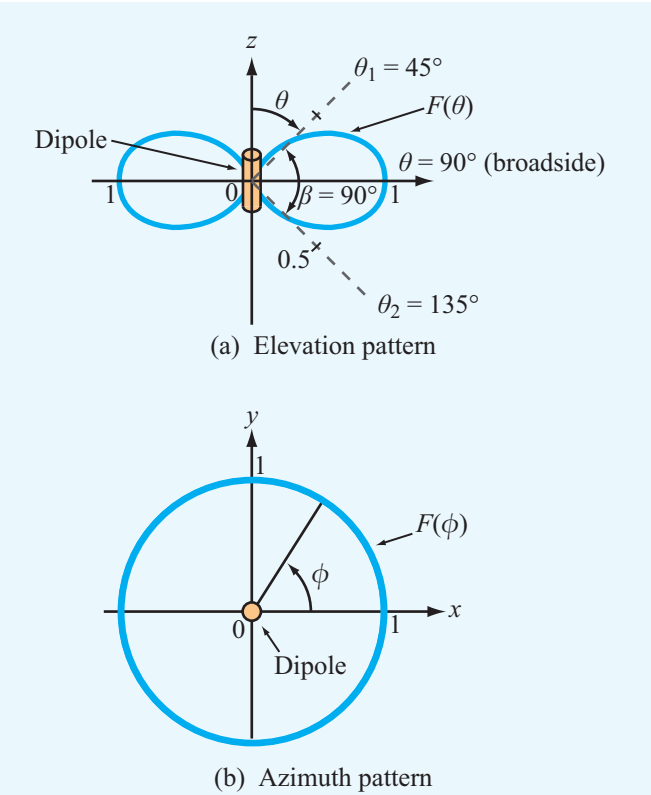


Figure 9-7 Radiation patterns of a short dipole.

► No energy is radiated by the short dipole along the direction of the dipole axis, and maximum radiation ( $F = 1$ ) occurs in the **broadside direction** ( $\theta = 90^\circ$ ). Since  $F(\theta)$  is independent of  $\phi$ , the pattern is doughnut-shaped in  $\theta$ - $\phi$  space. ◀

**Concept Question 9-1:** What does it mean to say that most antennas are reciprocal devices?

**Concept Question 9-2:** What is the radiated wave like in the far-field region of the antenna?

**Concept Question 9-3:** In a Hertzian dipole, what is the underlying assumption about the current flowing through the wire?

**Concept Question 9-4:** Outline the basic steps used to relate the current in a wire to the radiated power density.

**Exercise 9-1:** A 1-m long dipole is excited by a 5 MHz current with an amplitude of 5 A. At a distance of 2 km, what is the power density radiated by the antenna along its broadside direction?

**Answer:**  $S_0 = 8.2 \times 10^{-8} \text{ W/m}^2$ . (See [EM](#).)

9-2 Antenna Radiation Characteristics

An **antenna pattern** describes the far-field directional properties of an antenna when measured at a fixed distance from the antenna. In general, the antenna pattern is a three-dimensional plot that displays the strength of the radiated field or power density as a function of direction—with direction being specified by the zenith angle  $\theta$  and the azimuth angle  $\phi$ .

► By virtue of reciprocity, a receiving antenna has the same directional antenna pattern as the pattern that it exhibits when operated in the transmission mode. ◀

Consider a transmitting antenna placed at the origin of the observation sphere shown in **Fig. 9-8**. The differential power radiated by the antenna through an elemental area  $dA$  is

$$dP_{\text{rad}} = \mathbf{S}_{\text{av}} \cdot d\mathbf{A} = \mathbf{S}_{\text{av}} \cdot \hat{\mathbf{R}} dA = S dA \quad (\text{W}), \quad (9.16)$$

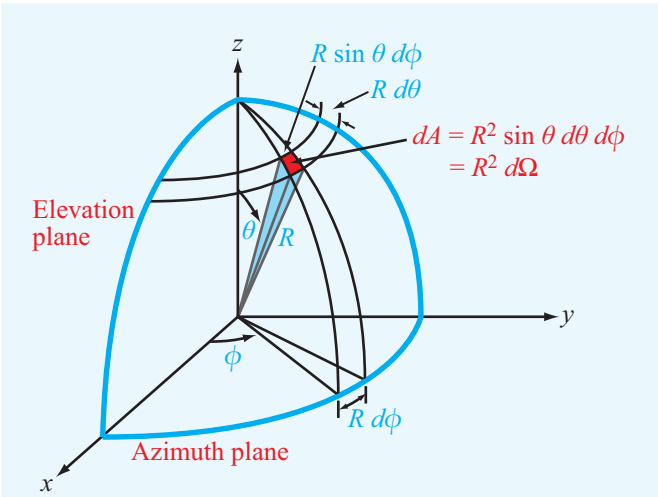


Figure 9-8 Definition of solid angle  $d\Omega = \sin \theta d\theta d\phi$ .

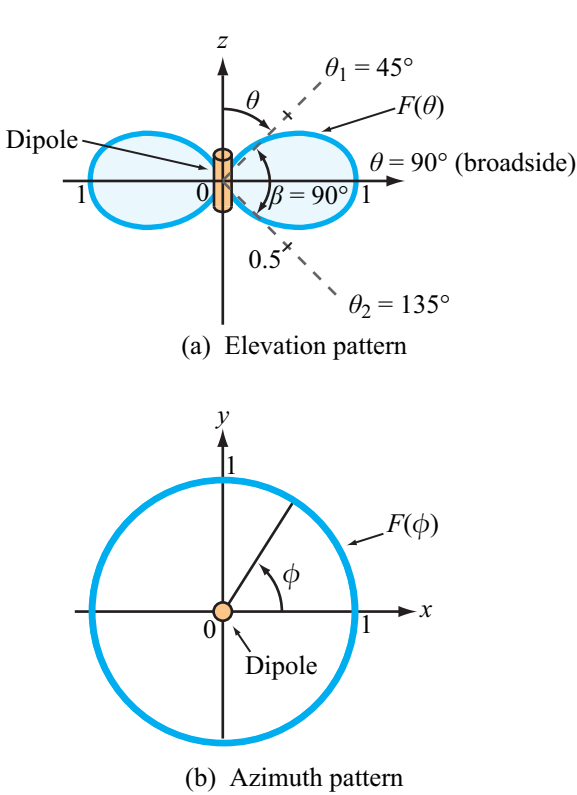


图 9-7 短偶极子的辐射方向图。

► 短偶极子沿偶极轴方向不辐射能量，最大辐射 ( $F = 1$ ) 出现在侧边方向 ( $\theta = 90^\circ$ )。由于  $F(\theta)$  独立于  $\phi$ ，因此该模式在  $\theta$ - $\phi$  空间中呈环形。 ◀

概念问题 9-1：大多数天线是互易设备是什么意思？

概念问题 9-2：天线远场区域的辐射波是什么样的？

概念问题 9-3：在赫兹偶极子中，关于流经电线的电流的基本假设是什么？

概念问题 9-4：概述用于将电线中的电流与辐射功率密度联系起来的步骤。

练习9-1：一个1米长的偶极子被5MHz、5A幅值的电流激励。在2km距离处，天线沿其侧面方向辐射的功率密度是多少？

答案：  $S_0 = 8.2 \times 10^{-8} \text{ W/m}^2$ 。（参见<sup>EM</sup>。）

9-2 天线辐射特性

天线方向图描述了在距天线固定距离处测量时天线的远场方向特性。一般来说，天线方向图是一个三维图，显示辐射场的强度或功率密度作为方向的函数，方向由天顶角  $\theta$  和方位角  $\phi$  指定。

► 由于互易性，接收天线具有与其在传输模式下操作时表现出的方向图相同的方向性天线方向图。◀

考虑如图 9-8 所示放置在观测球原点的发射天线。天线通过单元区域辐射的差分功率  $dA$  为

$$dP_{\text{rad}} = \mathbf{S}_{\text{av}} \cdot d\mathbf{A} = \mathbf{S}_{\text{av}} \cdot \hat{\mathbf{R}} dA = S dA \quad (\text{W}), \quad (9.16)$$

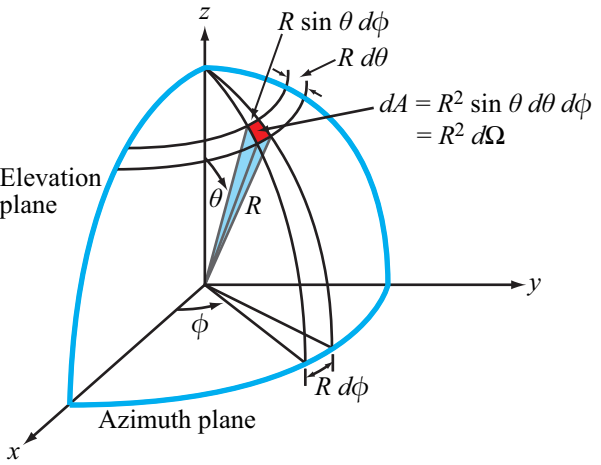
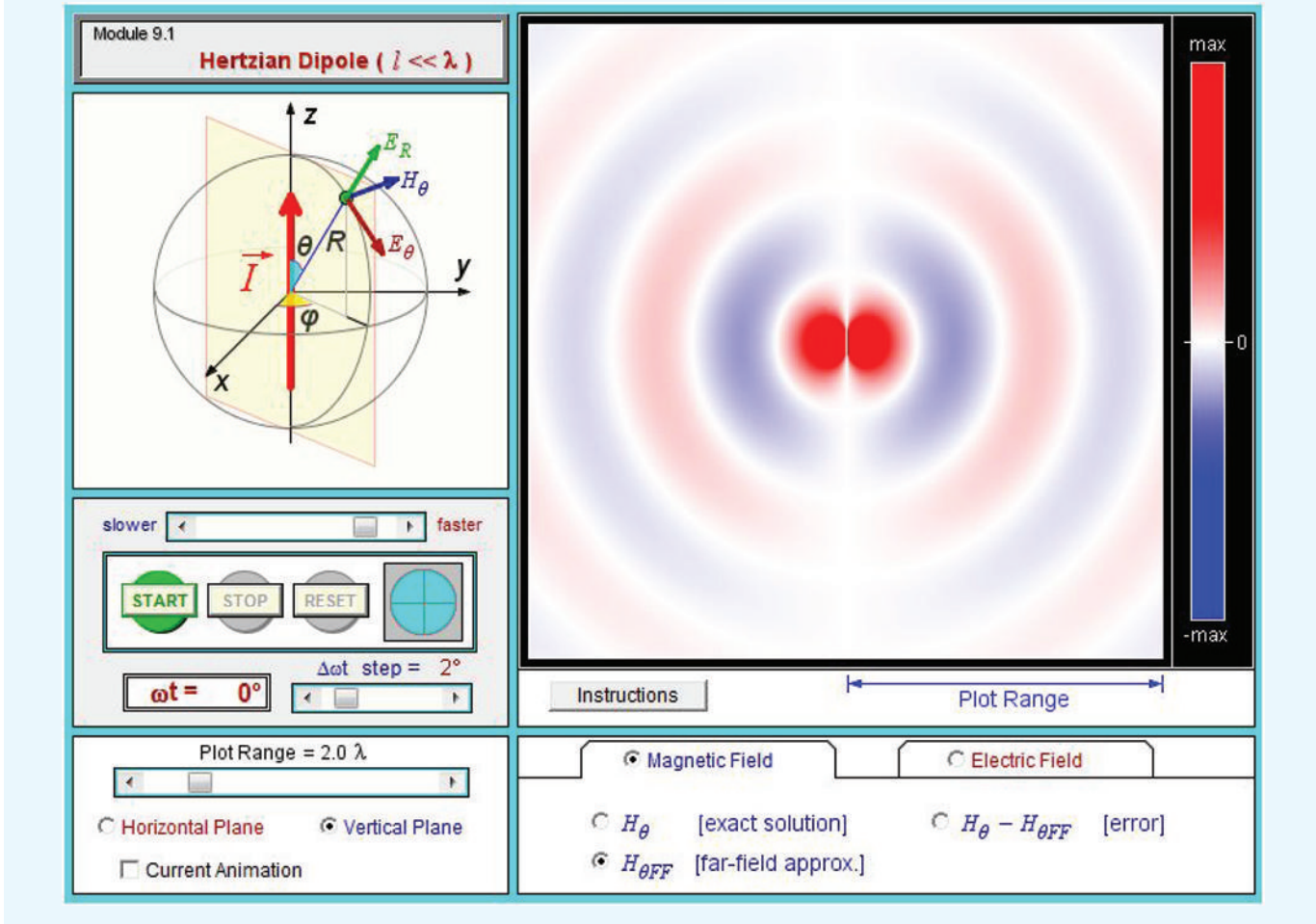


图 9-8 立体角  $d\Omega = \sin \theta d\theta d\phi$  的定义。

**Module 9.1 Hertzian Dipole ( $l \ll \lambda$ )** For a short dipole oriented along the  $z$  axis, this module displays the field distributions for  $\mathbf{E}$  and  $\mathbf{H}$  in both the horizontal and vertical planes. It can also animate the radiation process and current flow through the dipole.



where  $S$  is the radial component of the time-average Poynting vector  $\mathbf{S}_{av}$ . In the far-field region of any antenna,  $\mathbf{S}_{av}$  is always in the radial direction. In a spherical coordinate system,

$$dA = R^2 \sin \theta \, d\theta \, d\phi, \quad (9.17)$$

and the **solid angle**  $d\Omega$  associated with  $dA$ , which is defined as the subtended area divided by  $R^2$ , is given by

$$d\Omega = \frac{dA}{R^2} = \sin \theta \, d\theta \, d\phi \quad (\text{sr}). \quad (9.18)$$

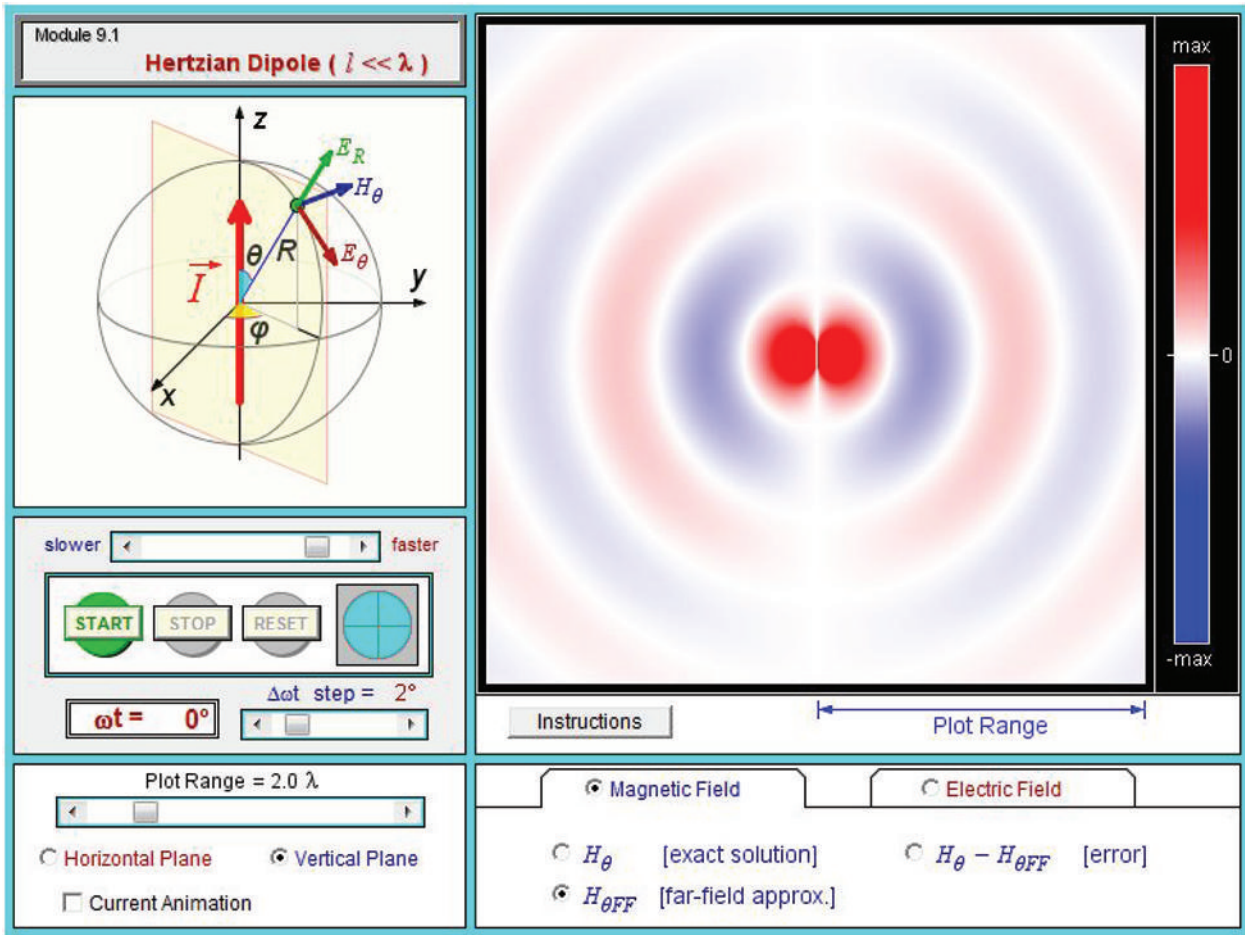
Note that, whereas a planar angle is measured in radians and the angular measure of a complete circle is  $2\pi$  (rad), a solid angle is measured in **steradians** (sr), and the angular measure for a spherical surface is  $\Omega = (4\pi R^2)/R^2 = 4\pi$  (sr). The solid angle of a hemisphere is  $2\pi$  (sr).

Using the relation  $dA = R^2 d\Omega$ ,  $dP_{rad}$  can be rewritten as

$$dP_{rad} = R^2 S(R, \theta, \phi) \, d\Omega. \quad (9.19)$$

The total power radiated by an antenna through a spherical surface at a fixed distance  $R$  is obtained by integrating

模块 9.1 赫兹偶极子 ( $l \ll \lambda$ ) 对于沿  $z$  轴定向的短偶极子，此模块显示  $\mathbf{E}$  和  $\mathbf{H}$  在水平和垂直平面上的场分布。它还可以动画化辐射过程和流经偶极子的电流。



其中  $S$  是时间平均坡印廷向量  $\mathbf{S}_{av}$  的径向分量。在任何天线的远场区域， $\mathbf{S}_{av}$  始终处于径向方向。在球坐标系中，

$$dA = R^2 \sin \theta \, d\theta \, d\phi, \quad (9.17)$$

与  $dA$  相关的立体角  $d\Omega$  定义为所对面积除以  $R^2$ ，由下式给出

$$d\Omega = \frac{dA}{R^2} = \sin \theta \, d\theta \, d\phi \quad (\text{sr}). \quad (9.18)$$

请注意，平面角以弧度为单位测量，完整圆的角度测量为  $2\pi$  (rad)，而立体角以球面度 (sr) 测量，球面的角度测量为  $\Omega = (4\pi R^2)/R^2 = 4\pi$  (sr)。半球的立体角为  $2\pi$  (sr)。

使用关系  $dA = R^2 d\Omega$ ， $dP_{rad}$  可以重写为

$$dP_{rad} = R^2 S(R, \theta, \phi) \, d\Omega. \quad (9.19)$$

天线在固定距离  $R$  处通过球面辐射的总功率可通过积分获得

Eq. (9.19) over that surface:

$$\begin{aligned}
 P_{\text{rad}} &= R^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} S(R, \theta, \phi) \sin \theta \, d\theta \, d\phi \\
 &= R^2 S_{\text{max}} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} F(\theta, \phi) \sin \theta \, d\theta \, d\phi \\
 &= R^2 S_{\text{max}} \iint_{4\pi} F(\theta, \phi) \, d\Omega \quad (\text{W}), \quad (9.20)
 \end{aligned}$$

where  $F(\theta, \phi)$  is the normalized radiation intensity defined by Eq. (9.13). The  $4\pi$  symbol under the integral sign is used as an abbreviation for the indicated limits on  $\theta$  and  $\phi$ . Formally,  $P_{\text{rad}}$  is called the **total radiated power**.

9-2.1 Antenna Pattern

Each specific combination of the zenith angle  $\theta$  and the azimuth angle  $\phi$  denotes a specific direction in the spherical coordinate system of Fig. 9-8. The normalized radiation intensity  $F(\theta, \phi)$  characterizes the directional pattern of the energy radiated by an antenna, and a plot of  $F(\theta, \phi)$  as a function of both  $\theta$  and  $\phi$  constitutes a three-dimensional pattern—an example of which is shown in Fig. 9-9.

Often, it is of interest to characterize the variation of  $F(\theta, \phi)$  in the form of two-dimensional plots in specific planes in the spherical coordinate system. The two planes most commonly specified for this purpose are the elevation and azimuth planes. The **elevation plane**, also called the  $\theta$  plane, is a plane corresponding to a constant value of  $\phi$ . For example,  $\phi = 0$  defines the  $x$ - $z$  plane and  $\phi = 90^\circ$  defines the  $y$ - $z$  plane, both of which are elevation planes (Fig. 9-8). A plot of  $F(\theta, \phi)$  versus  $\theta$  in either of these planes constitutes a two-dimensional pattern in the elevation plane. This is not to imply, however, that the elevation-plane pattern is necessarily the same in all elevation planes.

The **azimuth plane**, also called the  $\phi$  plane, is specified by  $\theta = 90^\circ$  and corresponds to the  $x$ - $y$  plane. The elevation and azimuth planes are often called the two **principal planes** of the spherical coordinate system.

Some antennas exhibit highly directive patterns with narrow beams, in which case it is often convenient to plot the antenna pattern on a decibel scale by expressing  $F$  in decibels:

$$F \text{ (dB)} = 10 \log F.$$

As an example, the antenna pattern shown in Fig. 9-10(a) is plotted on a decibel scale in polar coordinates with its intensity as the radial variable. This format permits a convenient visual interpretation of the directional distribution of the **radiation lobes**.

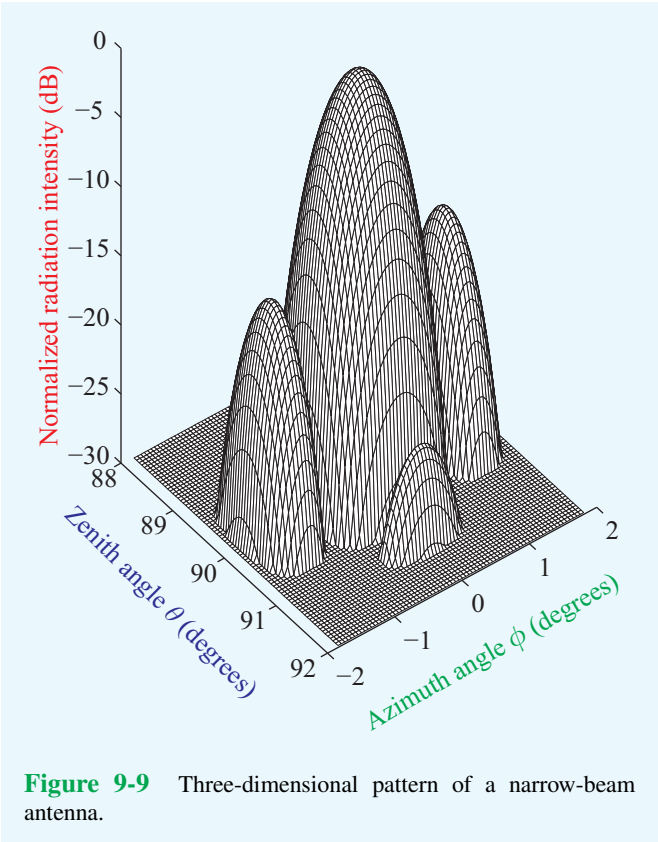


Figure 9-9 Three-dimensional pattern of a narrow-beam antenna.

Another format commonly used for inspecting the pattern of a narrow-beam antenna is the rectangular display shown in Fig. 9-10(b), which permits the pattern to be easily expanded by changing the scale of the horizontal axis. These plots represent the variation in only one plane in the observation sphere: the  $\phi = 0$  plane. Unless the pattern is symmetrical in  $\phi$ , additional patterns are required to define the overall variation of  $F(\theta, \phi)$  with  $\theta$  and  $\phi$ .

Strictly speaking, the polar angle  $\theta$  is always positive, being defined over the range from  $0^\circ$  ( $z$  direction) to  $180^\circ$  ( $-z$  direction), yet the  $\theta$  axis in Fig. 9-10(b) is shown to have both positive and negative values. This is not a contradiction, but rather a different form of plotting antenna patterns. The right-hand half of the plot represents the variation of  $F$  (dB) with  $\theta$  because  $\theta$  is increased in a clockwise direction in the  $x$ - $z$  plane [see inset in Fig. 9-10(b)], corresponding to  $\phi = 0$ . The left-hand half of the plot represents the variation of  $F$  (dB) with  $\theta$  as  $\theta$  is increased in a counterclockwise direction at  $\phi = 180^\circ$ . Thus, a negative  $\theta$  value simply denotes that the direction  $(\theta, \phi)$  is in the left-hand half of the  $x$ - $z$  plane.

The pattern shown in Fig. 9-10(a) indicates that the antenna is fairly directive, since most of the energy is radiated through

等式。(9.19)在该表面上：

$$\begin{aligned}
 P_{\text{rad}} &= R^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} S(R, \theta, \phi) \sin \theta \, d\theta \, d\phi \\
 &= R^2 S_{\text{max}} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} F(\theta, \phi) \sin \theta \, d\theta \, d\phi \\
 &= R^2 S_{\text{max}} \iint_{4\pi} F(\theta, \phi) \, d\Omega \quad (\text{W}), \quad (9.20)
 \end{aligned}$$

其中  $F(\theta, \phi)$  是由方程式定义的归一化辐射强度。(9.13)。积分符号下的  $4\pi$  符号用作  $\theta$  和  $\phi$  指示极限的缩写。形式上,  $P_{\text{rad}}$  称为总辐射功率。

9-2.1 天线方向图

天顶角 $\theta$ 和方位角 $\phi$ 的每个特定组合表示图9-8的球面坐标系中的特定方向。归一化辐射强度  $F(\theta, \phi)$  表征了天线辐射能量的方向图,  $F(\theta, \phi)$  作为  $\theta$  和  $\phi$  的函数的绘图构成了一个三维图, 其示例如图 9-9 所示。

通常, 人们感兴趣的是在球坐标系中的特定平面上以二维图的形式描述  $F(\theta, \phi)$  的变化。为此目的最常指定的两个平面是仰角平面和方位角平面。高程平面也称为 $\theta$ 平面, 是与常数 $\phi$ 相对应的平面。例如,  $\phi = 0$  定义  $x$ - $z$  平面,  $\phi = 90^\circ$  定义  $y$ - $z$  平面, 这两个平面都是高程平面(图 9-8)。这些平面中任一平面中的  $F(\theta, \phi)$  与  $\theta$  的关系图构成了高程平面中的二维图案。然而, 这并不意味着所有高程平面中的高程平面图案必须相同。

方位角平面, 也称为  $\phi$  平面, 由  $\theta = 90^\circ$  指定, 对应于  $x$ - $y$  平面。仰角平面和方位角平面通常被称为球坐标系的两个主平面。

一些天线表现出具有窄波束的高度定向方向图, 在这种情况下, 通过以分贝表示  $F$ , 通常可以方便地在分贝刻度上绘制天线方向图:

$$F \text{ (dB)} = 10 \log F.$$

作为一个例子, 图9-10(a)所示的天线方向图是在极坐标中以分贝刻度绘制的, 其强度作为径向变量。这种格式可以方便地直观地解释辐射瓣的方向分布。

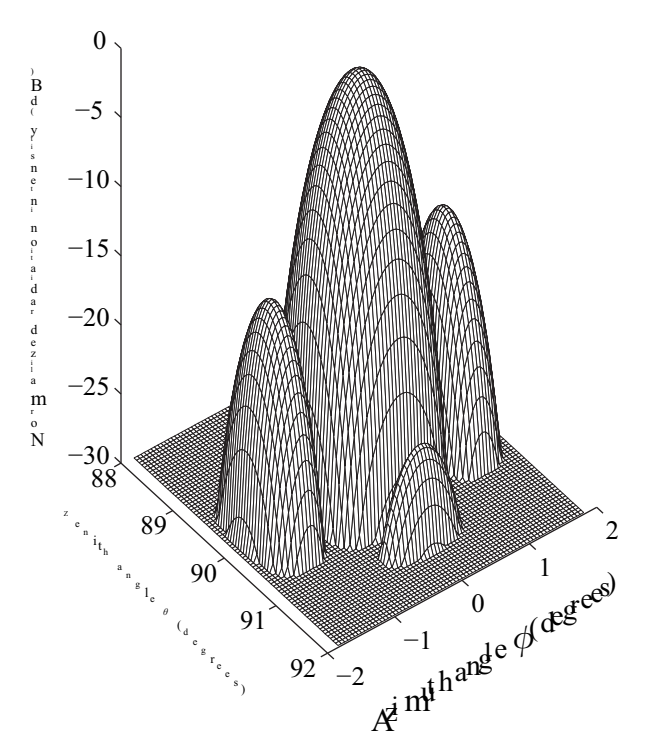


图9-9 窄波束天线的三维方向图。

另一种常用的检查窄波束天线方向图的格式是图 9-10(b)所示的矩形显示, 它允许通过改变水平轴的比例来轻松扩展方向图。这些图仅表示观测范围内一个平面的变化:  $\phi = 0$  平面。除非模式在  $\phi$  中是对称的, 否则需要额外的模式来定义  $F(\theta, \phi)$  与  $\theta$  和  $\phi$  的整体变化。

严格来说, 极角 $\theta$ 始终为正, 定义范围为 $0^\circ$  ( $z$ 方向)到 $180^\circ$  ( $-z$ 方向), 但图9-10(b)中的 $\theta$ 轴显示为既有正值又有负值。这并不矛盾, 而是绘制天线方向图的不同形式。图的右半部分表示  $F$  (dB) 随  $\theta$  的变化, 因为  $\theta$  在  $x$ - $z$  平面中沿顺时针方向增加[参见图 9-10(b)中的插图], 对应于  $\phi = 0$ 。图的左半部分表示  $F$  (dB) 随  $\theta$  的变化, 因为  $\theta$  在  $\phi = 180^\circ$  处沿逆时针方向增加 $180^\circ$ 。因此, 负  $\theta$  值仅表示方向  $(\theta, \phi)$  位于  $x$ - $z$  平面的左半部分。

图 9-10(a) 所示的图案表明天线具有相当的方向性, 因为大部分能量是通过

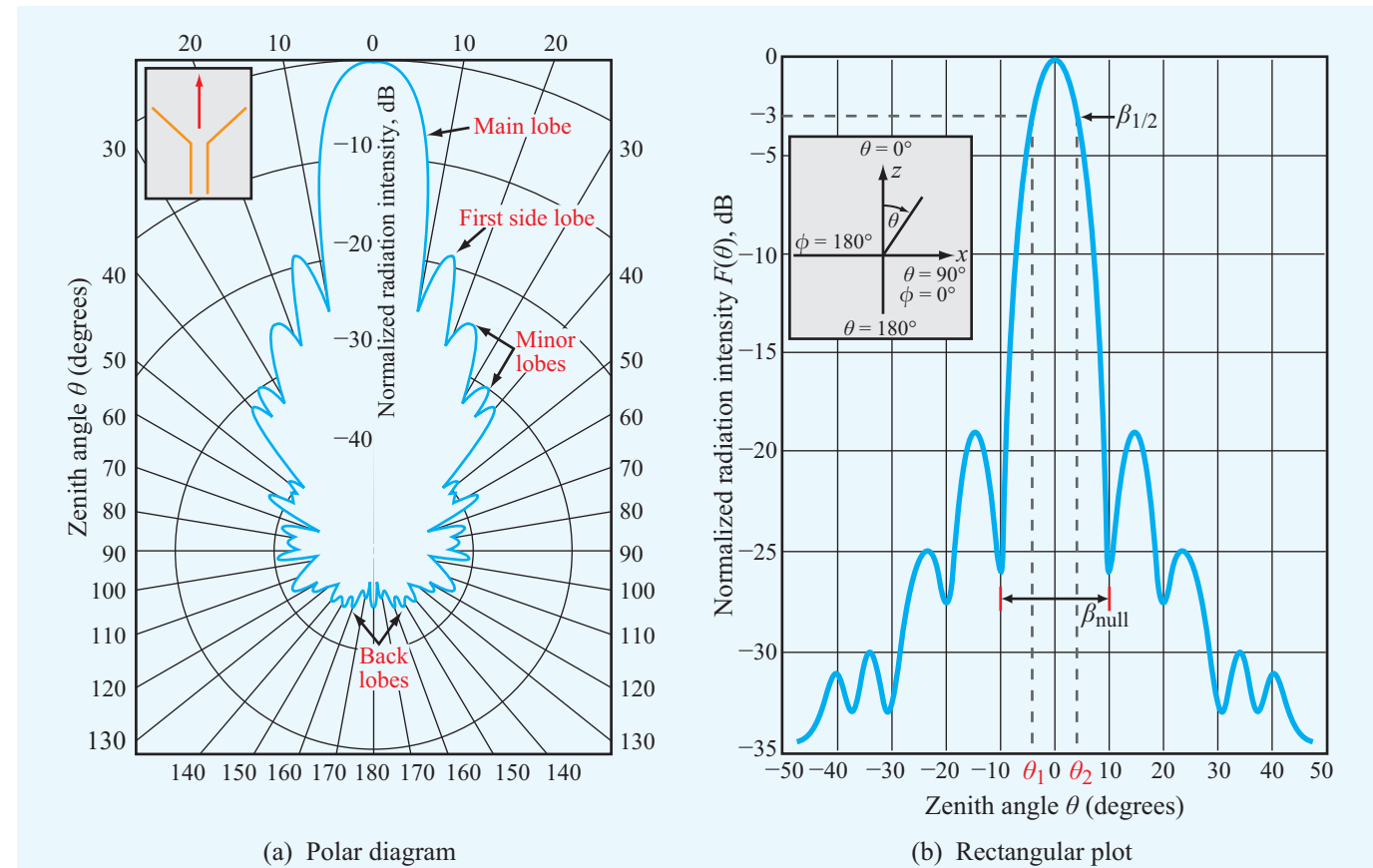


Figure 9-10 Representative plots of the normalized radiation pattern of a microwave antenna in (a) polar form and (b) rectangular form.

a narrow sector called the **main lobe**. In addition to the main lobe, the pattern exhibits several **side lobes** and **back lobes** as well. For most applications, these extra lobes are considered undesirable because they represent wasted energy for transmitting antennas and potential interference directions for receiving antennas.

### 9-2.2 Beam Dimensions

For an antenna with a single main lobe, the **pattern solid angle**  $\Omega_p$  describes the equivalent width of the main lobe of the antenna pattern (Fig. 9-11). It is defined as the integral of the normalized radiation intensity  $F(\theta, \phi)$  over a sphere:

$$\Omega_p = \iint_{4\pi} F(\theta, \phi) d\Omega \quad (\text{sr}). \quad (9.21)$$

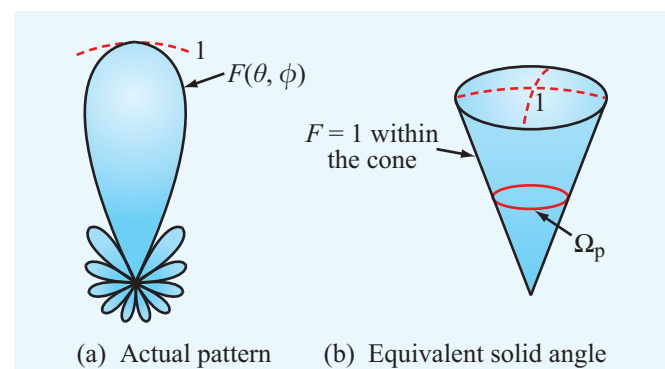


Figure 9-11 The pattern solid angle  $\Omega_p$  defines an equivalent cone over which all of the radiation of the actual antenna is concentrated with uniform intensity equal to the maximum of the actual pattern.

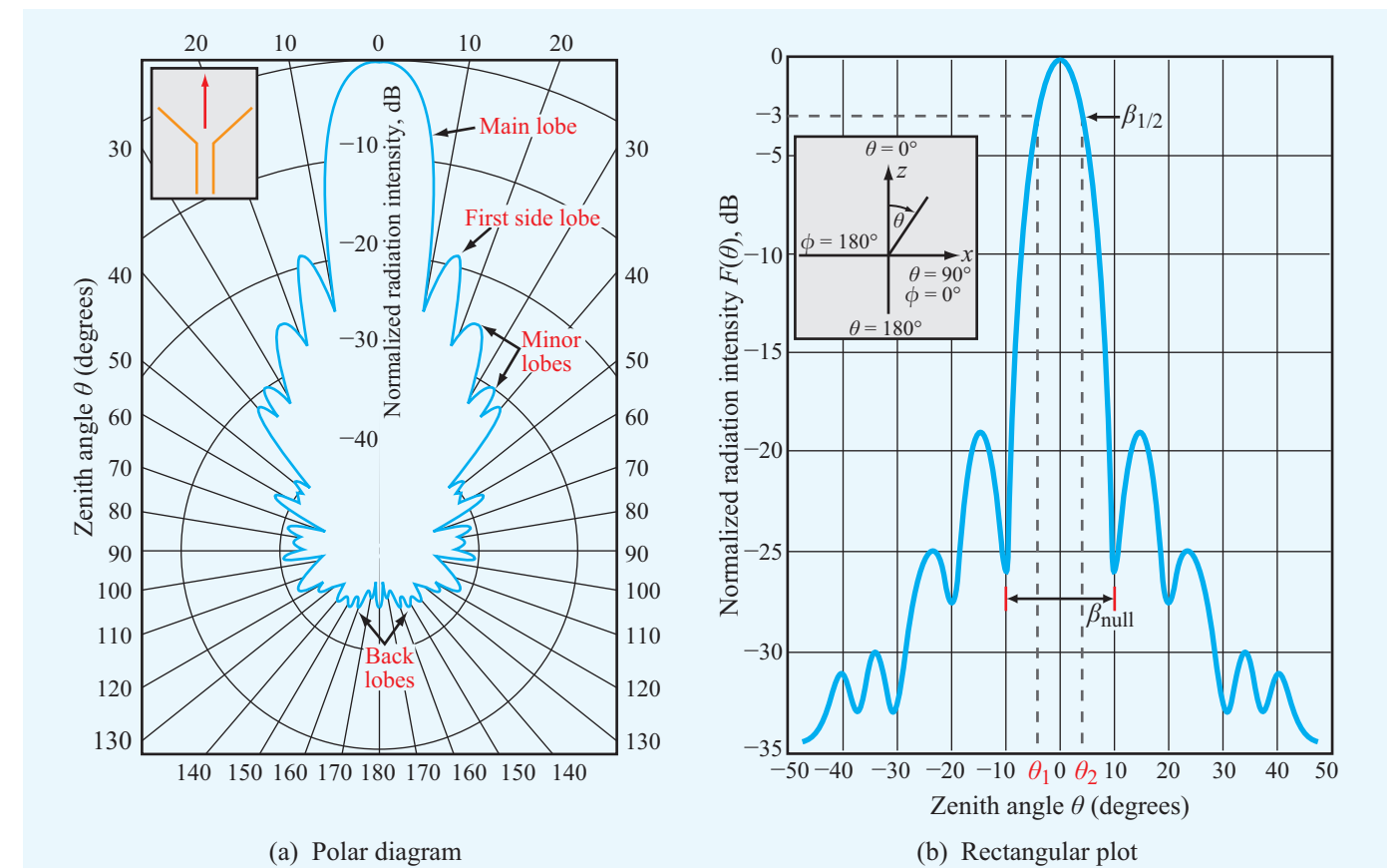


图9-10 微波天线归一化辐射方向图两种表示：(a) 极坐标图；(b) 直角坐标图。

称为主瓣的狭窄部分。除了主瓣之外，该模式还表现出多个旁瓣和后瓣。对于大多数应用来说，这些额外的波瓣被认为是不受欢迎的，因为它们代表了发射天线的能量浪费和接收天线的潜在干扰方向。

### 9-2.2 波束立体角

对于单主瓣天线，波束立体角  $\Omega_p$  描述了天线方向图主瓣的等效宽度（图9-11）。它被定义为球面上归一化辐射强度  $F(\theta, \phi)$  的积分：

$$\Omega_p = \iint_{4\pi} F(\theta, \phi) d\Omega \quad (\text{sr}). \quad (9.21)$$

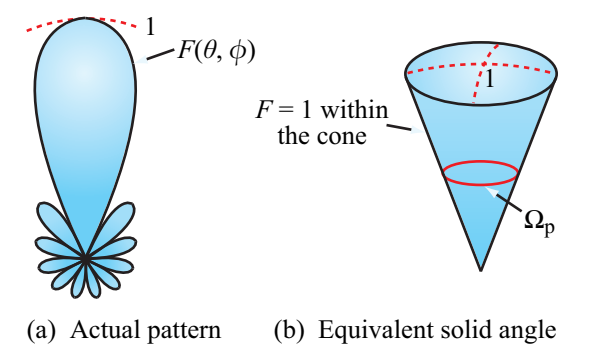


图 9-11 波束立体角  $\Omega_p$  定义了一个等效圆锥体，实际天线的所有辐射都集中在该圆锥体上，其均匀强度等于实际方向图的最大值。

► For an isotropic antenna with  $F(\theta, \phi) = 1$  in all directions,  $\Omega_p = 4\pi$  (sr). ◀

The pattern solid angle characterizes the directional properties of the three-dimensional radiation pattern. To characterize the width of the main lobe in a given plane, the term used is **beamwidth**. The **half-power beamwidth**, or simply the beamwidth  $\beta$ , is defined as the angular width of the main lobe between the two angles at which the magnitude of  $F(\theta, \phi)$  is equal to half of its peak value (or  $-3$  dB on a decibel scale). For example, for the pattern displayed in Fig. 9-10(b),  $\beta$  is given by

$$\beta = \theta_2 - \theta_1, \quad (9.22)$$

where  $\theta_1$  and  $\theta_2$  are the **half-power angles** at which  $F(\theta, 0) = 0.5$  (with  $\theta_2$  denoting the larger value and  $\theta_1$  denoting the smaller one, as shown in the figure). If the pattern is symmetrical and the peak value of  $F(\theta, \phi)$  is at  $\theta = 0$ , then  $\beta = 2\theta_2$ . For the short-dipole pattern shown earlier in Fig. 9-7(a),  $F(\theta)$  is maximum at  $\theta = 90^\circ$ ,  $\theta_2$  is at  $135^\circ$ , and  $\theta_1$  is at  $45^\circ$ . Hence,  $\beta = 135^\circ - 45^\circ = 90^\circ$ . The beamwidth  $\beta$  is also known as the **3 dB beamwidth**. In addition to the half-power beamwidth, other beam dimensions may be of interest for certain applications, such as the **null beamwidth**  $\beta_{\text{null}}$ , which is the angular width between the first nulls on the two sides of the peak (Fig. 9-10(b)).

9-2.3 Antenna Directivity

The **directivity**  $D$  of an antenna is defined as the ratio of its maximum normalized radiation intensity,  $F_{\text{max}}$  (which by definition is equal to 1), to the average value of  $F(\theta, \phi)$  over all directions ( $4\pi$  space):

$$D = \frac{F_{\text{max}}}{F_{\text{av}}} = \frac{1}{\frac{1}{4\pi} \iint_{4\pi} F(\theta, \phi) d\Omega} = \frac{4\pi}{\Omega_p}. \quad (9.23)$$

(dimensionless)

Here  $\Omega_p$  is the pattern solid angle defined by Eq. (9.21). Thus, the narrower  $\Omega_p$  of an antenna pattern is, the greater is the directivity. For an isotropic antenna,  $\Omega_p = 4\pi$ ; hence, its directivity  $D_{\text{iso}} = 1$ .

By using Eq. (9.20) in Eq. (9.23),  $D$  can be expressed as

$$D = \frac{4\pi R^2 S_{\text{max}}}{P_{\text{rad}}} = \frac{S_{\text{max}}}{S_{\text{av}}}, \quad (9.24)$$

where  $S_{\text{av}} = P_{\text{rad}}/(4\pi R^2)$  is the average value of the radiated power density and is equal to the total power radiated by

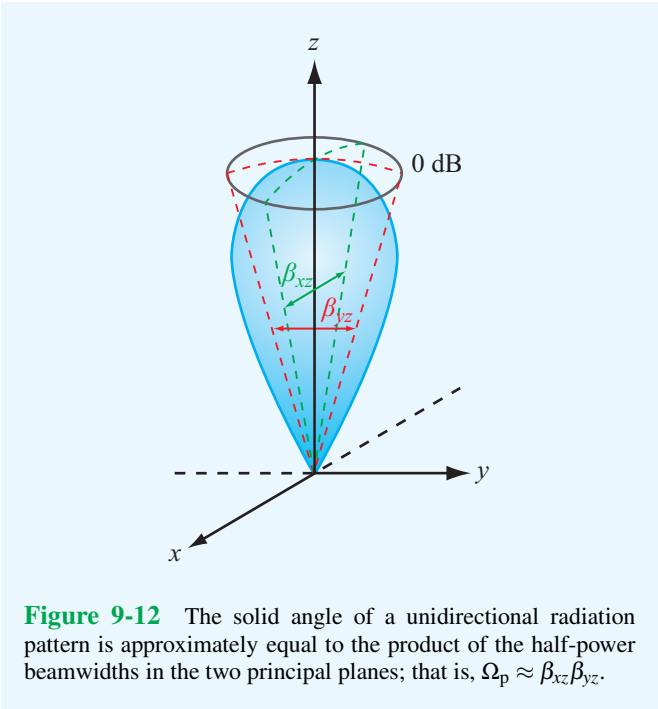


Figure 9-12 The solid angle of a unidirectional radiation pattern is approximately equal to the product of the half-power beamwidths in the two principal planes; that is,  $\Omega_p \approx \beta_{xz}\beta_{yz}$ .

the antenna,  $P_{\text{rad}}$ , divided by the surface area of a sphere of radius  $R$ .

► Since  $S_{\text{av}} = S_{\text{iso}}$ , where  $S_{\text{iso}}$  is the power density radiated by an isotropic antenna,  $D$  represents the ratio of the maximum power density radiated by the antenna to the power density radiated by an isotropic antenna, where both are measured at the same range  $R$  and excited by the same amount of input power. ◀

Usually,  $D$  is expressed in decibels:  $*D$  (dB) =  $10 \log D$ . For an antenna with a single main lobe pointing in the  $z$  direction shown in Fig. 9-12,  $\Omega_p$  may be approximated as the product of the half-power beamwidths  $\beta_{xz}$  and  $\beta_{yz}$  (in radians):

$$\Omega_p \approx \beta_{xz}\beta_{yz}, \quad (9.25)$$

and

$$D = \frac{4\pi}{\Omega_p} \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \quad \text{(single main lobe)} \quad (9.26)$$

\*A note of caution: Even though we often express certain dimensionless quantities in decibels, we should always convert their decibel values to natural values before using them in the relations given in this chapter.

► 对于各方向  $F(\theta, \phi) = 1$  的各向同性天线,  $\Omega_p = 4\pi$  (sr). ◀

方向图立体角表征了三维辐射方向图的方向特性。为了表征给定平面中主瓣的宽度, 使用的术语是波束宽度。半功率波束宽度, 或简称波束宽度  $\beta$ , 定义为两个角度之间主瓣的角宽度, 在该角度处  $F(\theta, \phi)$  的幅度等于其峰值的一半 (或分贝标度上的  $-3$  dB)。例如, 对于图 9-10(b) 中显示的模式,  $\beta$  由下式给出

$$\beta = \theta_2 - \theta_1, \quad (9.22)$$

其中,  $\theta_1$ 和 $\theta_2$ 是 $F(\theta, 0) = 0.5$ 时的半功率角 (其中 $\theta_2$ 表示较大值,  $\theta_1$ 表示较小值, 如图所示)。如果图案是对称的并且  $F(\theta, \phi)$  的峰值位于  $\theta = 0$ , 则  $\beta = 2\theta_2$ 。对于图 9-7(a) 中所示的短偶极子模式,  $F(\theta)$  在  $\theta = 90^\circ$  处最大,  $\theta_2$  在  $135^\circ$  处最大,  $\theta_1$  在  $45^\circ$  处最大。因此,  $\beta = 135^\circ - 45^\circ = 90^\circ$ 。波束宽度  $\beta$  也称为 3 dB 波束宽度。除了半功率波束宽度之外, 某些应用还可能关注其他波束尺寸, 例如零波束宽度  $\beta_{\text{null}}$ , 它是峰两侧第一个零点之间的角宽度 (图 9-10(b))。

9-2.3 天线方向性

天线的方向性  $D$  定义为其最大归一化辐射强度  $F_{\text{max}}$  ( (根据定义等于 1)) 与所有方向 ( $4\pi$  空间) 上  $F(\theta, \phi)$  平均值的比率:

$$D = \frac{F_{\text{max}}}{F_{\text{av}}} = \frac{1}{\frac{1}{4\pi} \iint_{4\pi} F(\theta, \phi) d\Omega} = \frac{4\pi}{\Omega_p}. \quad (9.23)$$

(dimensionless)

这里 $\Omega_p$ 是由方程定义的图案立体角。 (9.21) 。因此, 天线方向图 $\Omega_p$ 越窄, 方向性越大。对于各向同性天线,  $\Omega_p = 4\pi$ ; 因此, 其方向性  $D_{\text{iso}} = 1$ 。

通过使用等式。 (9.20) 式(9.23),  $D$  可表示为

$$D = \frac{4\pi R^2 S_{\text{max}}}{P_{\text{rad}}} = \frac{S_{\text{max}}}{S_{\text{av}}}, \quad (9.24)$$

其中  $S_{\text{av}} = P_{\text{rad}}/(4\pi R^2)$  是辐射功率密度的平均值, 等于辐射的总功率

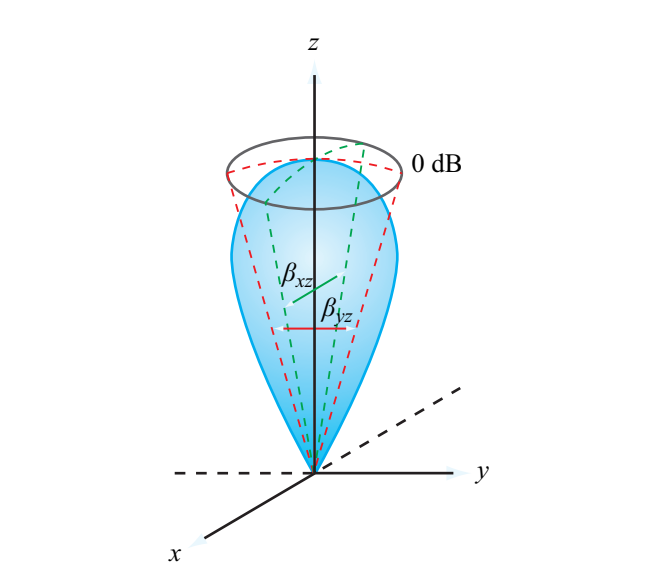


图9-12 单向辐射方向图的立体角约等于两个主平面内半功率波束宽度的乘积; 即 $\Omega_p \approx \beta_{xz}\beta_{yz}$ 。

天线  $P_{\text{rad}}$  除以半径为  $R$  的球体的表面积。

► 由于  $S_{\text{av}} = S_{\text{iso}}$  (其中  $S_{\text{iso}}$  是各向同性天线辐射的功率密度),  $D$  表示天线辐射的最大功率密度与各向同性天线辐射的功率密度之比, 其中两者都是在相同的距离  $R$  处测量并由相同的输入功率激励。 ◀

通常,  $D$ 以分贝表示:  $*D$  (dB) =  $10 \log D$ 。对于图 9-12 所示的具有指向  $z$  方向的单主瓣的天线,  $\Omega_p$  可以近似为半功率波束宽度  $\beta_{xz}$  和  $\beta_{yz}$  ( (以弧度表示) ) 的乘积:

$$\Omega_p \approx \beta_{xz}\beta_{yz}, \quad (9.25)$$

和

$$D = \frac{4\pi}{\Omega_p} \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \quad \text{(single main lobe)} \quad (9.26)$$

\*A note of caution: Even though we often express certain dimensionless quantities in decibels, we should always convert their decibel values to natural values before using them in the relations given in this chapter.

## 9-2 ANTENNA RADIATION CHARACTERISTICS

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Although approximate, this relation provides a useful method for estimating the antenna directivity from measurements of the beamwidths in the two orthogonal planes whose intersection is the axis of the main lobe.

### Example 9-1: Antenna Radiation Properties

Determine (a) the direction of maximum radiation, (b) pattern solid angle, (c) directivity, and (d) half-power beamwidth in the  $y$ - $z$  plane for an antenna that radiates only into the upper hemisphere with normalized radiation intensity given by  $F(\theta, \phi) = \cos^2 \theta$ .

**Solution:** The statement that the antenna radiates through only the upper hemisphere is equivalent to

$$F(\theta, \phi) = F(\theta) = \begin{cases} \cos^2 \theta & \text{for } 0 \leq \theta \leq \pi/2 \\ & \text{and } 0 \leq \phi \leq 2\pi, \\ 0 & \text{elsewhere.} \end{cases}$$

(a) The function  $F(\theta) = \cos^2 \theta$  is independent of  $\phi$  and is maximum when  $\theta = 0^\circ$ . A polar plot of  $F(\theta)$  is shown in Fig. 9-13.

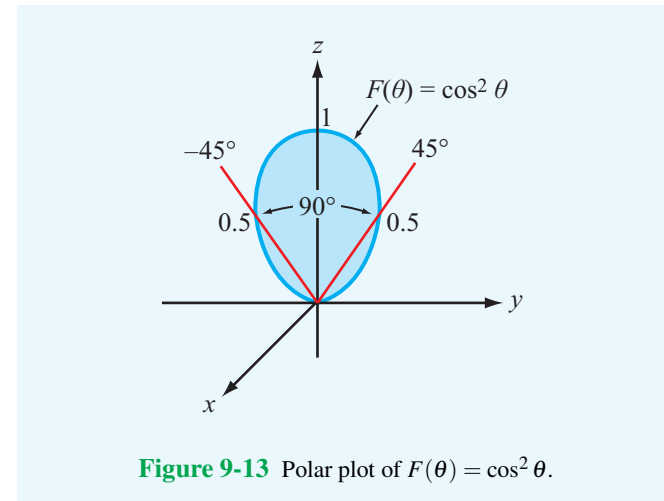


Figure 9-13 Polar plot of  $F(\theta) = \cos^2 \theta$ .

(b) From Eq. (9.21), the pattern solid angle  $\Omega_p$  is given by

$$\begin{aligned} \Omega_p &= \iint_{4\pi} F(\theta, \phi) d\Omega = \int_{\phi=0}^{2\pi} \left[ \int_{\theta=0}^{\pi/2} \cos^2 \theta \sin \theta d\theta \right] d\phi \\ &= \int_{\phi=0}^{2\pi} \left[ -\frac{\cos^3 \theta}{3} \right]_0^{\pi/2} d\phi \\ &= \int_0^{2\pi} \frac{1}{3} d\phi = \frac{2\pi}{3} \quad (\text{sr}). \end{aligned}$$

(c) Application of Eq. (9.23) gives

$$D = \frac{4\pi}{\Omega_p} = 4\pi \left( \frac{3}{2\pi} \right) = 6,$$

which corresponds to  $D$  (dB) =  $10 \log 6 = 7.78$  dB.

(d) The half-power beamwidth  $\beta$  is obtained by setting  $F(\theta) = 0.5$ . That is,

$$F(\theta) = \cos^2 \theta = 0.5,$$

which gives the half-power angles  $\theta_1 = -45^\circ$  and  $\theta_2 = 45^\circ$ . Hence,

$$\beta = \theta_2 - \theta_1 = 90^\circ.$$

### Example 9-2: Directivity of a Hertzian Dipole

Calculate the directivity of a Hertzian dipole.

**Solution:** Application of Eq. (9.23) with  $F(\theta) = \sin^2 \theta$  [from Eq. (9.15)] gives

$$\begin{aligned} D &= \frac{4\pi}{\iint_{4\pi} F(\theta, \phi) \sin \theta d\theta d\phi} \\ &= \frac{4\pi}{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin^3 \theta d\theta d\phi} = \frac{4\pi}{8\pi/3} = 1.5 \end{aligned}$$

or, equivalently, 1.76 dB.

### 9-2.4 Antenna Gain

Of the total power  $P_t$  (transmitter power) supplied to the antenna, a part,  $P_{\text{rad}}$ , is radiated out into space, and the remainder,  $P_{\text{loss}}$ , is dissipated as heat in the antenna structure. The **radiation efficiency**  $\xi$  is defined as the ratio of  $P_{\text{rad}}$  to  $P_t$ :

$$\xi = \frac{P_{\text{rad}}}{P_t}. \quad (\text{dimensionless}) \quad (9.27)$$

The **gain** of an antenna is defined as

$$G = \frac{4\pi R^2 S_{\text{max}}}{P_t}, \quad (9.28)$$

which is similar in form to the expression given by Eq. (9.24) for the directivity  $D$  except that it is referenced to the input

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虽然是近似的，但这种关系提供了一种有用的方法，可以通过测量两个正交平面（其交点为主瓣轴）的波束宽度来估计天线方向性。

### 例 9-1：天线辐射特性

确定仅辐射到上半球的天线的 (a) 最大辐射方向、(b) 方向图立体角、(c) 方向性和 (d)  $y$ - $z$  平面中的半功率波束宽度，归一化辐射强度由  $F(\theta, \phi) = \cos^2 \theta$  给出。

解：天线仅通过上半球辐射的说法等价于

$$F(\theta, \phi) = F(\theta) = \begin{cases} \cos^2 \theta & \text{for } 0 \leq \theta \leq \pi/2 \\ & \text{and } 0 \leq \phi \leq 2\pi, \\ 0 & \text{elsewhere.} \end{cases}$$

(a) 函数  $F(\theta) = \cos^2 \theta$  与  $\phi$  无关，并且在  $\theta = 0^\circ$  时最大。  $F(\theta)$  的极坐标图如图 9-13 所示。

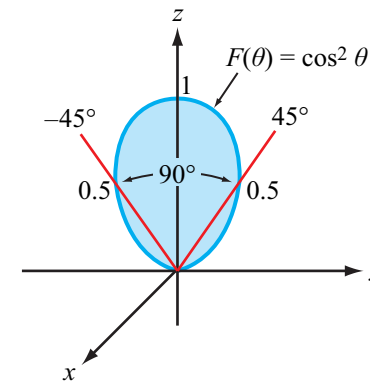


图 9-13  $F(\theta) = \cos^2 \theta$  的极坐标图。

(b) 从方程式 (9.21)，图案立体角  $\Omega_p$  由下式给出

$$\begin{aligned} \Omega_p &= \iint_{4\pi} F(\theta, \phi) d\Omega = \int_{\phi=0}^{2\pi} \left[ \int_{\theta=0}^{\pi/2} \cos^2 \theta \sin \theta d\theta \right] d\phi \\ &= \int_{\phi=0}^{2\pi} \left[ -\frac{\cos^3 \theta}{3} \right]_0^{\pi/2} d\phi \\ &= \int_0^{2\pi} \frac{1}{3} d\phi = \frac{2\pi}{3} \quad (\text{sr}). \end{aligned}$$

(c) 等式的应用(9.23) 给出

$$D = \frac{4\pi}{\Omega_p} = 4\pi \left( \frac{3}{2\pi} \right) = 6,$$

对应于  $D$  (dB) =  $10 \log 6 = 7.78$  dB。

(d) 通过将  $F(\theta)$  设置为 0.5 来获得半功率波束宽度  $\beta$ 。那是

$$F(\theta) = \cos^2 \theta = 0.5,$$

给出半功率角  $\theta_1 = -45^\circ$  和  $\theta_2 = 45^\circ$ 。因此，

$$\beta = \theta_2 - \theta_1 = 90^\circ.$$

### 例 9-2：赫兹偶极子的方向性

计算赫兹偶极子的方向性。

解决方案：应用方程。(9.23) 与  $F(\theta) = \sin^2 \theta$  [来自方程。(9.15)] 给出

$$\begin{aligned} D &= \frac{4\pi}{\iint_{4\pi} F(\theta, \phi) \sin \theta d\theta d\phi} \\ &= \frac{4\pi}{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin^3 \theta d\theta d\phi} = \frac{4\pi}{8\pi/3} = 1.5 \end{aligned}$$

或等效为 1.76 dB。

### 9-2.4 天线增益

在提供给天线的总功率  $P_t$  (发射机功率) 中，一部分  $P_{\text{rad}}$  辐射到空间中，其余部分  $P_{\text{loss}}$  在天线结构中作为热量耗散。辐射效率  $\xi$  定义为  $P_{\text{rad}}$  与  $P_t$  之比：

$$\xi = \frac{P_{\text{rad}}}{P_t}. \quad (\text{dimensionless}) \quad (9.27)$$

天线的增益定义为

$$G = \frac{4\pi R^2 S_{\text{max}}}{P_t}, \quad (9.28)$$

其形式与等式给出的表达式类似。(9.24) 对于方向性  $D$ ，除了它参考输入

power supplied to the antenna,  $P_{\text{t}}$ , rather than to the radiated power  $P_{\text{rad}}$ . In view of Eq. (9.27),

$$G = \xi D. \quad (\text{dimensionless}) \tag{9.29}$$

► The gain accounts for ohmic losses in the antenna material, whereas the directivity does not. For a lossless antenna,  $\xi = 1$ , and  $G = D$ . ◀

9-2.5 Radiation Resistance

To a transmission line connected between a generator supplying power  $P_{\text{t}}$  on one end and an antenna on the other end, the antenna is merely a load with **input impedance**  $Z_{\text{in}}$ . If the line is lossless and properly matched to the antenna, all of  $P_{\text{t}}$  is transferred to the antenna. In general,  $Z_{\text{in}}$  consists of a resistive component  $R_{\text{in}}$  and a reactive component  $X_{\text{in}}$ :

$$Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}}. \tag{9.30}$$

The resistive component is defined as equivalent to a resistor  $R_{\text{in}}$  that would consume an average power  $P_{\text{t}}$  when the amplitude of the ac current flowing through it is  $I_0$ ,

$$P_{\text{t}} = \tfrac{1}{2} I_0^2 R_{\text{in}}. \tag{9.31}$$

Since  $P_{\text{t}} = P_{\text{rad}} + P_{\text{loss}}$ , it follows that  $R_{\text{in}}$  can be defined as the sum of a **radiation resistance**  $R_{\text{rad}}$  and a **loss resistance**  $R_{\text{loss}}$ ,

$$R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}, \tag{9.32}$$

with

$$P_{\text{rad}} = \tfrac{1}{2} I_0^2 R_{\text{rad}}, \tag{9.33a}$$

$$P_{\text{loss}} = \tfrac{1}{2} I_0^2 R_{\text{loss}}, \tag{9.33b}$$

where  $I_0$  is the amplitude of the sinusoidal current exciting the antenna. As defined earlier, the **radiation efficiency** is the ratio of  $P_{\text{rad}}$  to  $P_{\text{t}}$ , or

$$\xi = \frac{P_{\text{rad}}}{P_{\text{t}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}. \tag{9.34}$$

The radiation resistance  $R_{\text{rad}}$  can be calculated by integrating the far-field power density over a sphere to obtain  $P_{\text{rad}}$  and then equating the result to Eq. (9.33a).

Example 9-3: Radiation Resistance and Efficiency of a Hertzian Dipole

A 4-cm long center-fed dipole is used as an antenna at 75 MHz. The antenna wire is made of copper and has a radius  $a = 0.4$  mm. From Eqs. (7.92a) and (7.94), the loss resistance of a circular wire of length  $l$  is given by

$$R_{\text{loss}} = \frac{l}{2\pi a} \sqrt{\frac{\pi f \mu_{\text{c}}}{\sigma_{\text{c}}}}, \tag{9.35}$$

where  $\mu_{\text{c}}$  and  $\sigma_{\text{c}}$  are the magnetic permeability and conductivity of the wire, respectively. Calculate the radiation resistance and the radiation efficiency of the dipole antenna.

**Solution:** At 75 MHz,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{7.5 \times 10^7} = 4 \text{ m}.$$

The length to wavelength ratio is  $l/\lambda = 4 \text{ cm}/4 \text{ m} = 10^{-2}$ . Hence, this is a short dipole. From Eq. (9.24),

$$P_{\text{rad}} = \frac{4\pi R^2}{D} S_{\text{max}}. \tag{9.36}$$

For the Hertzian dipole,  $S_{\text{max}}$  is given by Eq. (9.14), and from Example 9-2, we established that  $D = 1.5$ . Hence,

$$P_{\text{rad}} = \frac{4\pi R^2}{1.5} \times \frac{15\pi I_0^2}{R^2} \left(\frac{l}{\lambda}\right)^2 = 40\pi^2 I_0^2 \left(\frac{l}{\lambda}\right)^2. \tag{9.37}$$

Equating this result to Eq. (9.33a) and then solving for the radiation resistance  $R_{\text{rad}}$  leads to

$$R_{\text{rad}} = 80\pi^2 (l/\lambda)^2 \quad (\Omega). \quad (\text{short dipole}) \tag{9.38}$$

For  $l/\lambda = 10^{-2}$ ,  $R_{\text{rad}} = 0.08 \, \Omega$ .

Next, we determine the loss resistance  $R_{\text{loss}}$ . For copper, Appendix B gives  $\mu_{\text{c}} \approx \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$  and  $\sigma_{\text{c}} = 5.8 \times 10^7 \text{ S/m}$ . Hence,

$$\begin{aligned} R_{\text{loss}} &= \frac{l}{2\pi a} \sqrt{\frac{\pi f \mu_{\text{c}}}{\sigma_{\text{c}}}} \\ &= \frac{4 \times 10^{-2}}{2\pi \times 4 \times 10^{-4}} \left( \frac{\pi \times 75 \times 10^6 \times 4\pi \times 10^{-7}}{5.8 \times 10^7} \right)^{1/2} \\ &= 0.036 \, \Omega. \end{aligned}$$

Therefore, the radiation efficiency is

$$\xi = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{0.08}{0.08 + 0.036} = 0.69.$$

提供给天线的功率  $P_{\text{t}}$ , 而不是辐射功率  $P_{\text{rad}}$ 。鉴于方程。(9.27),

$$G = \xi D. \quad (\text{dimensionless}) \tag{9.29}$$

► 增益考虑了天线材料中的欧姆损耗，而方向性则不考虑。对于无损天线， $\xi = 1$ ,  $G = D$ . ◀

9-2.5 耐辐射性

对于连接在一端提供功率  $P_{\text{t}}$  的发生器和另一端天线之间的传输线，天线仅仅是一个输入阻抗为  $Z_{\text{in}}$  的负载。如果线路是无损的并且与天线正确匹配，则所有  $P_{\text{t}}$  都会传输到天线。一般来说， $Z_{\text{in}}$ 由电阻分量 $R_{\text{in}}$ 和电抗分量 $X_{\text{in}}$ 组成：

$$Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}}. \tag{9.30}$$

电阻组件被定义为等效于电阻器  $R_{\text{in}}$ ，当流过它的交流电流的幅度为  $I_0$  时，该电阻器将消耗平均功率  $P_{\text{t}}$ ,

$$P_{\text{t}} = \tfrac{1}{2} I_0^2 R_{\text{in}}. \tag{9.31}$$

由于  $P_{\text{t}} = P_{\text{rad}} + P_{\text{loss}}$ ，因此  $R_{\text{in}}$  可以定义为辐射电阻  $R_{\text{rad}}$  和损耗电阻  $R_{\text{loss}}$  之和，

$$R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}, \tag{9.32}$$

和

$$P_{\text{rad}} = \tfrac{1}{2} I_0^2 R_{\text{rad}}, \tag{9.33a}$$

$$P_{\text{loss}} = \tfrac{1}{2} I_0^2 R_{\text{loss}}, \tag{9.33b}$$

其中  $I_0$  是激励天线的正弦电流的幅度。如前所述，辐射效率是  $P_{\text{rad}}$  与  $P_{\text{t}}$  之比，或者

$$\xi = \frac{P_{\text{rad}}}{P_{\text{t}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}. \tag{9.34}$$

辐射电阻  $R_{\text{rad}}$  可以通过对球体上的远场功率密度进行积分来计算，以获得  $P_{\text{rad}}$ ，然后将结果等同于式 (1)。(9.33a)。

例 9-3：赫兹偶极子的辐射电阻和效率

使用 4 厘米长的中心馈电偶极子作为 75 MHz 的天线。天线导线由铜制成，半径为= 0.4 mm。从方程式。(7.92a) 和 (7.94) 中，长度为 l 的圆形导线的损耗电阻由下式给出

$$R_{\text{loss}} = \frac{l}{2\pi a} \sqrt{\frac{\pi f \mu_{\text{c}}}{\sigma_{\text{c}}}}, \tag{9.35}$$

其中  $\mu_{\text{c}}$  和  $\sigma_{\text{c}}$  分别是导线的磁导率和电导率。计算偶极子天线的辐射电阻和辐射效率。

解决方案：在 75 MHz 时，

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{7.5 \times 10^7} = 4 \text{ m}.$$

长度与波长之比为  $l/\lambda = 4 \text{ cm}/4 \text{ m} = 10^{-2}$ 。因此，这是一个短偶极子。从方程。(9.24),

$$P_{\text{rad}} = \frac{4\pi R^2}{D} S_{\text{max}}. \tag{9.36}$$

对于赫兹偶极子， $S_{\text{max}}$  由等式给出。(9.14)，并根据例 9-2，我们确定 D = 1.5。因此，

$$P_{\text{rad}} = \frac{4\pi R^2}{1.5} \times \frac{15\pi I_0^2}{R^2} \left(\frac{l}{\lambda}\right)^2 = 40\pi^2 I_0^2 \left(\frac{l}{\lambda}\right)^2. \tag{9.37}$$

将此结果等同于方程式。(9.33a) 然后求解辐射电阻  $R_{\text{rad}}$  可得出

$$R_{\text{rad}} = 80\pi^2 (l/\lambda)^2 \quad (\Omega). \quad (\text{short dipole}) \tag{9.38}$$

For  $l/\lambda = 10^{-2}$ ,  $R_{\text{rad}} = 0.08 \, \Omega$ .

接下来，我们确定损耗电阻  $R_{\text{loss}}$ 。对于铜，附录 B 给出了  $\mu_{\text{c}} \approx \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$  和  $\sigma_{\text{c}} = 5.8 \times 10^7 \text{ S/m}$ 。因此，

$$\begin{aligned} R_{\text{loss}} &= \frac{l}{2\pi a} \sqrt{\frac{\pi f \mu_{\text{c}}}{\sigma_{\text{c}}}} \\ &= \frac{4 \times 10^{-2}}{2\pi \times 4 \times 10^{-4}} \left( \frac{\pi \times 75 \times 10^6 \times 4\pi \times 10^{-7}}{5.8 \times 10^7} \right)^{1/2} \\ &= 0.036 \, \Omega. \end{aligned}$$

因此，辐射效率为

$$\xi = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{0.08}{0.08 + 0.036} = 0.69.$$

### 9-3 HALF-WAVE DIPOLE ANTENNA

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Thus, the dipole is 69% efficient.

**Concept Question 9-5:** What does the pattern solid angle represent?

**Concept Question 9-6:** What is the magnitude of the directivity of an isotropic antenna?

**Concept Question 9-7:** What physical and material properties affect the radiation efficiency of a fixed-length Hertzian dipole antenna?

**Exercise 9-2:** An antenna has a conical radiation pattern with a normalized radiation intensity  $F(\theta) = 1$  for  $\theta$  between  $0^\circ$  and  $45^\circ$  and zero for  $\theta$  between  $45^\circ$  and  $180^\circ$ . The pattern is independent of the azimuth angle  $\phi$ . Find (a) the pattern solid angle and (b) the directivity.

**Answer:** (a)  $\Omega_p = 1.84$  sr, (b)  $D = 6.83$  or, equivalently, 8.3 dB. (See [EM](#).)

**Exercise 9-3:** The maximum power density radiated by a short dipole at a distance of 1 km is  $60$  (nW/m<sup>2</sup>). If  $I_0 = 10$  A, find the radiation resistance.

**Answer:**  $R_{\text{rad}} = 10$  m $\Omega$ . (See [EM](#).)

### 9-3 Half-Wave Dipole Antenna

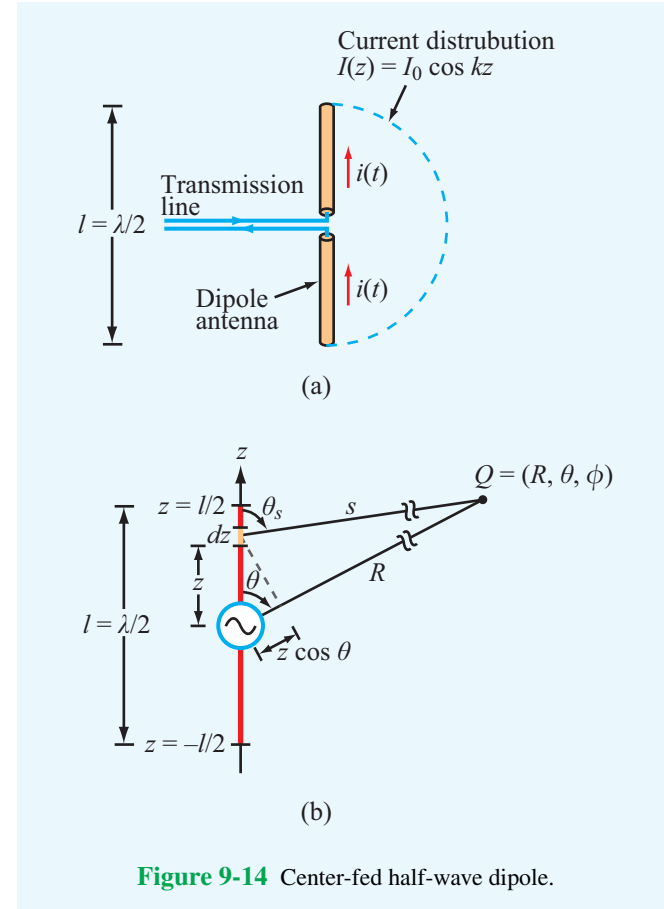
In Section 9-1, we developed expressions for the electric and magnetic fields radiated by a Hertzian dipole of length  $l \ll \lambda$ . We now use these expressions as building blocks to obtain expressions for the fields radiated by a half-wave dipole antenna, so named because its length  $l = \lambda/2$ . As shown in **Fig. 9-14**, the half-wave dipole consists of a thin wire fed at its center by a generator connected to the antenna terminals via a transmission line. The current flowing through the wire has a symmetrical distribution with respect to the center of the dipole, and the current is zero at its ends. Mathematically,  $i(t)$  is given by

$$i(t) = I_0 \cos \omega t \cos kz = \Re [I_0 \cos kz e^{j\omega t}], \quad (9.39a)$$

whose phasor is

$$\tilde{I}(z) = I_0 \cos kz, \quad -\lambda/4 \leq z \leq \lambda/4, \quad (9.39b)$$

and  $k = 2\pi/\lambda$ . Equation (9.9a) gives an expression for  $\tilde{E}_\theta$ , which is the far field radiated by a Hertzian dipole of length  $l$  when excited by a current  $I_0$ . Let us adapt that expression to an



**Figure 9-14** Center-fed half-wave dipole.

infinitesimal dipole segment of length  $dz$ , which is excited by a current  $\tilde{I}(z)$  and located at a distance  $s$  from the observation point  $Q$  (**Fig. 9-14(b)**). Thus,

$$d\tilde{E}_\theta(z) = \frac{jk\eta_0}{4\pi} \tilde{I}(z) dz \left( \frac{e^{-jks}}{s} \right) \sin \theta_s, \quad (9.40a)$$

and the associated magnetic field is

$$d\tilde{H}_\phi(z) = \frac{d\tilde{E}_\theta(z)}{\eta_0}. \quad (9.40b)$$

The far field due to radiation by the entire antenna is obtained by integrating the fields from all of the Hertzian dipoles making up the antenna:

$$\tilde{E}_\theta = \int_{z=-\lambda/4}^{\lambda/4} d\tilde{E}_\theta. \quad (9.41)$$

Before we calculate this integral, we make the following two approximations. The first relates to the magnitude part

### 9-3 HALF-WAVE DIPOLE ANTENNA

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因此，偶极子的效率为 69%。

**Concept Question 9-5:** What does the pattern solid angle represent?

**Concept Question 9-6:** What is the magnitude of the directivity of an isotropic antenna?

**Concept Question 9-7:** What physical and material properties affect the radiation efficiency of a fixed-length Hertzian dipole antenna?

**Exercise 9-2:** An antenna has a conical radiation pattern with a normalized radiation intensity  $F(\theta) = 1$  for  $\theta$  between  $0^\circ$  and  $45^\circ$  and zero for  $\theta$  between  $45^\circ$  and  $180^\circ$ . The pattern is independent of the azimuth angle  $\phi$ . Find (a) the pattern solid angle and (b) the directivity.

**Answer:** (a)  $\Omega_p = 1.84$  sr, (b)  $D = 6.83$  or, equivalently, 8.3 dB. (See [EM](#).)

**Exercise 9-3:** The maximum power density radiated by a short dipole at a distance of 1 km is  $60$  (nW/m<sup>2</sup>). If  $I_0 = 10$  A, find the radiation resistance.

**Answer:**  $R_{\text{rad}} = 10$  m $\Omega$ . (See [EM](#).)

### 9-3 半波偶极子天线

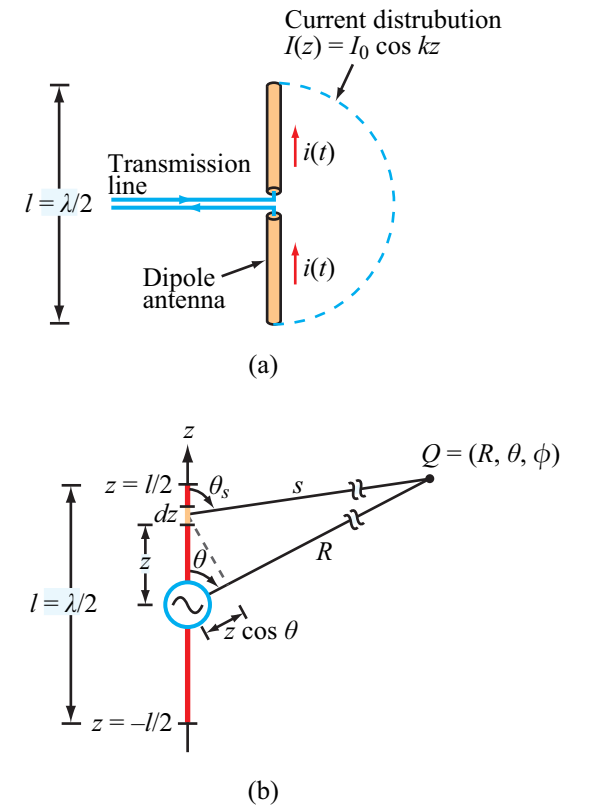
在第 9-1 节中，我们开发了长度为  $l \ll \lambda$  的赫兹偶极子辐射的电场和磁场的表达式。现在，我们使用这些表达式作为构建块来获得半波偶极子天线辐射场的表达式，之所以如此命名是因为其长度为  $l = \lambda/2$ 。如图 9-14 所示，半波偶极子由一根细线组成，该细线由通过传输线连接到天线端子的发生器在其中心馈电。流过导线的电流相对于偶极子中心对称分布，两端电流为零。数学上， $i(t)$  由下式给出

$$i(t) = I_0 \cos \omega t \cos kz = \Re [I_0 \cos kz e^{j\omega t}], \quad (9.39a)$$

谁的相量是

$$\tilde{I}(z) = I_0 \cos kz, \quad -\lambda/4 \leq z \leq \lambda/4, \quad (9.39b)$$

和  $k = 2\pi/\lambda$ 。方程 (9.9a) 给出了  $\tilde{E}_\theta$  的表达式，它是长度为  $l$  的赫兹偶极子在受到电流  $I_0$  激励时辐射的远场。让我们将该表达式改编成



**图 9-14** 中心馈电半波偶极子。

长度为  $dz$  的无穷小偶极子段，由电流  $\tilde{I}(z)$  激发，位于距观察点  $Q$  距离  $s$  处（图 9-14 (b)）。因此，

$$d\tilde{E}_\theta(z) = \frac{jk\eta_0}{4\pi} \tilde{I}(z) dz \left( \frac{e^{-jks}}{s} \right) \sin \theta_s, \quad (9.40a)$$

相关的磁场是

$$d\tilde{H}_\phi(z) = \frac{d\tilde{E}_\theta(z)}{\eta_0}. \quad (9.40b)$$

整个天线辐射产生的远场是通过对构成天线的所有赫兹偶极子的场进行积分获得的：

$$\tilde{E}_\theta = \int_{z=-\lambda/4}^{\lambda/4} d\tilde{E}_\theta. \quad (9.41)$$

在计算这个积分之前，我们进行以下两个近似。第一个与幅度部分有关

of the spherical propagation factor,  $1/s$ . In **Fig. 9-14(b)**, the distance  $s$  between the current element and the observation point  $Q$  is considered so large in comparison with the length of the dipole that the difference between  $s$  and  $R$  may be neglected in terms of its effect on  $1/s$ . Hence, we may set  $1/s \approx 1/R$ , and by the same argument, we set  $\theta_s \approx \theta$ . The error  $\Delta$  between  $s$  and  $R$  is a maximum when the observation point is along the  $z$  axis and it is equal to  $\lambda/4$  (corresponding to half of the antenna length). If  $R \gg \lambda$ , this error will have an insignificant effect on  $1/s$ . The second approximation is associated with the phase factor  $e^{-jks}$ . An error in distance  $\Delta$  corresponds to an error in phase  $k\Delta = (2\pi/\lambda)(\lambda/4) = \pi/2$ . As a rule of thumb, a phase error greater than  $\pi/8$  is considered unacceptable because it may lead to a significant error in the computed value of the field  $\tilde{E}_\theta$ . Hence, the approximation  $s \approx R$  is too crude for the phase factor and cannot be used. A more tolerable option is to use the **parallel-ray approximation** given by

$$s \approx R - z \cos \theta, \quad (9.42)$$

as illustrated in **Fig. 9-14(b)**.

Substituting Eq. (9.42) for  $s$  in the phase factor of Eq. (9.40a) and replacing  $s$  with  $R$  and  $\theta_s$  with  $\theta$  elsewhere in the expression, we obtain

$$d\tilde{E}_\theta = \frac{jk\eta_0}{4\pi} \tilde{I}(z) dz \left( \frac{e^{-jkR}}{R} \right) \sin \theta e^{jkz \cos \theta}. \quad (9.43)$$

After (1) inserting Eq. (9.43) into Eq. (9.41), (2) using the expression for  $\tilde{I}(z)$  given by Eq. (9.39b), and (3) carrying out the integration, the following expressions are obtained:

$$\tilde{E}_\theta = j60I_0 \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\} \left( \frac{e^{-jkR}}{R} \right), \quad (9.44a)$$

$$\tilde{H}_\phi = \frac{\tilde{E}_\theta}{\eta_0}. \quad (9.44b)$$

The corresponding time-average power density is

$$\begin{aligned} S(R, \theta) &= \frac{|\tilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2} \left\{ \frac{\cos^2[(\pi/2) \cos \theta]}{\sin^2 \theta} \right\} \\ &= S_0 \left\{ \frac{\cos^2[(\pi/2) \cos \theta]}{\sin^2 \theta} \right\} \quad (\text{W/m}^2). \end{aligned} \quad (9.45)$$

Examination of Eq. (9.45) reveals that  $S(R, \theta)$  is maximum at  $\theta = \pi/2$ , and its value is

$$S_{\max} = S_0 = \frac{15I_0^2}{\pi R^2}.$$

Hence, the normalized radiation intensity is

$$F(\theta) = \frac{S(R, \theta)}{S_0} = \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\}^2. \quad (9.46)$$

The radiation pattern of the half-wave dipole exhibits roughly the same doughnut-like shape shown earlier in **Fig. 9-7** for the short dipole. *Its directivity is slightly larger (1.64 compared with 1.5 for the short dipole), but its radiation resistance is  $73 \, \Omega$  (as shown later in Section 9-3.2), which is orders of magnitude larger than that of a short dipole.*

### 9-3.1 Directivity of $\lambda/2$ Dipole

To evaluate both the directivity  $D$  and the radiation resistance  $R_{\text{rad}}$  of the half-wave dipole, we first need to calculate the total radiated power  $P_{\text{rad}}$  by applying Eq. (9.20):

$$\begin{aligned} P_{\text{rad}} &= R^2 \iint_{4\pi} S(R, \theta) d\Omega \\ &= \frac{15I_0^2}{\pi} \int_0^{2\pi} \int_0^\pi \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\}^2 \sin \theta d\theta d\phi. \end{aligned} \quad (9.47)$$

The integration over  $\phi$  is equal to  $2\pi$ , and numerical evaluation of the integration over  $\theta$  gives the value 1.22. Consequently,

$$P_{\text{rad}} = 36.6I_0^2 \quad (\text{W}). \quad (9.48)$$

From Eq. (9.45), we found that  $S_{\max} = 15I_0^2/(\pi R^2)$ . Using this in Eq. (9.24) gives the following result for the directivity  $D$  of the half-wave dipole:

$$D = \frac{4\pi R^2 S_{\max}}{P_{\text{rad}}} = \frac{4\pi R^2}{36.6I_0^2} \left( \frac{15I_0^2}{\pi R^2} \right) = 1.64 \quad (9.49)$$

or, equivalently, 2.15 dB.

### 9-3.2 Radiation Resistance of $\lambda/2$ Dipole

From Eq. (9.33a),

$$R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2} = \frac{2 \times 36.6I_0^2}{I_0^2} \approx 73 \, \Omega. \quad (9.50)$$

As was noted earlier in Example 9-3, because the radiation resistance of a Hertzian dipole is comparable in magnitude to that of its loss resistance  $R_{\text{loss}}$ , its radiation efficiency  $\xi$  is rather small. For the 4-cm long dipole of Example 9-3,  $R_{\text{rad}} = 0.08 \, \Omega$  (at 75 MHz) and  $R_{\text{loss}} = 0.036 \, \Omega$ . If we keep

球面传播因子  $1/s$ 。在图9-14(b)中，与偶极子的长度相比，电流微元和观察点Q之间的距离s被认为是如此之大，以至于s和R之间的差异对于 $1/s$ 的影响可以忽略不计。因此，我们可以设置  $1/s \approx 1/R$ ，并且通过相同的论证，我们设置  $\theta_s \approx \theta$ 。当观测点沿z轴时，s和R之间的误差 $\Delta$ 最大，等于 $\lambda/4$ （相当于天线长度的一半）。如果  $R \gg \lambda$ ，则该误差对  $1/$  的影响微不足道。第二个近似值与相位因子  $e^{-jks}$  相关。距离 $\Delta$ 的误差对应于相位 $k\Delta = (2\pi/\lambda)(\lambda/4) = \pi/2$ 的误差。根据经验，大于  $\pi/8$  的相位误差被认为是不可接受的，因为它可能会导致场  $\tilde{E}_\theta$  的计算值出现重大错误。因此，近似值  $s \approx R$  对于相位因子来说过于粗略，无法使用。更可容忍的选择是使用由下式给出的平行射线近似

$$s \approx R - z \cos \theta, \quad (9.42)$$

如图9-14(b)所示。

代入方程式(9.42)对于方程相位因子中的  $s$ 。(9.40a)并将表达式中其他地方的  $s$  替换为  $R$ ，将  $\theta_s$  替换为  $\theta$ ，我们得到

$$d\tilde{E}_\theta = \frac{jk\eta_0}{4\pi} \tilde{I}(z) dz \left( \frac{e^{-jkR}}{R} \right) \sin \theta e^{jkz \cos \theta}. \quad (9.43)$$

(1) 插入方程后：(9.43)代入方程。(9.41)，(2) 使用方程给出的  $\tilde{I}(z)$  表达式。(9.39b)和(3)进行积分，得到以下表达式：

$$\tilde{E}_\theta = j60I_0 \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\} \left( \frac{e^{-jkR}}{R} \right), \quad (9.44a)$$

$$\tilde{H}_\phi = \frac{\tilde{E}_\theta}{\eta_0}. \quad (9.44b)$$

相应的时均功率密度为

$$\begin{aligned} S(R, \theta) &= \frac{|\tilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2} \left\{ \frac{\cos^2[(\pi/2) \cos \theta]}{\sin^2 \theta} \right\} \\ &= S_0 \left\{ \frac{\cos^2[(\pi/2) \cos \theta]}{\sin^2 \theta} \right\} \quad (\text{W/m}^2). \end{aligned} \quad (9.45)$$

方程的检验(9.45)表明  $S(R, \theta)$  在  $\theta = \pi/2$  处最大，其值为

$$S_{\max} = S_0 = \frac{15I_0^2}{\pi R^2}.$$

因此，归一化辐射强度为

$$F(\theta) = \frac{S(R, \theta)}{S_0} = \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\}^2. \quad (9.46)$$

半波偶极子的辐射图表现出与图 9-7 中短偶极子大致相同的甜甜圈形状。它的方向性稍大（1.64，而短偶极子为1.5），但其辐射电阻为  $73 \, \Omega$ （，如后面第9-3.2)节所示，比短偶极子大几个数量级。

#### 9-3.1 $\lambda/2$ 偶极子的方向性

为了评估半波偶极子的方向性  $D$  和辐射电阻  $R_{\text{rad}}$ ，我们首先需要通过应用方程（1）计算总辐射功率  $P_{\text{rad}}$ 。（9.20）：

$$\begin{aligned} P_{\text{rad}} &= R^2 \iint_{4\pi} S(R, \theta) d\Omega \\ &= \frac{15I_0^2}{\pi} \int_0^{2\pi} \int_0^\pi \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\}^2 \sin \theta d\theta d\phi. \end{aligned} \quad (9.47)$$

$\phi$  上的积分等于  $2\pi$ ，并且  $\theta$  上积分的数值计算给出值 1.22。最后，

$$P_{\text{rad}} = 36.6I_0^2 \quad (\text{W}). \quad (9.48)$$

从方程。(9.45)，我们发现  $S_{\max} = 15I_0^2/(\pi R^2)$ 。在等式中使用它。(9.24) 给出半波偶极子方向性  $D$  的以下结果：

$$D = \frac{4\pi R^2 S_{\max}}{P_{\text{rad}}} = \frac{4\pi R^2}{36.6I_0^2} \left( \frac{15I_0^2}{\pi R^2} \right) = 1.64 \quad (9.49)$$

或等效的 2.15 dB。

#### 9-3.2 $\lambda/2$ 偶极子的辐射电阻

从方程。(9.33a),

$$R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2} = \frac{2 \times 36.6I_0^2}{I_0^2} \approx 73 \, \Omega. \quad (9.50)$$

正如前面例 9-3 中所指出的，由于赫兹偶极子的辐射电阻在大小上与其损耗电阻  $R_{\text{loss}}$  相当，因此其辐射效率  $\xi$  相当小。对于示例 9-3 的 4 厘米长偶极子，75 MHz) 下的  $R_{\text{rad}} = 0.08 \, \Omega$ （和  $R_{\text{loss}} = 0.036 \, \Omega$ 。如果我们保留

the frequency the same and increase the length of the dipole to 2 m ( $\lambda = 4$  m at  $f = 75$  MHz),  $R_{\text{rad}}$  becomes  $73\ \Omega$  and  $R_{\text{loss}}$  increases to  $1.8\ \Omega$ . The radiation efficiency increases from 69% for the short dipole to 98% for the half-wave dipole. More significant is the fact that it is practically impossible to match a transmission line to an antenna with a resistance on the order of  $0.1\ \Omega$ , while it is quite easy to do so when  $R_{\text{rad}} = 73\ \Omega$ . Moreover, since  $R_{\text{loss}} \ll R_{\text{rad}}$  for the half-wave dipole,  $R_{\text{in}} \approx R_{\text{rad}}$  and Eq. (9.30) becomes

$$Z_{\text{in}} \approx R_{\text{rad}} + jX_{\text{in}}. \quad (9.51)$$

Deriving an expression for  $X_{\text{in}}$  for the half-wave dipole is fairly complicated and beyond the scope of this book. However, it is significant to note that  $X_{\text{in}}$  is a strong function of  $l/\lambda$  and that it decreases from  $42\ \Omega$  at  $l/\lambda = 0.5$  to zero at  $l/\lambda = 0.48$ , whereas  $R_{\text{rad}}$  remains approximately unchanged. Hence, by reducing the length of the half-wave dipole by 4%,  $Z_{\text{in}}$  becomes purely real and equal to  $73\ \Omega$ , thereby making it possible to match the dipole to a  $75\ \Omega$  transmission line without resorting to the use of a matching network.

9-3.3 Quarter-Wave Monopole Antenna

► When placed over a conducting ground plane, a quarter-wave monopole antenna excited by a source at its base (Fig. 9-15(a)) exhibits the same radiation pattern in the region above the ground plane as a half-wave dipole in free space. ◀

This is because, from image theory (Section 4-11), the conducting plane can be replaced with the image of the  $\lambda/4$  monopole, as illustrated in Fig. 9-15(b). Thus, the  $\lambda/4$  monopole radiates an electric field identical to that given by Eq. (9.44a), and its normalized radiation intensity is given by Eq. (9.46); but the radiation is limited to the upper half-space defined by  $0 \leq \theta \leq \pi/2$ . Hence, a monopole radiates only half as much power as the dipole. Consequently, for a  $\lambda/4$  monopole,  $P_{\text{rad}} = 18.3I_0^2$ , and its radiation resistance is  $R_{\text{rad}} = 36.5\ \Omega$ . The approach used with the quarter-wave monopole is also valid for any vertical wire antenna placed above a conducting plane, including a Hertzian monopole.

**Concept Question 9-8:** What is the physical length of a half-wave dipole operating at (a) 1 MHz (in the AM broadcast band), (b) 100 MHz (FM broadcast band), and (c) 10 GHz (microwave band)?

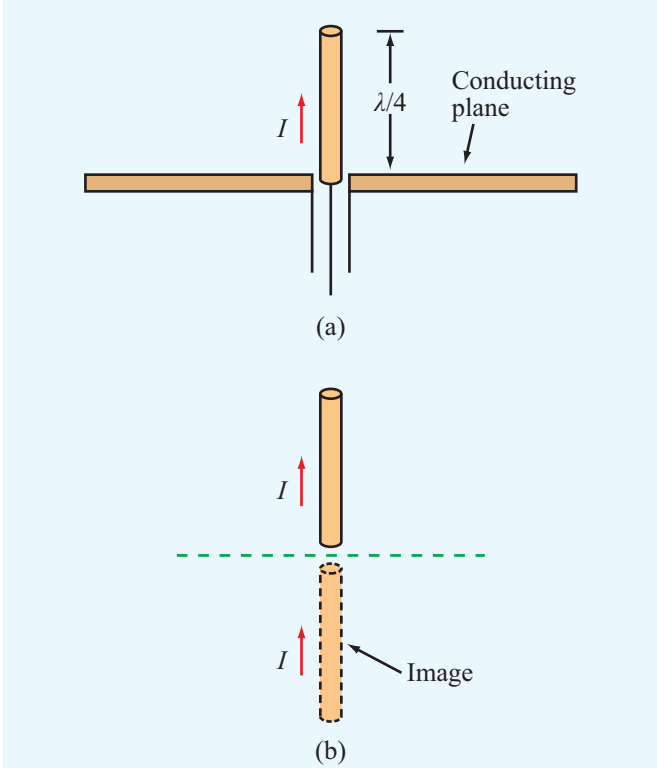


Figure 9-15 A quarter-wave monopole above a conducting plane is equivalent to a full half-wave dipole in free space.

**Concept Question 9-9:** How does the radiation pattern of a half-wave dipole compare with that of a Hertzian dipole? How do their directivities, radiation resistances, and radiation efficiencies compare?

**Concept Question 9-10:** How does the radiation efficiency of a quarter-wave monopole compare with that of a half-wave dipole, assuming that both are made of the same material and have the same cross section?

**Exercise 9-4:** For the half-wave dipole antenna, evaluate  $F(\theta)$  versus  $\theta$  to determine the half-power beamwidth in the elevation plane (the plane containing the dipole axis).

**Answer:**  $\beta = 78^\circ$ . (See EM.)

频率相同并将偶极子长度增加到 2 m ( $f = 75$  MHz 时  $\lambda = 4$  m),  $R_{\text{rad}}$  变为  $73\ \Omega$ ,  $R_{\text{loss}}$  增加到  $1.8\ \Omega$ 。辐射效率从短偶极子的 69% 增加到半波偶极子的 98%。更重要的是, 实际上不可能将传输线与电阻约为  $0.1\ \Omega$  的天线相匹配, 而当  $R_{\text{rad}} = 73\ \Omega$  时则很容易做到这一点。

此外, 由于  $R_{\text{loss}} \ll R_{\text{rad}}$  对于半波偶极子, 因此  $R_{\text{in}} \approx R_{\text{rad}}$  和等式: (9.30) 变为

$$Z_{\text{in}} \approx R_{\text{rad}} + jX_{\text{in}}. \quad (9.51)$$

推导半波偶极子  $X_{\text{in}}$  的表达式相当复杂, 超出了本书的范围。然而, 值得注意的是,  $X_{\text{in}}$  是  $1/\lambda$  的强函数, 并且它从  $1/\lambda = 0.5$  时的  $42\ \Omega$  减少到  $1/\lambda = 0.48$  时的零, 而  $R_{\text{rad}}$  几乎保持不变。因此, 通过将半波偶极子的长度减少 4%,  $Z_{\text{in}}$  变成纯实数并且等于  $73\ \Omega$ , 从而使得可以将偶极子与  $75\ \Omega$  传输线匹配, 而无需使用匹配网络。

9-3.3 四分之一波单极天线

► 当放置在导电接地平面上时, 由底部源激励的四分之一波单极天线 (图 9-15 (a)) 在接地平面上方的区域中表现出与自由空间中的半波偶极子相同的辐射图。 ◀

这是因为, 根据镜像理论 (第 4-11 节), 传导平面可以用  $\lambda/4$  单极子的镜像代替, 如图 9-15 (b) 所示。因此,  $\lambda/4$  单极子辐射的电场与方程 1 给出的电场相同。 (9.44a), 其归一化辐射强度由式 (9.44a) 给出。 (9.46); 但辐射仅限于  $0 \leq \theta \leq \pi/2$  定义的上半空间。因此, 单极子辐射的功率仅为偶极子的一半。因此, 对于  $\lambda/4$  单极子,  $P_{\text{rad}} = 18.3I_0^2$ , 其辐射电阻为  $R_{\text{rad}} = 36.5\ \Omega$ 。

使用四分之一波单极子的方法也适用于放置在导电平面上方的任何垂直线天线, 包括赫兹单极子。

概念问题 9-8: 在 (a) 1 MHz (AM 广播频段)、(b) 100 MHz (FM 广播频段) 和 (c) 10 GHz (微波频段) 工作的半波偶极子的物理长度是多少?

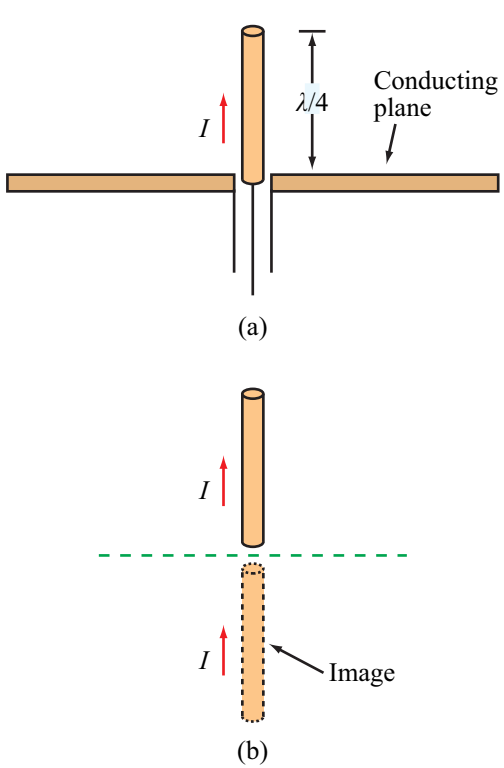


图 9-15 导电平面上方的四分之一波单极子相当于自由空间中的全半波偶极子。

概念问题 9-9: 半波偶极子的辐射图与赫兹偶极子的辐射图相比如何? 它们的方向性、辐射电阻和辐射效率如何比较?

概念问题 9-10: 假设四分之一波长单极子和半波偶极子由相同的材料制成并且具有相同的横截面, 那么它们的辐射效率如何?

练习 9-4: 对于半波偶极子天线, 计算  $F(\theta)$  与  $\theta$  的关系, 以确定仰角平面 (包含偶极子轴的平面) 中的半功率波束宽度。

答案:  $\beta = 78^\circ$ . (参见 EM.)

**Exercise 9-5:** If the maximum power density radiated by a half-wave dipole is  $50 \mu\text{W}/\text{m}^2$  at a range of 1 km, what is the current amplitude  $I_0$ ?

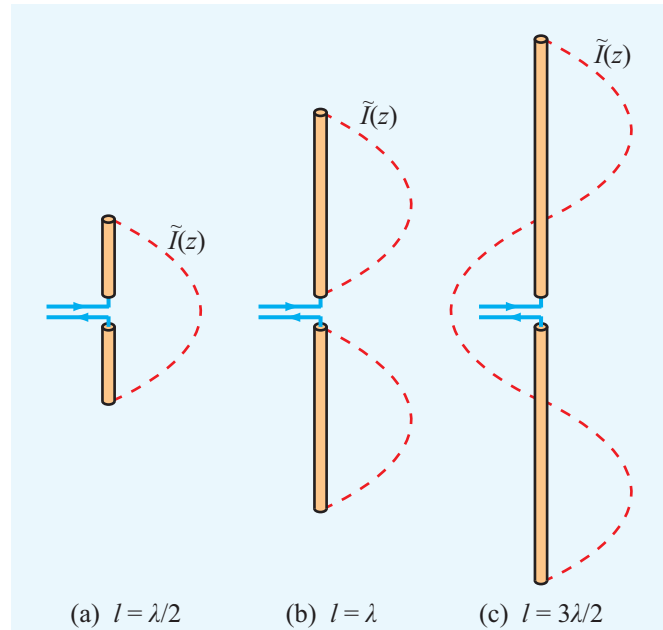
**Answer:**  $I_0 = 3.24 \text{ A}$ . (See [EM](#).)

## 9-4 Dipole of Arbitrary Length

So far, we examined the radiation properties of the Hertzian and half-wave dipoles. We now consider the more general case of a linear dipole of arbitrary length  $l$  relative to  $\lambda$ . For a center-fed dipole, as depicted in **Fig. 9-16**, the currents flowing through its two halves are symmetrical and must go to zero at its ends. Hence, the current phasor  $\tilde{I}(z)$  can be expressed as a sine function with an argument that goes to zero at  $z = \pm l/2$ :

$$\tilde{I}(z) = \begin{cases} I_0 \sin[k(l/2 - z)], & \text{for } 0 \leq z \leq l/2, \\ I_0 \sin[k(l/2 + z)], & \text{for } -l/2 \leq z < 0, \end{cases} \quad (9.52)$$

where  $I_0$  is the current amplitude. The procedure for calculating the electric and magnetic fields and the associated power density of the wave radiated by such an antenna is basically the same as that used previously in connection with the half-wave dipole antenna. The only difference is the current distribution  $\tilde{I}(z)$ . If we insert the expression for  $\tilde{I}(z)$  given by



**Figure 9-16** Current distribution for three center-fed dipoles.

Eq. (9.52) into Eq. (9.43), we obtain the following expression for the differential electric field  $d\tilde{E}_\theta$  of the wave radiated by an elemental length  $dz$  at location  $z$  along the dipole:

$$d\tilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi} \left( \frac{e^{-jkR}}{R} \right) \sin\theta e^{jkz\cos\theta} dz \times \begin{cases} \sin[k(l/2 - z)] & \text{for } 0 \leq z \leq l/2, \\ \sin[k(l/2 + z)] & \text{for } -l/2 \leq z < 0. \end{cases} \quad (9.53)$$

The total field radiated by the dipole is

$$\begin{aligned} \tilde{E}_\theta &= \int_{-l/2}^{l/2} d\tilde{E}_\theta = \int_0^{l/2} d\tilde{E}_\theta + \int_{-l/2}^0 d\tilde{E}_\theta \\ &= \frac{jk\eta_0 I_0}{4\pi} \left( \frac{e^{-jkR}}{R} \right) \sin\theta \\ &\quad \times \left\{ \int_0^{l/2} e^{jkz\cos\theta} \sin[k(l/2 - z)] dz \right. \\ &\quad \left. + \int_{-l/2}^0 e^{jkz\cos\theta} \sin[k(l/2 + z)] dz \right\}. \end{aligned} \quad (9.54)$$

If we apply Euler's identity to express  $e^{jkz\cos\theta}$  as

$$e^{jkz\cos\theta} = \cos(kz\cos\theta) + j\sin(kz\cos\theta),$$

we can integrate the two integrals and obtain the result

$$\tilde{E}_\theta = j60I_0 \left( \frac{e^{-jkR}}{R} \right) \left[ \frac{\cos(\frac{kl}{2}\cos\theta) - \cos(\frac{kl}{2})}{\sin\theta} \right]. \quad (9.55)$$

The corresponding time-average power density radiated by the dipole antenna is given by

$$S(\theta) = \frac{|\tilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos(\frac{\pi l}{\lambda}\cos\theta) - \cos(\frac{\pi l}{\lambda})}{\sin\theta} \right]^2, \quad (9.56)$$

where we have used the relations  $\eta_0 \approx 120\pi \text{ } (\Omega)$  and  $k = 2\pi/\lambda$ . For  $l = \lambda/2$ , Eq. (9.56) reduces to the expression given by Eq. (9.45) for the half-wave dipole. Plots of the normalized radiation intensity,  $F(\theta) = S(R, \theta)/S_{\max}$ , are shown in **Fig. 9-17** for dipoles of lengths  $\lambda/2$ ,  $\lambda$ , and  $3\lambda/2$ . The dipoles with  $l = \lambda/2$  and  $l = \lambda$  have similar radiation patterns with both maxima along  $\theta = 90^\circ$ , but the half-power beamwidth of the wavelength-long dipole is narrower than that of the half-wave dipole, and  $S_{\max} = 60I_0^2/(\pi R^2)$  for the wavelength-long dipole, which is four times that for the half-wave dipole. The pattern of the dipole with length  $l = 3\lambda/2$  exhibits a structure with multiple lobes, and its direction of maximum radiation is not along  $\theta = 90^\circ$ .

练习9-5：如果半波偶极子在1公里范围内辐射的最大功率密度为 $50 \mu\text{W}/\text{m}^2$ ，则电流幅度 $I_0$ 是多少？

答案： $I_0 = 3.24 \text{ A}$ 。（参见<sup>EM</sup>。）

## 9-4 任意长度的偶极子

到目前为止，我们研究了赫兹偶极子和半波偶极子的辐射特性。现在我们考虑更一般的情况，即相对于 $\lambda$ 具有任意长度 $l$ 的线性偶极子。对于中心馈电偶极子，如图9-16所示，流过其两半的电流是对称的，并且在其两端必须为零。因此，电流相量 $\tilde{I}(z)$ 可以表示为一个正弦函数，其参数在 $z = \pm l/2$ 处为零：

$$\tilde{I}(z) = \begin{cases} I_0 \sin[k(l/2 - z)], & \text{for } 0 \leq z \leq l/2, \\ I_0 \sin[k(l/2 + z)], & \text{for } -l/2 \leq z < 0, \end{cases} \quad (9.52)$$

其中 $I_0$ 是电流幅度。计算电场和磁场以及由这种天线辐射的波的相关功率密度的过程基本上与先前用于半波偶极子天线的过程相同。唯一的区别是电流分布 $\tilde{I}(z)$ 。如果我们插入由下式给出的 $\tilde{I}(z)$ 表达式

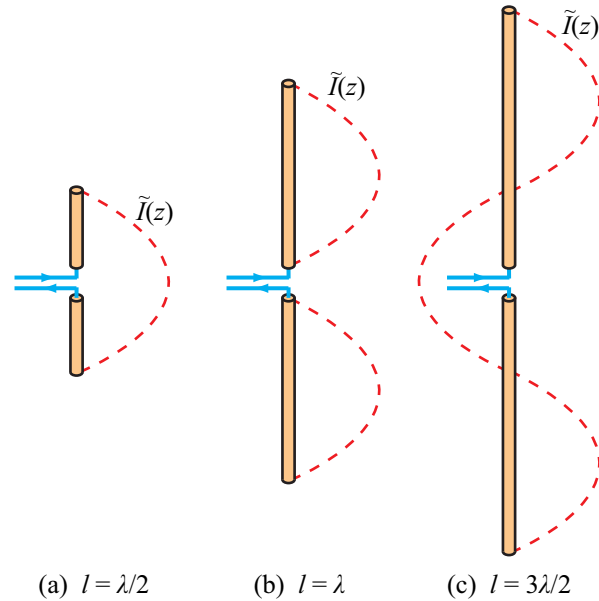


图 9-16 三个中心馈电偶极子的电流分布。

等式。(9.52) 代入方程。根据方程 (9.43)，我们得到了沿偶极子位置  $z$  处的元素长度  $dz$  辐射的波的微分电场  $d\tilde{E}_\theta$  的表达式：

$$d\tilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi} \left( \frac{e^{-jkR}}{R} \right) \sin\theta e^{jkz\cos\theta} dz \times \begin{cases} \sin[k(l/2 - z)] & \text{for } 0 \leq z \leq l/2, \\ \sin[k(l/2 + z)] & \text{for } -l/2 \leq z < 0. \end{cases} \quad (9.53)$$

偶极子辐射的总场为

$$\begin{aligned} \tilde{E}_\theta &= \int_{-l/2}^{l/2} d\tilde{E}_\theta = \int_0^{l/2} d\tilde{E}_\theta + \int_{-l/2}^0 d\tilde{E}_\theta \\ &= \frac{jk\eta_0 I_0}{4\pi} \left( \frac{e^{-jkR}}{R} \right) \sin\theta \\ &\quad \times \left\{ \int_0^{l/2} e^{jkz\cos\theta} \sin[k(l/2 - z)] dz \right. \\ &\quad \left. + \int_{-l/2}^0 e^{jkz\cos\theta} \sin[k(l/2 + z)] dz \right\}. \end{aligned} \quad (9.54)$$

如果我们应用欧拉恒等式将  $e^{jkz\cos\theta}$  表示为

$$e^{jkz\cos\theta} = \cos(kz\cos\theta) + j\sin(kz\cos\theta),$$

我们可以将两个积分积分并得到结果

$$\tilde{E}_\theta = j60I_0 \left( \frac{e^{-jkR}}{R} \right) \left[ \frac{\cos(\frac{kl}{2}\cos\theta) - \cos(\frac{kl}{2})}{\sin\theta} \right]. \quad (9.55)$$

偶极子天线辐射的相应时间平均功率密度由下式给出

$$S(\theta) = \frac{|\tilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos(\frac{\pi l}{\lambda}\cos\theta) - \cos(\frac{\pi l}{\lambda})}{\sin\theta} \right]^2, \quad (9.56)$$

其中我们使用了关系  $\eta_0 \approx 120\pi \text{ } (\Omega)$  和  $k = 2\pi/\lambda$ 。对于  $l = \lambda/2$ ，方程。(9.56) 简化为方程给出的表达式。(9.45) 为半波偶极子。图 9-17 显示了长度为  $\lambda/2$ 、 $\lambda$  和  $3\lambda/2$  偶极子的归一化辐射强度  $F(\theta) = S(R, \theta)/S_{\max}$  的图。具有  $l = \lambda/2$  和  $l = \lambda$  的偶极子具有类似的辐射方向图，最大值均沿  $\theta = 90^\circ$ ，但长波长偶极子的半功率波束宽度比半波偶极子窄，并且长波长偶极子的  $S_{\max} = 60I_0^2/(\pi R^2)$  是半波偶极子的四倍。长度为  $l = 3\lambda/2$  的偶极子的图案呈现出多瓣结构，其最大辐射方向不是沿着  $\theta = 90^\circ$ 。

9-5 Effective Area of a Receiving Antenna

So far, antennas have been treated as directional radiators of energy. Now, we examine the reverse process, namely how a receiving antenna extracts energy from an incident wave and delivers it to a load. The ability of an antenna to capture energy from an incident wave of power density  $S_i$  ( $\text{W}/\text{m}^2$ ) and convert it into an **intercepted power**  $P_{\text{int}}$  (W) for delivery to a matched load is characterized by the **effective area**  $A_e$ :

$$A_e = \frac{P_{\text{int}}}{S_i} \quad (\text{m}^2). \quad (9.57)$$

Other commonly used names for  $A_e$  include **effective aperture** and **receiving cross section**. The antenna receiving process may be modeled in terms of a Thévenin equivalent circuit (Fig. 9-18) consisting of a voltage  $\tilde{V}_{\text{oc}}$  in series with the antenna input impedance  $Z_{\text{in}}$ . Here,  $\tilde{V}_{\text{oc}}$  is the open-circuit voltage induced by the incident wave at the antenna terminals, and  $Z_L$  is the impedance of the load connected to the antenna (representing a receiver or some other circuit). In general, both  $Z_{\text{in}}$  and  $Z_L$  are complex:

$$Z_{\text{in}} = R_{\text{rad}} + jX_{\text{in}}, \quad (9.58a)$$

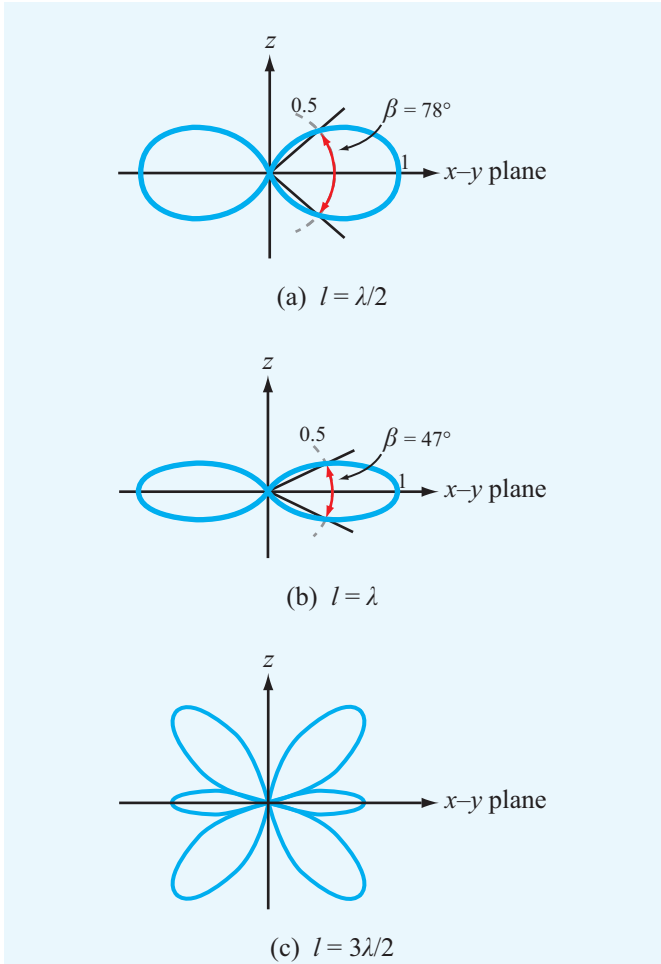


Figure 9-17 Radiation patterns of dipoles with lengths of  $\lambda/2$ ,  $\lambda$ , and  $3\lambda/2$ .

**Module 9.2 Linear Dipole Antenna** For a linear antenna of any specified length (in units of  $\lambda$ ), this module displays the current distribution along the antenna and the far-field radiation patterns in the horizontal and elevation planes. It also calculates the total power radiated by the antenna, the radiation resistance, and the antenna directivity.

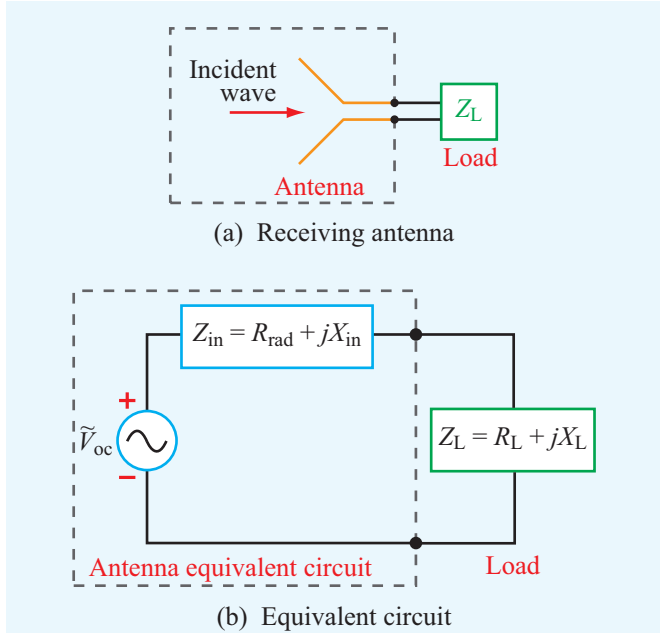


Figure 9-18 Receiving antenna represented by an equivalent circuit.

9-5 接收天线的有效面积

到目前为止，天线已被视为能量的定向辐射器。现在，我们研究相反的过程，即接收天线如何从入射波中提取能量并将其传递给负载。天线从功率密度为  $S_i$  ( $\text{W}/\text{m}^2$ ) 的入射波捕获能量并将其转换为截获功率  $P_{\text{int}}$  (W) 以传送到匹配负载的能力由有效面积  $A_e$  来表征：

$$A_e = \frac{P_{\text{int}}}{S_i} \quad (\text{m}^2). \quad (9.57)$$

$A_e$  的其他常用名称包括有效孔径和接收横截面。天线接收过程可以根据Thévenin等效电路（图9-18）进行建模，该电路由与天线输入阻抗 $Z_{\text{in}}$ 串联的电压 $\tilde{V}_{\text{oc}}$ 组成。这里， $\tilde{V}_{\text{oc}}$ 是天线端子处的入射波感应的开路电压， $Z_L$ 是连接到天线的负载的阻抗（代表接收器或其他电路）。一般来说， $Z_{\text{in}}$ 和 $Z_L$ 都是复数：

$$Z_{\text{in}} = R_{\text{rad}} + jX_{\text{in}}, \quad (9.58a)$$

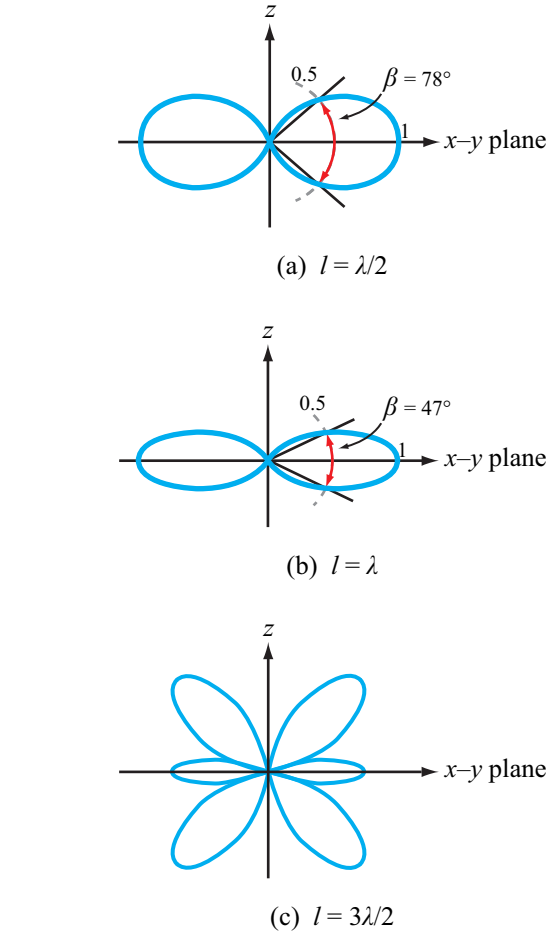


图 9-17 长度为  $\lambda/2$ 、 $\lambda$  和  $3\lambda/2$  的偶极子的辐射方向图。

模块 9.2 线性偶极子天线 对于任何指定长度（以  $\lambda$  为单位）的线性天线，该模块显示沿天线的电流分布以及水平面和仰角面中的远场辐射方向图。它还计算天线辐射的总功率、辐射电阻和天线方向性。

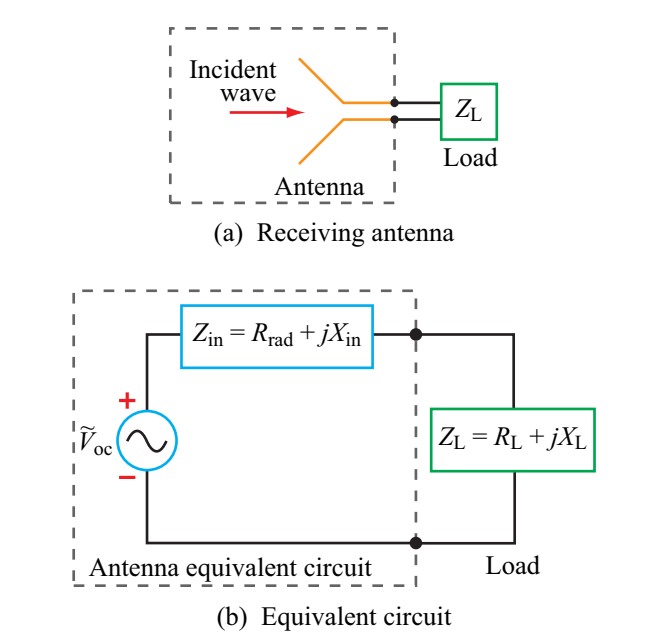
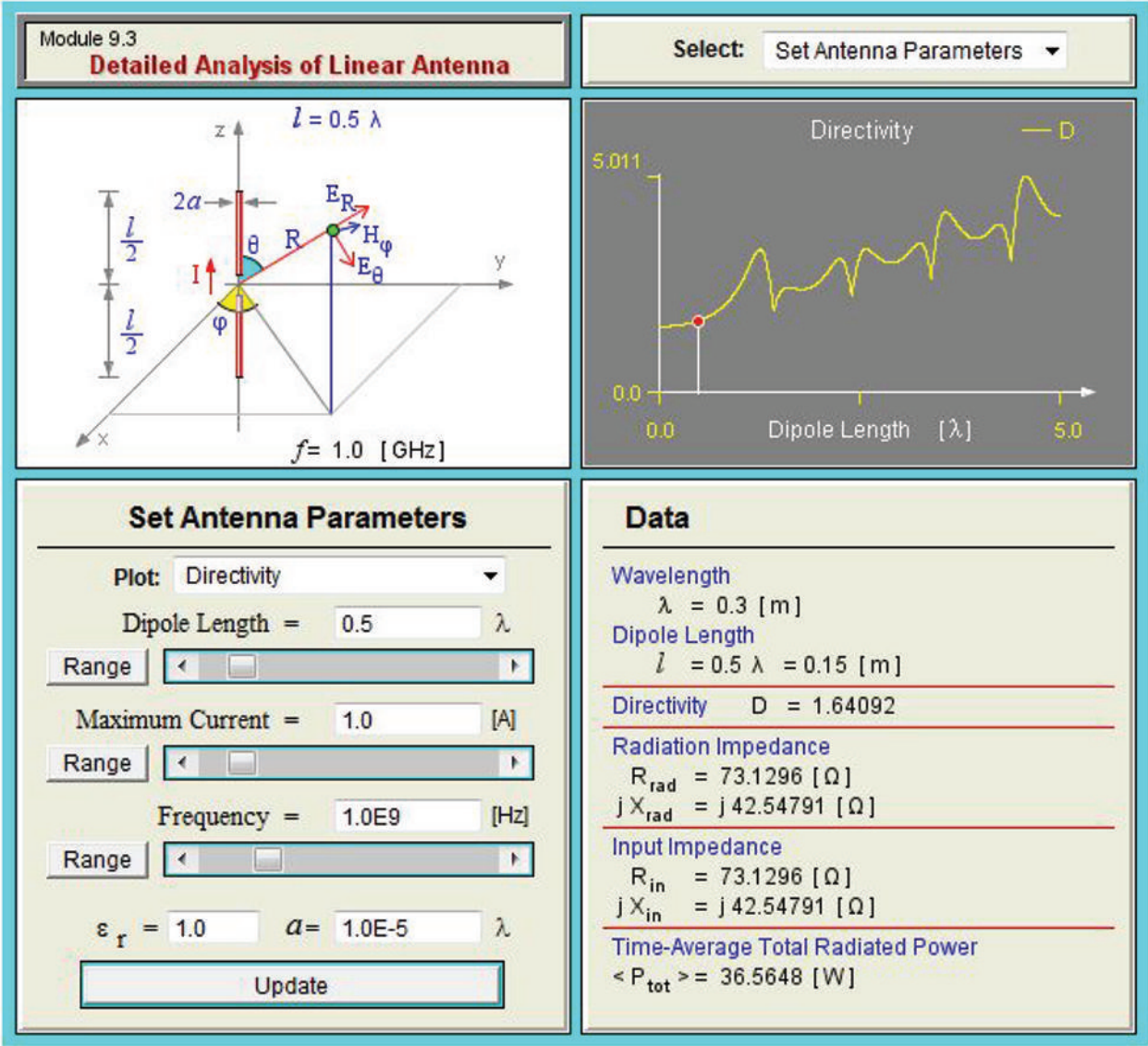


图 9-18 用等效电路表示的接收天线。

**Module 9.3 Detailed Analysis of Linear Antenna** This module complements Module 9.2 by offering extensive information about the specified linear antenna, including its directivity and plots of its current and field distributions.



$Z_L = R_L + jX_L,$  (9.58b)

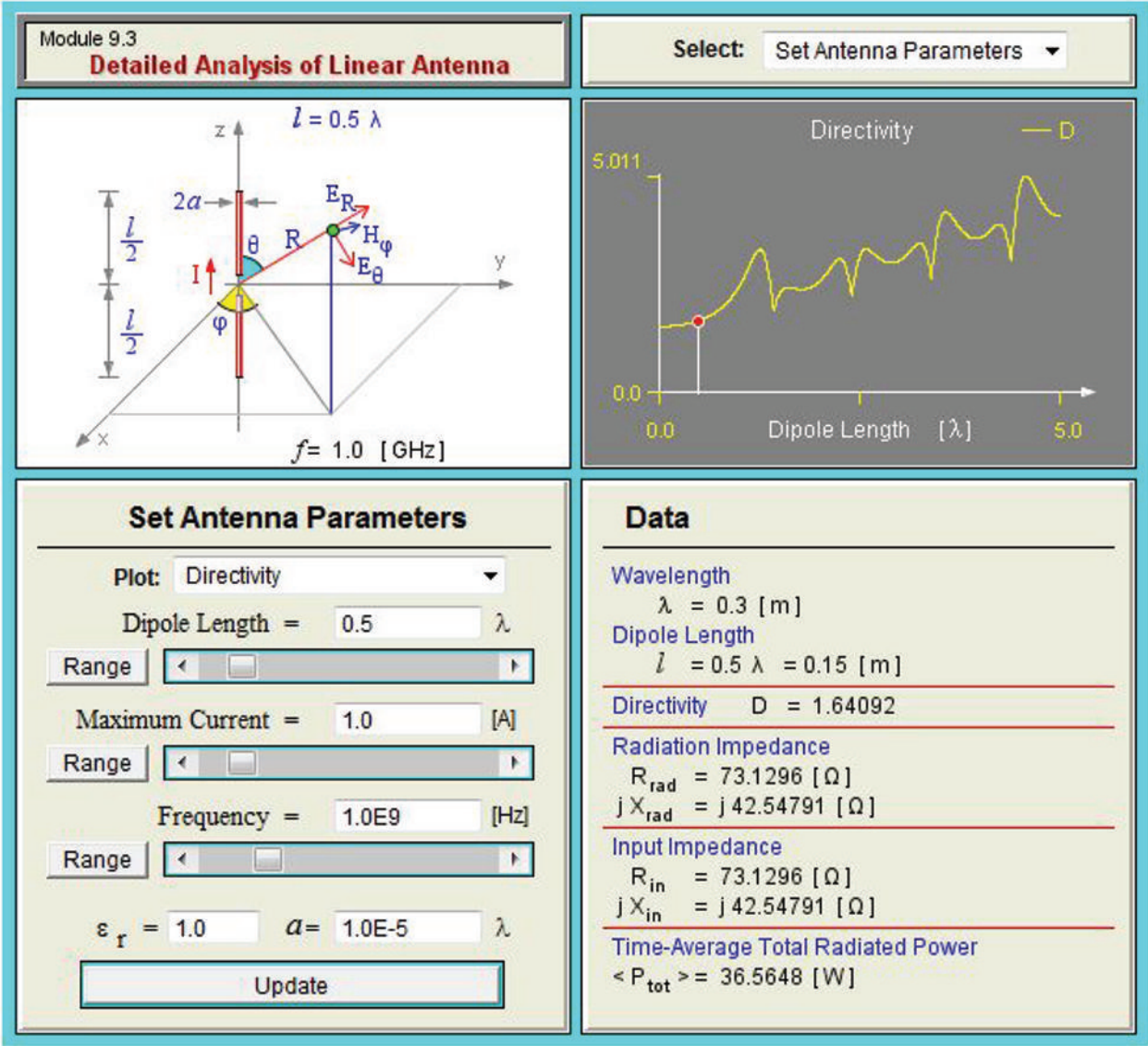
phasor, the time-average power delivered to the load is

$$P_L = \frac{1}{2} |\tilde{I}_L|^2 R_{\text{rad}} = \frac{1}{2} \left[ \frac{|\tilde{V}_{\text{oc}}|}{2R_{\text{rad}}} \right]^2 R_{\text{rad}} = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}}, \quad (9.59)$$

where  $\tilde{I}_L = \tilde{V}_{\text{oc}}/(2R_{\text{rad}})$  is the phasor current flowing through the circuit. Since the antenna is lossless, all of the intercepted

where  $R_{\text{rad}}$  denotes the radiation resistance of the antenna (assuming  $R_{\text{loss}} \ll R_{\text{rad}}$ ). To maximize power transfer to the load, the load impedance must be chosen so that either  $Z_L = Z_{\text{in}}^*$ , or equivalently,  $R_L = R_{\text{rad}}$  and  $X_L = -X_{\text{in}}$ . In that case, the circuit reduces to a source  $\tilde{V}_{\text{oc}}$  that is connected across a resistance equal to  $2R_{\text{rad}}$ . Since  $\tilde{V}_{\text{oc}}$  is a sinusoidal voltage

模块 9.3 线性天线的详细分析 该模块通过提供有关指定线性天线的广泛信息（包括其方向性及其电流和场分布图）来补充模块 9.2。



$Z_L = R_L + jX_L,$  (9.58b)

相量，输送到负载的时间平均功率为

$$P_L = \frac{1}{2} |\tilde{I}_L|^2 R_{\text{rad}} = \frac{1}{2} \left[ \frac{|\tilde{V}_{\text{oc}}|}{2R_{\text{rad}}} \right]^2 R_{\text{rad}} = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}}, \quad (9.59)$$

其中 $R_{\text{rad}}$ 表示天线的辐射电阻（假设 $R_{\text{loss}} \ll R_{\text{rad}}$ ）。为了最大限度地提高向负载传输的功率，必须选择负载阻抗，以便  $Z_L = Z_{\text{in}}^*$  或等效的  $R_L = R_{\text{rad}}$  和  $X_L = -X_{\text{in}}$ 。在这种情况下，电路简化为源极  $\tilde{V}_{\text{oc}}$ ，其跨接等于  $2R_{\text{rad}}$  的电阻。由于  $\tilde{V}_{\text{oc}}$  是正弦电压

其中 $\tilde{I}_L = \tilde{V}_{\text{oc}}/(2R_{\text{rad}})$ 是流过电路的相量电流。由于天线是无损的，所有截获的

power  $P_{\text{int}}$  ends up in the load resistance  $R_L$ . Hence,

$$P_{\text{int}} = P_L = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}}. \quad (9.60)$$

For an incident wave with electric field  $\tilde{E}_i$  parallel to the antenna polarization direction, the power density carried by the wave is

$$S_i = \frac{|\tilde{E}_i|^2}{2\eta_0} = \frac{|\tilde{E}_i|^2}{240\pi}. \quad (9.61)$$

The ratio of the results provided by Eqs. (9.60) and (9.61) gives

$$A_e = \frac{P_{\text{int}}}{S_i} = \frac{30\pi|\tilde{V}_{\text{oc}}|^2}{R_{\text{rad}}|\tilde{E}_i|^2}. \quad (9.62)$$

The open-circuit voltage  $\tilde{V}_{\text{oc}}$  induced in the receiving antenna is due to the incident field  $\tilde{E}_i$ , but the relation between them depends on the specific antenna under consideration. By way of illustration, let us consider the case of the short-dipole antenna of Section 9-1. Because the length  $l$  of the short dipole is small compared with  $\lambda$ , the current induced by the incident field is uniform across its length, and the open-circuit voltage is simply  $\tilde{V}_{\text{oc}} = \tilde{E}_i l$ . Noting that  $R_{\text{rad}} = 80\pi^2(l/\lambda)^2$  for the short dipole (see Eq. (9.38)) and using  $\tilde{V}_{\text{oc}} = \tilde{E}_i l$ , Eq. (9.62) simplifies to

$$A_e = \frac{3\lambda^2}{8\pi} \quad (\text{m}^2). \quad \textbf{(short dipole)} \quad (9.63)$$

In Example 9-2 it was shown that for the Hertzian dipole the directivity  $D = 1.5$ . In terms of  $D$ , Eq. (9.63) can be rewritten in the form

$$A_e = \frac{\lambda^2 D}{4\pi} \quad (\text{m}^2). \quad \textbf{(any antenna)} \quad (9.64)$$

► Despite the fact that the relation between  $A_e$  and  $D$  given by Eq. (9.64) was derived for a Hertzian dipole, it can be shown that it is also valid *for any antenna* under matched-impedance conditions. ◀

**Exercise 9-6:** The effective area of an antenna is  $9 \text{ m}^2$ . What is its directivity in decibels at 3 GHz?

**Answer:**  $D = 40.53 \text{ dB}$ . (See ◉.)

**Exercise 9-7:** At 100 MHz, the pattern solid angle of an antenna is 1.3 sr. Find (a) the antenna directivity  $D$  and (b) its effective area  $A_e$ .

**Answer:** (a)  $D = 9.67$ , (b)  $A_e = 6.92 \text{ m}^2$ . (See ◉.)

9-6 Friis Transmission Formula

The two antennas shown in Fig. 9-19 are part of a free-space communication link with the separation between the antennas,  $R$ , being large enough for each to be in the far-field region of the other. The transmitting and receiving antennas have effective areas  $A_t$  and  $A_r$  and radiation efficiencies  $\xi_t$  and  $\xi_r$ , respectively. Our objective is to find a relationship between  $P_t$ , which is the power supplied to the transmitting antenna, and  $P_{\text{rec}}$ , which is the power delivered to the receiver. As always, we assume that both antennas are impedance-matched to their respective transmission lines. Initially, we consider the case where the two antennas are oriented so that the peak of the radiation pattern of each antenna points in the direction of the other.

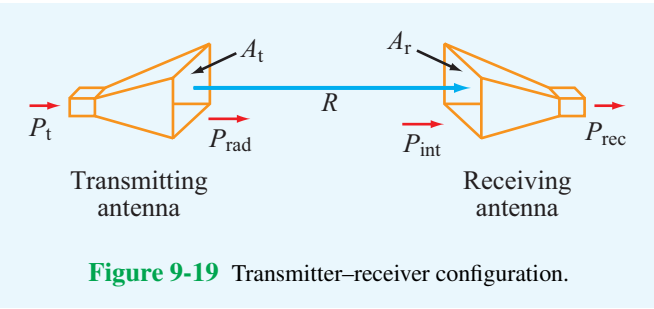
We start by treating the transmitting antenna as a lossless isotropic radiator. The power density incident upon the receiving antenna at a distance  $R$  from an isotropic transmitting antenna is simply equal to the transmitter power  $P_t$  divided by the surface area of a sphere of radius  $R$ :

$$S_{\text{iso}} = \frac{P_t}{4\pi R^2}. \quad (9.65)$$

The real transmitting antenna is neither lossless nor isotropic. Hence, the power density  $S_r$  due to the real antenna is

$$S_r = G_t S_{\text{iso}} = \xi_t D_t S_{\text{iso}} = \frac{\xi_t D_t P_t}{4\pi R^2}. \quad (9.66)$$

Through the gain  $G_t = \xi_t D_t$ ,  $\xi_t$  accounts for the fact that only part of the power  $P_t$  supplied to the antenna is radiated out into space, and  $D_t$  accounts for the directivity of the transmitting



功率  $P_{\text{int}}$  最终变为负载电阻  $R_L$ 。因此,

$$P_{\text{int}} = P_L = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}}. \quad (9.60)$$

对于电场  $\tilde{E}_i$  平行于天线极化方向的入射波, 波所携带的功率密度为

$$S_i = \frac{|\tilde{E}_i|^2}{2\eta_0} = \frac{|\tilde{E}_i|^2}{240\pi}. \quad (9.61)$$

等式提供的结果的比率。(9.60) 和 (9.61) 给出

$$A_e = \frac{P_{\text{int}}}{S_i} = \frac{30\pi|\tilde{V}_{\text{oc}}|^2}{R_{\text{rad}}|\tilde{E}_i|^2}. \quad (9.62)$$

接收天线中感应的开路电压  $\tilde{V}_{\text{oc}}$  是由入射场  $\tilde{E}_i$  引起的, 但它们之间的关系取决于所考虑的特定天线。作为说明, 让我们考虑第 9-1 节的短偶极子天线的情况。由于短偶极子的长度  $l$  与  $\lambda$  相比较小, 因此入射场感应的电流在其长度上是均匀的, 开路电压仅为  $\tilde{V}_{\text{oc}} = \tilde{E}_i l$ 。注意到  $R_{\text{rad}} = 80\pi^2(l/\lambda)^2$  对于短偶极子 (参见方程 (9.38)) 并使用  $\tilde{V}_{\text{oc}} = \tilde{E}_i l$ , 方程(9.62) 简化为

$$A_e = \frac{3\lambda^2}{8\pi} \quad (\text{m}^2). \quad \textbf{(short dipole)} \quad (9.63)$$

在实施例9-2中表明, 对于赫兹偶极子, 方向性 $D$ =为1.5。就  $D$  而言, 方程。(9.63) 可以重写为以下形式

$$A_e = \frac{\lambda^2 D}{4\pi} \quad (\text{m}^2). \quad \textbf{(any antenna)} \quad (9.64)$$

► 尽管事实上  $A_e$  和  $D$  之间的关系由方程给出。式 (9.64) 是针对赫兹偶极子导出的, 可以证明它对于匹配阻抗条件下的任何天线也有效。 ◀

练习9-6: 天线的有效面积为 $9 \text{ m}^2$ 。3 GHz 时的方向性 (以分贝为单位) 是多少?

**Answer:**  $D = 40.53 \text{ dB}$ . (See ◉.)

练习 9-7: 在 100 MHz 时, 天线的方向图立体角为 1.3 sr。求 (a) 天线方向性  $D$  和 (b) 其有效面积  $A_e$ 。

**Answer:** (a)  $D = 9.67$ , (b)  $A_e = 6.92 \text{ m}^2$ . (See ◉.)

9-6 弗里斯传输公式

图 9-19 中所示的两个天线是自由空间通信链路的一部分, 天线之间的间隔  $R$  足够大, 足以使每个天线位于另一个天线的远场区域中。发射和接收天线的有效面积分别为  $A_t$  和  $A_r$ , 辐射效率分别为  $\xi_t$  和  $\xi_r$ 。我们的目标是找到  $P_t$  (提供给发射天线的功率) 和  $P_{\text{rec}}$  (提供给接收器的功率) 之间的关系。一如既往, 我们假设两个天线都与其各自的传输线阻抗匹配。最初, 我们考虑两个天线定向的情况, 使得每个天线的辐射方向图的峰值指向另一个天线的方向。

我们首先将发射天线视为无损各向同性辐射器。距各向同性发射天线距离  $R$  处入射到接收天线的功率密度简单地等于发射器功率  $P_t$  除以半径为  $R$  的球体表面积:

$$S_{\text{iso}} = \frac{P_t}{4\pi R^2}. \quad (9.65)$$

真正的发射天线既不是无损的, 也不是各向同性的。因此, 实际天线的功率密度  $S_r$  为

$$S_r = G_t S_{\text{iso}} = \xi_t D_t S_{\text{iso}} = \frac{\xi_t D_t P_t}{4\pi R^2}. \quad (9.66)$$

通过增益  $G_t = \xi_t D_t$ ,  $\xi_t$  说明了只有部分提供给天线的功率  $P_t$  辐射到空间中的事实, 而  $D_t$  则说明了发射的方向性

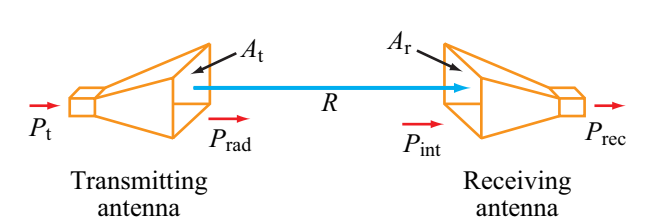


图 9-19 发射器-接收器配置。

antenna (in the direction of the receiving antenna). Moreover, by Eq. (9.64),  $D_t$  is related to  $A_t$  by  $D_t = 4\pi A_t/\lambda^2$ . Hence, Eq. (9.66) becomes

$$S_r = \frac{\xi_t A_t P_t}{\lambda^2 R^2} . \quad (9.67)$$

On the receiving-antenna side, the power intercepted by the receiving antenna is equal to the product of the incident power density  $S_r$  and the effective area  $A_r$ :

$$P_{\text{int}} = S_r A_r = \frac{\xi_t A_t A_r P_t}{\lambda^2 R^2} . \quad (9.68)$$

The power delivered to the receiver,  $P_{\text{rec}}$ , is equal to the intercepted power  $P_{\text{int}}$  multiplied by the radiation efficiency of the receiving antenna,  $\xi_r$ . Hence,  $P_{\text{rec}} = \xi_r P_{\text{int}}$ , which leads to the result

$$\frac{P_{\text{rec}}}{P_t} = \frac{\xi_t \xi_r A_t A_r}{\lambda^2 R^2} = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 . \quad (9.69)$$

► This relation is known as the *Friis transmission formula*, and  $P_{\text{rec}}/P_t$  is called the *power transfer ratio*. ◀

If the two antennas are not oriented in the direction of maximum power transfer, Eq. (9.69) assumes the general form

$$\frac{P_{\text{rec}}}{P_t} = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r), \quad (9.70)$$

where  $F_t(\theta_t, \phi_t)$  is the normalized radiation intensity of the transmitting antenna at angles  $(\theta_t, \phi_t)$  corresponding to the direction of the receiving antenna (as seen by the antenna pattern of the transmitting antenna), and a similar definition applies to  $F_r(\theta_r, \phi_r)$  for the receiving antenna.

Example 9-4: Satellite Communication System

A 6 GHz direct-broadcast TV satellite system transmits 100 W through a 2 m diameter parabolic dish antenna from a distance of approximately 40,000 km above Earth’s surface. Each TV channel occupies a bandwidth of 5 MHz. Due to electromagnetic noise picked up by the antenna as well as noise generated by the receiver electronics, a home TV receiver has a noise level given by

$$P_n = K T_{\text{sys}} B \quad (\text{W}), \quad (9.71)$$

where  $T_{\text{sys}}$  [measured in kelvins (K)] is a figure of merit called the *system noise temperature* that characterizes the noise

performance of the receiver–antenna combination,  $K$  is Boltzmann’s constant [ $1.38 \times 10^{-23}$  (J/K)], and  $B$  is the receiver bandwidth in Hz.

The *signal-to-noise ratio*  $S_n$  (which should not be confused with the power density  $S$ ) is defined as the ratio of  $P_{\text{rec}}$  to  $P_n$ :

$$S_n = P_{\text{rec}}/P_n \quad (\text{dimensionless}). \quad (9.72)$$

For a receiver with  $T_{\text{sys}} = 580$  K, what minimum diameter of a parabolic dish receiving antenna is required for high-quality TV reception with  $S_n = 40$  dB? The satellite and ground receiving antennas may be assumed lossless, and their effective areas may be assumed equal to their physical apertures.

**Solution:** The following quantities are given:

$$\begin{aligned} P_t &= 100 \text{ W}, \quad f = 6 \text{ GHz} = 6 \times 10^9 \text{ Hz}, \quad S_n = 10^4, \\ &\text{transmit antenna diameter } d_t = 2 \text{ m}, \\ T_{\text{sys}} &= 580 \text{ K}, \quad R = 40,000 \text{ km} = 4 \times 10^7 \text{ m}, \\ B &= 5 \text{ MHz} = 5 \times 10^6 \text{ Hz}. \end{aligned}$$

The wavelength  $\lambda = c/f = 5 \times 10^{-2}$  m, and the area of the transmitting satellite antenna is  $A_t = (\pi d_t^2/4) = \pi$  (m<sup>2</sup>). From Eq. (9.71), the receiver noise power is

$$P_n = K T_{\text{sys}} B = 1.38 \times 10^{-23} \times 580 \times 5 \times 10^6 = 4 \times 10^{-14} \text{ W}.$$

Using Eq. (9.69) with  $\xi_t = \xi_r = 1$ ,

$$P_{\text{rec}} = \frac{P_t A_t A_r}{\lambda^2 R^2} = \frac{100\pi A_r}{(5 \times 10^{-2})^2 (4 \times 10^7)^2} = 7.85 \times 10^{-11} A_r.$$

The area of the receiving antenna,  $A_r$ , can now be determined by equating the ratio  $P_{\text{rec}}/P_n$  to  $S_n = 10^4$ :

$$10^4 = \frac{7.85 \times 10^{-11} A_r}{4 \times 10^{-14}},$$

which yields the value  $A_r = 5.1$  m<sup>2</sup>. The required minimum diameter is  $d_r = \sqrt{4A_r/\pi} = 2.55$  m.

**Exercise 9-8:** If the operating frequency of the communication system described in Example 9-4 is doubled to 12 GHz, what would then be the minimum required diameter of a home receiving TV antenna?

**Answer:**  $d_r = 1.27$  m. (See [EM](#).)

**Exercise 9-9:** A 3 GHz microwave link consists of two identical antennas each with a gain of 30 dB. Determine the received power given that the transmitter output power is 1 kW and the two antennas are 10 km apart.

**Answer:**  $P_{\text{rec}} = 6.33 \times 10^{-4}$  W. (See [EM](#).)

天线（接收天线方向）。此外，通过等式。(9.64)， $D_t$  通过  $D_t = 4\pi A_t/\lambda^2$  与  $A_t$  相关。因此，等式。(9.66) 变为

$$S_r = \frac{\xi_t A_t P_t}{\lambda^2 R^2} . \quad (9.67)$$

在接收天线侧，接收天线截获的功率等于入射功率密度 $S_r$ 与有效面积 $A_r$ 的乘积：

$$P_{\text{int}} = S_r A_r = \frac{\xi_t A_t A_r P_t}{\lambda^2 R^2} . \quad (9.68)$$

传送到接收器的功率  $P_{\text{rec}}$  等于拦截功率  $P_{\text{int}}$  乘以接收天线的辐射效率  $\xi_r$ 。因此， $P_{\text{rec}} = \xi_r P_{\text{int}}$ ，导致结果

$$\frac{P_{\text{rec}}}{P_t} = \frac{\xi_t \xi_r A_t A_r}{\lambda^2 R^2} = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 . \quad (9.69)$$

► 这种关系称为弗里斯传输公式， $P_{\text{rec}}/P_t$ 称为功率传输比。◀

如果两个天线没有朝向最大功率传输的方向，则方程：(9.69) 呈现一般形式

$$\frac{P_{\text{rec}}}{P_t} = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r), \quad (9.70)$$

其中  $F_t(\theta_t, \phi_t)$  是发射天线在与接收天线方向（如发射天线的天线方向图所示）相对应的角度  $(\theta_t, \phi_t)$  处的归一化辐射强度，类似的定义适用于接收天线的  $F_r(\theta_r, \phi_r)$ 。

例9-4：卫星通信系统

6 GHz 直播电视卫星系统通过直径 2 m 的抛物面碟形天线从距地球表面约 40,000 公里的距离传输 100 W 功率。每个电视频道占用 5 MHz 的带宽。由于天线接收到的电磁噪声以及接收器电子设备产生的噪声，家庭电视接收器的噪声水平为：

$$P_n = K T_{\text{sys}} B \quad (\text{W}), \quad (9.71)$$

其中  $T_{\text{sys}}$  [以开尔文 (K) 为单位测量] 是一个品质因数，称为系统噪声温度，它表征了噪声

接收器-天线组合的性能， $K$  是玻尔兹曼常数 [ $1.38 \times 10^{-23}$  (J/K)],  $B$  是接收器带宽（以 Hz 为单位）。

信噪比  $S_n$ （不应与功率密度  $S$ ）混淆，其定义为  $P_{\text{rec}}$  与  $P_n$  之比：

$$S_n = P_{\text{rec}}/P_n \quad (\text{dimensionless}). \quad (9.72)$$

对于  $T_{\text{sys}} = 580$  K 的接收器，要实现  $S_n = 40$  dB 的高质量电视接收，抛物面碟形接收天线的最小直径需要多少？可以假设卫星和地面接收天线是无损的，并且可以假设它们的有效面积等于它们的物理孔径。

解：给出下列数量：

$$\begin{aligned} P_t &= 100 \text{ W}, \quad f = 6 \text{ GHz} = 6 \times 10^9 \text{ Hz}, \quad S_n = 10^4, \\ &\text{transmit antenna diameter } d_t = 2 \text{ m}, \\ T_{\text{sys}} &= 580 \text{ K}, \quad R = 40,000 \text{ km} = 4 \times 10^7 \text{ m}, \\ B &= 5 \text{ MHz} = 5 \times 10^6 \text{ Hz}. \end{aligned}$$

波长为 $\lambda = c/f=5\times 10^{-2}$ m，发射卫星天线面积为 $A_t = (\pi d_t^2/4) = \pi$  (m<sup>2</sup>)。从方程。(9.71)，接收器噪声功率为

$$P_n = K T_{\text{sys}} B = 1.38 \times 10^{-23} \times 580 \times 5 \times 10^6 = 4 \times 10^{-14} \text{ W}.$$

使用方程式(9.69) 与  $\xi_t = \xi_r = 1$ ,

$$P_{\text{rec}} = \frac{P_t A_t A_r}{\lambda^2 R^2} = \frac{100\pi A_r}{(5 \times 10^{-2})^2 (4 \times 10^7)^2} = 7.85 \times 10^{-11} A_r.$$

现在可以通过将  $P_{\text{rec}}/P_n$  与  $S_n = 10^4$  的比率相等来确定接收天线的面积  $A_r$ ：

$$10^4 = \frac{7.85 \times 10^{-11} A_r}{4 \times 10^{-14}},$$

得出值  $A_r = 5.1$  m<sup>2</sup>。所需的最小直径为  $d_r = \sqrt{4A_r/\pi} = 2.55$  m。

练习9-8：如果例9-4中描述的通信系统的工作频率加倍到12 GHz，那么家庭接收电视天线所需的最小直径是多少？

**Answer:**  $d_r = 1.27$  m. (See [EM](#).)

练习 9-9：3 GHz 微波链路由两个相同的天线组成，每个天线的增益均为 30 dB。假定发射机输出功率为 1 kW，两个天线相距 10 km，确定接收功率。

**Answer:**  $P_{\text{rec}} = 6.33 \times 10^{-4}$  W. (See [EM](#).)

Technology Brief 17:  
Health Risks of EM Fields

Can the use of cell phones cause cancer? Does exposure to the electromagnetic fields (EMFs) associated with power lines pose health risks to humans? Are we endangered by EMFs generated by home appliances, telephones, electrical wiring, and the myriad of electronic gadgets we use every day (Fig. TF17-1)? Despite reports in some of the popular media alleging a causative relationship between low-level EMFs and many diseases, according to reports issued by governmental and professional boards in the U.S. and Europe, the answer is given here.

► NO, we are not at risk, so long as manufacturers adhere to the approved governmental standards for **maximum permissible exposure (MPE)** levels. With regard to cell phones, the official reports caution that their conclusions are limited to phone use of less than 15 years, since data for longer use is not yet available. ◀



Figure TF17-1 Electromagnetic fields are emitted by power lines, cell phones, TV towers, and many other electronic circuits and devices.

Physiological Effects of EMFs

The energy carried by a photon with an EM frequency  $f$  is given by  $E = hf$ , where  $h$  is Planck's constant. The mode of interaction between a photon passing through a material and the material's atoms or molecules is very much dependent on  $f$ . If  $f$  is greater than about  $10^{15}$  Hz (which falls in the ultraviolet (UV) band of the EM spectrum), the photon's energy is sufficient to free an electron and remove it completely, thereby ionizing the affected atom or molecule. Consequently, the energy carried by such EM waves is called **ionizing radiation**, in contrast with **non-ionizing radiation** (Fig. TF17-2), where photons may be able to cause an electron to move to a higher energy level but not eject it from its host atom or molecule.

Assessing health risks associated with exposure to EMFs is complicated by the number of variables involved, including: (1) the frequency  $f$ , (2) the intensities of the electric and magnetic fields, (3) the exposure duration—whether continuous or discontinuous, pulsed or uniform—and (4) the specific part of the body that is getting exposed. We know that intense laser illumination can cause corneal burn, high-level X-rays can damage living tissue and cause cancer, and in fact, any form of EM energy can be dangerous if the exposure level and/or duration exceed certain safety limits. Governmental and professional safety boards are tasked with establishing maximum permissible exposure (MPE) levels that protect human beings against adverse health effects associated with EMFs. In the U.S., the relevant standards are IEEE Std C95.6 (dated 2002), which addresses EM fields in the 1 Hz to 3 kHz range, and IEEE Std 95.1 (dated 2005), which deals with the frequency range from 3 kHz to 300 GHz. On the European side of the Atlantic, responsibility for establishing MPE levels resides with the Scientific Committee on Emerging and Newly Identified Health Risks (SCENIHR) of the European Commission.

► At frequencies below 100 kHz, the goal is to minimize adverse effects of exposure to electric fields that can cause **electrostimulation** of nerve and muscle cells. Above 5 MHz, the main concern is excessive tissue heating, and in the transition region of 100 kHz to 5 MHz, safety standards are designed to protect against both electrostimulation and excessive heating. ◀

技术简介 17: 电磁场的健康风险

使用手机会致癌吗？暴露于与电源线相关的电磁场 (EMF) 会对人类造成健康风险吗？我们是否受到家用电器、电话、电线以及我们每天使用的无数电子产品产生的电磁场的威胁 (图TF17-1)？尽管一些流行媒体报道称低水平电磁场与许多疾病之间存在因果关系，但根据美国和欧洲政府和专业委员会发布的报告，这里给出了答案。

► 不，只要制造商遵守政府批准的最大允许暴露 (MPE) 水平标准，我们就不会面临风险。关于手机，官方报告警告说，他们的结论仅限于手机使用时间少于 15 年的情况，因为尚无更长时间使用的数据。 ◀



图 TF17-1 电磁场由电源线、手机、电视塔以及许多其他电子电路和设备发射。

电磁场的生理效应

电磁频率为  $f$  的光子携带的能量由  $E = hf$  给出，其中  $h$  是普朗克常数。穿过材料的光子与材料的原子或分子之间的相互作用模式很大程度上取决于  $f$ 。如果  $f$  大于约  $10^{15}$  Hz (落在电磁光谱的紫外 (UV) 波段)，则光子的能量足以释放电子并将其完全移除，从而电离受影响的原子或分子。因此，这种电磁波携带的能量称为电离辐射，与非电离辐射相反 (图TF17-2)，其中光子可能使电子移动到更高的能级，但不会将其从其主体原子或分子中喷射出来。

评估与暴露于电磁场相关的健康风险因涉及的变量数量而变得复杂，包括：(1) 频率  $f$ 、(2) 电场和磁场的强度、(3) 暴露持续时间 (无论是连续还是不连续、脉冲还是均匀) 以及 (4) 暴露的身体特定部位。我们知道，强烈的激光照射会导致角膜烧伤，高强度 X 射线会损伤活体组织并导致癌症，事实上，如果暴露水平和/或持续时间超过一定的安全限度，任何形式的电磁能量都可能是危险的。政府和专业安全委员会的任务是确定最大允许暴露 (MPE) 水平，以保护人类免受与电磁场相关的不良健康影响。在美国，相关标准包括 IEEE Std C95.6 (2002 年) 和 IEEE Std 95.1 (2005 年)，前者涉及 1 Hz 至 3 kHz 范围内的电磁场，后者涉及 3 kHz 至 300 GHz 频率范围。在大西洋的欧洲一侧，欧盟委员会新兴和新发现的健康风险科学委员会 (SCENIHR) 负责制定 MPE 水平。

► 在低于 100 kHz 的频率下，目标是最大限度地减少暴露于电场中可能导致神经和肌肉细胞电刺激的不利影响。高于 5 MHz 时，主要问题是组织过度加热，而在 100 kHz 至 5 MHz 的过渡区域，安全标准旨在防止电刺激和过度加热。 ◀

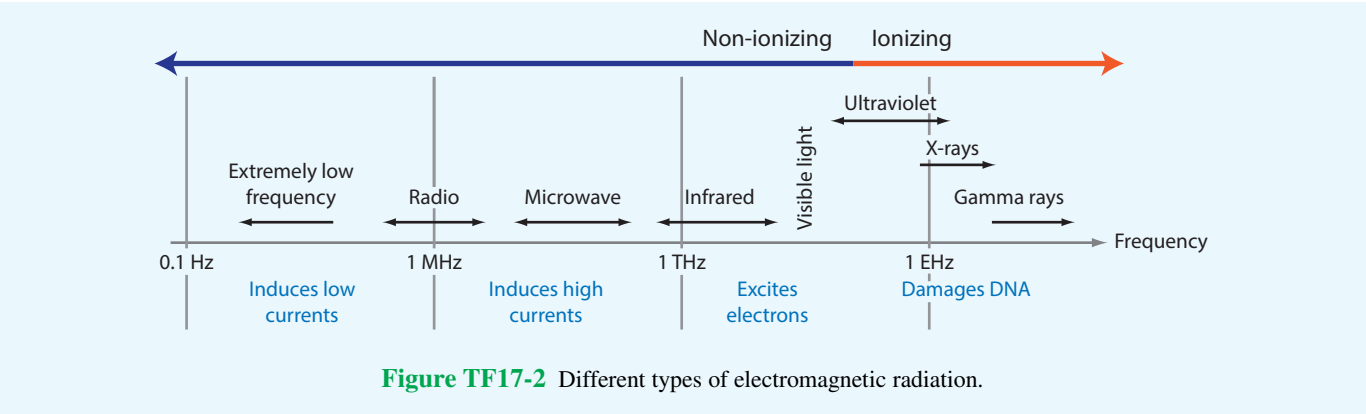


Figure TF17-2 Different types of electromagnetic radiation.

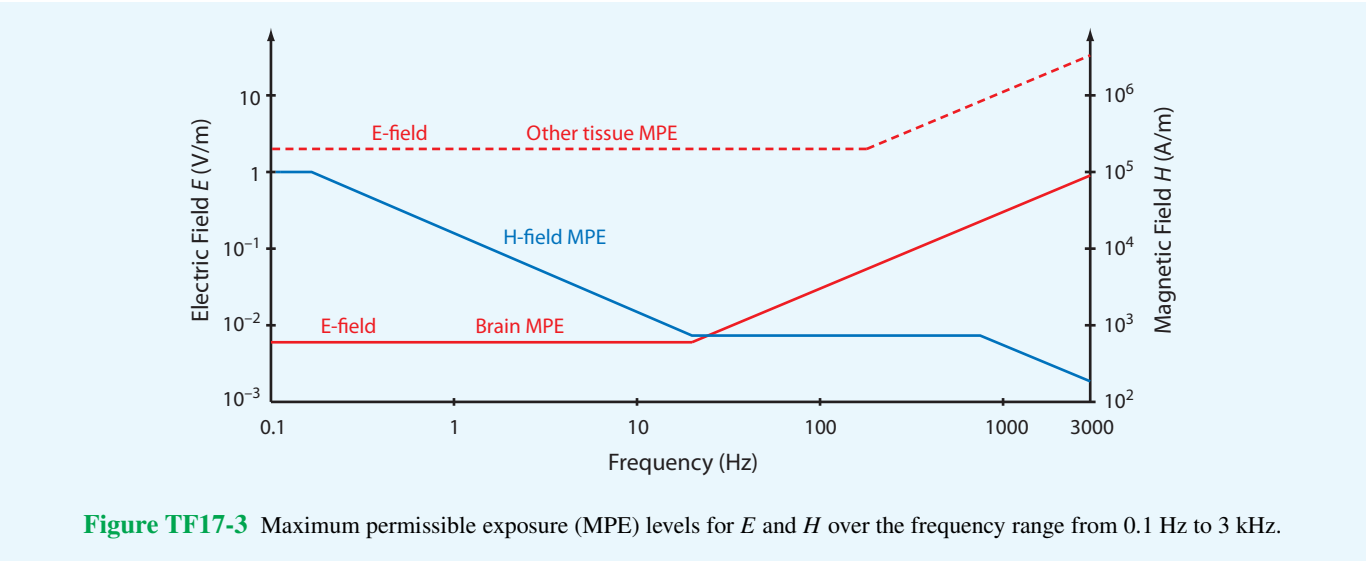


Figure TF17-3 Maximum permissible exposure (MPE) levels for  $E$  and  $H$  over the frequency range from 0.1 Hz to 3 kHz.

**Frequency Range  $0 \leq f \leq 3$  kHz:** The plots in **Fig. TF17-3** display the values of MPE for electric and magnetic fields over the frequency range below 3 kHz. According to IEEE Std C95.6, it is sufficient to demonstrate compliance with the MPE levels for either the electric field  $E$  or the magnetic field  $H$ . According to the plot for  $H$ , exposure at 60 Hz should not exceed 720 A/m. The magnetic field due to power lines is typically in the range of 2 to 6 A/m underneath the lines, which is at least two orders of magnitude smaller than the established safe level for  $H$ .

**Frequency Range  $3 \text{ kHz} \leq f \leq 300 \text{ GHz}$ :** At frequencies below 500 MHz, MPE is specified in terms of the electric and magnetic field strengths of the EM energy (**Fig. TF17-4**). From 100 MHz to 300 GHz (and beyond), MPE is specified in terms of the product of  $E$  and  $H$ , namely the power density  $S$ . Cell phones operate in the 1 to 2 GHz band; the specified MPE is 1 W/m<sup>2</sup> (or equivalently 0.1 mW/cm<sup>2</sup>).

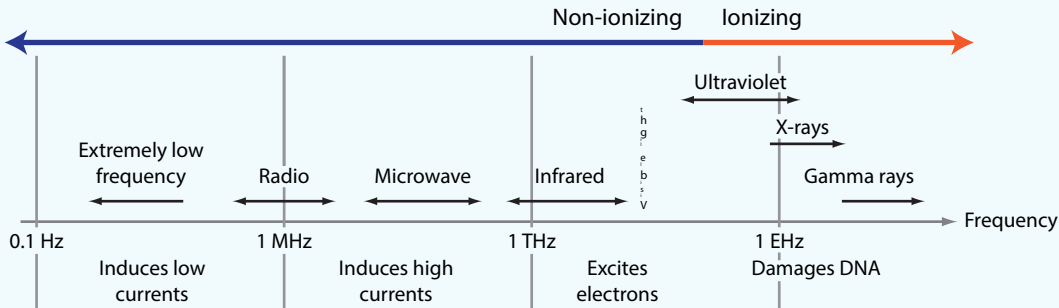


图 TF17-2 不同类型的电磁辐射。

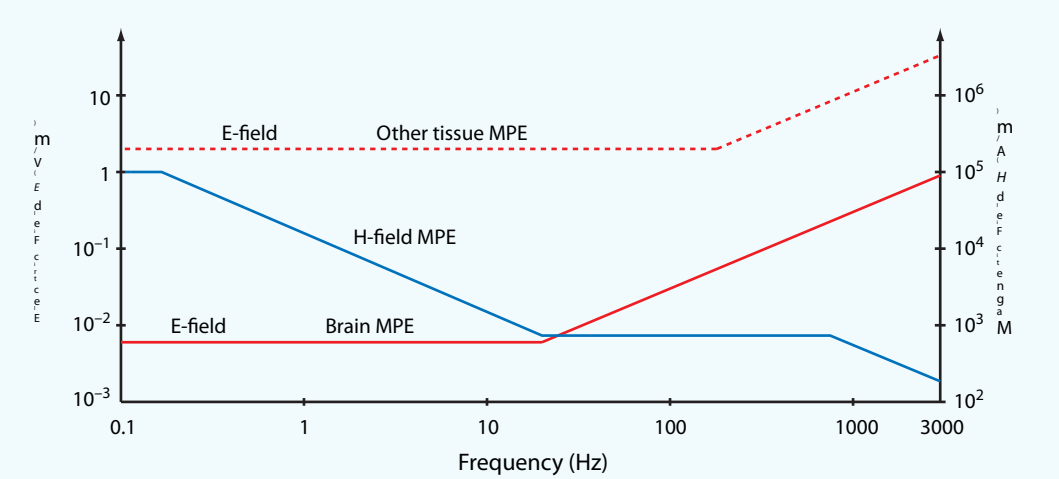
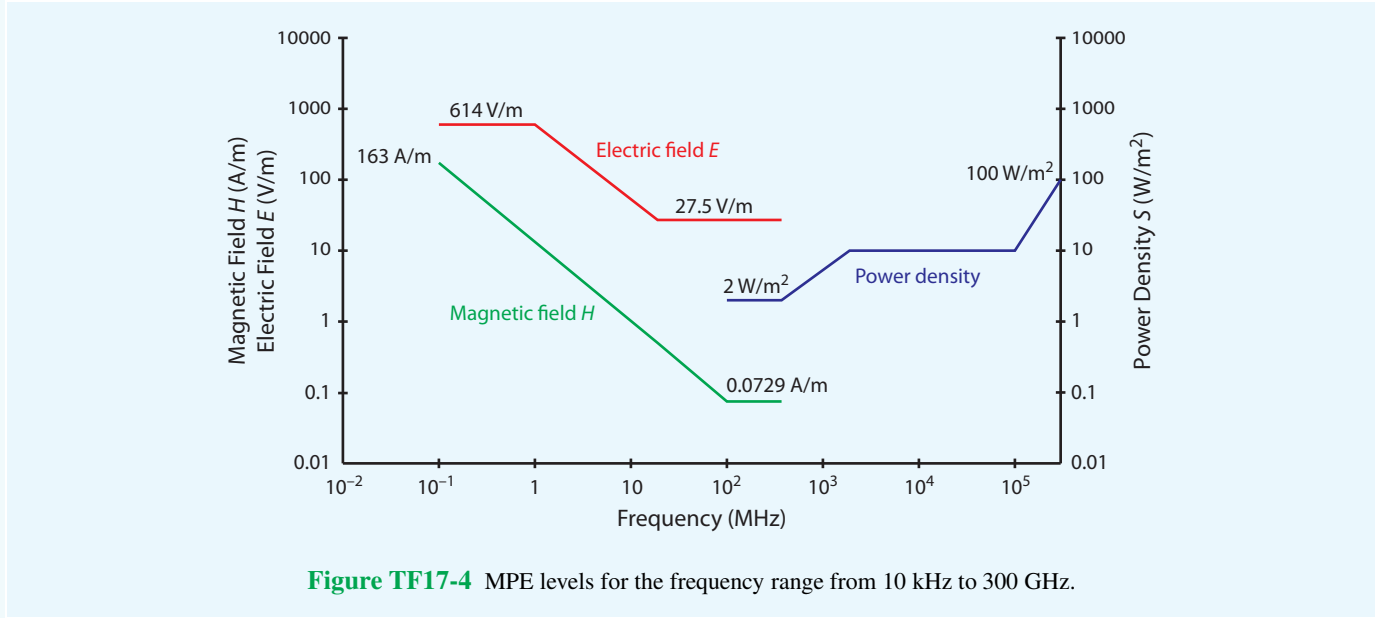


图 TF17-3 E 和 H 在 0.1 Hz 至 3 kHz 频率范围内的最大允许暴露 (MPE) 水平。

频率范围  $0 \leq f \leq 3$  kHz: 图 TF17-3 中的图显示了 3 kHz 以下频率范围内电场和磁场的 MPE 值。根据 IEEE 标准 C95.6, 足以证明电场  $E$  或磁场  $H$  符合 MPE 水平。根据  $H$  绘图, 60 Hz 下的暴露不应超过 720 A/m。电源线产生的磁场通常在线路下方 2 至 6 A/m 的范围内, 这比既定的  $H$  安全水平至少小两个数量级。

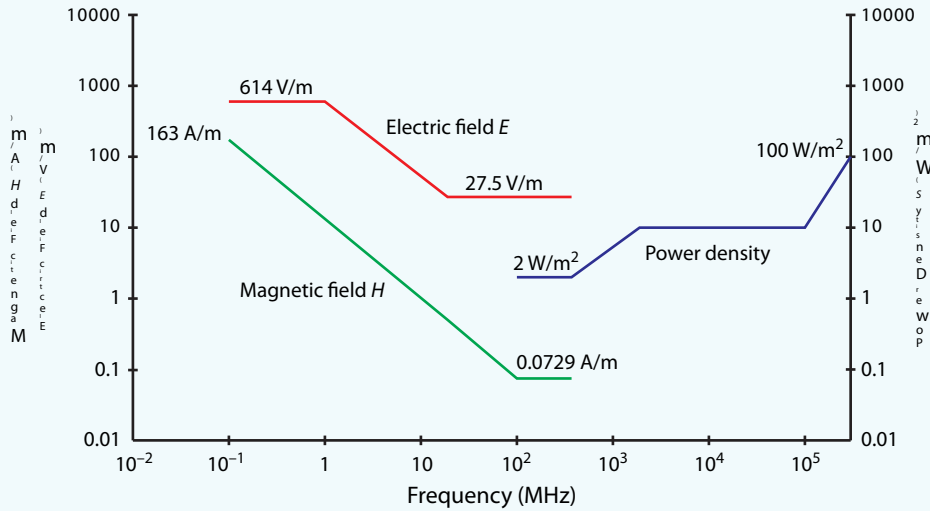
频率范围  $3 \text{ kHz} \leq f \leq 300 \text{ GHz}$ : 在低于 500 MHz 的频率下, MPE 根据 EM 能量的电场和磁场强度来指定 (图 TF 17-4)。从 100 MHz 到 300 GHz (及以上), MPE 以  $E$  和  $H$  的乘积来指定, 即功率密度  $S$ 。手机工作在 1 到 2 GHz 频段; 指定的 MPE 为 1 W/m<sup>2</sup> (或等效的 0.1 mW/cm<sup>2</sup>)。



Bottom Line

We are constantly bombarded by EM energy—from solar illumination to blackbody radiation emitted by all matter. Our bodies absorb, reflect, and emit EM energy all the time. Living organisms, including humans, require exposure to EM radiation to survive, but excessive exposure can cause adverse effects. The term *excessive exposure* connotes a complicated set of relationships among such variables as field strength, exposure duration and mode (continuous, pulsed, etc.), body part, etc. The emission

standards established by the Federal Communications Commission in the U.S. and similar governmental bodies in other countries are based on a combination of epidemiological studies, experimental observations, and theoretical understanding of how EM energy interacts with biological material. Generally speaking, the maximum permissible exposure levels specified by these standards are typically two orders of magnitude lower than the levels known to cause adverse effects, but in view of the multiplicity of variables involved, there is no guarantee that adhering to the standards will avoid health risks absolutely. The bottom line is: Use common sense!



底线

我们不断受到电磁能量的轰炸——从太阳光到所有物质发出的黑体辐射。我们的身体一直在吸收、反射和发射电磁能量。包括人类在内的生物体需要暴露于电磁辐射才能生存，但过度暴露会造成不利影响。术语“过度暴露”意味着诸如场强、暴露持续时间和模式（连续、脉冲等）、身体部位等变量之间的一组复杂关系。

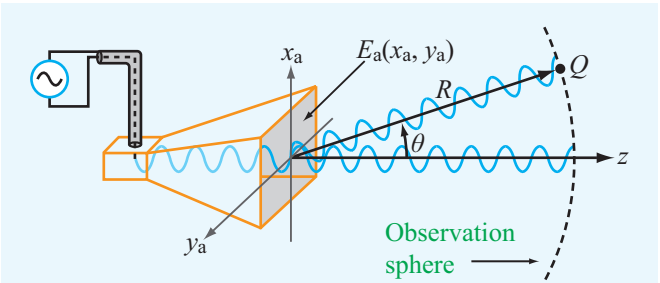
美国联邦通信委员会和其他国家类似政府机构制定的标准基于流行病学研究、实验观察以及对电磁能如何与生物材料相互作用的理论理解。一般来说，这些标准规定的最大允许暴露水平通常比已知会造成不利影响的水平低两个数量级，但考虑到涉及的变量的多样性，不能保证遵守这些标准就绝对避免健康风险。底线是：运用常识！

**Exercise 9-10:** The effective area of a parabolic dish antenna is approximately equal to its physical aperture. If its directivity is 30 dB at 10 GHz, what is its effective area? If the frequency is increased to 30 GHz, what will be its new directivity?

**Answer:**  $A_e = 0.07 \text{ m}^2$ ,  $D = 39.44 \text{ dB}$ . (See [EM](#).)

### 9-7 Radiation by Large-Aperture Antennas

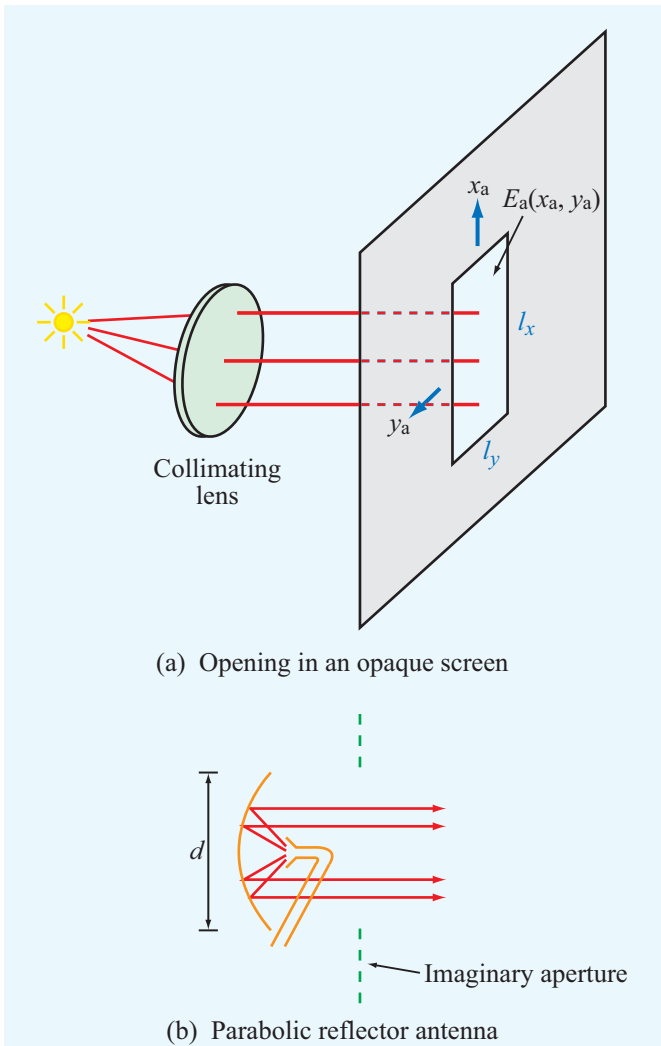
For wire antennas, the sources of radiation are the infinitesimal current elements comprising the current distribution along the wire and the total radiated field at a given point in space is equal to the sum, or integral, of the fields radiated by all the elements. A parallel scenario applies to aperture antennas, except that now the source of radiation is the electric-field distribution across the aperture. Consider the horn antenna shown in [Fig. 9-20](#). It is connected to a source through a coaxial transmission line with the outer conductor of the line connected to the metal body of the horn and the inner conductor made to protrude through a small hole partially into the throat end of the horn. The protruding conductor acts as a monopole antenna, generating waves that radiate outwardly toward the horn's aperture. The electric field of the wave arriving at the aperture, which may vary as a function of  $x_a$  and  $y_a$  over the horn's aperture, is called the **electric-field aperture distribution** or **illumination**,  $E_a(x_a, y_a)$ . Inside the horn, wave propagation is guided by the horn's geometry, but as the wave transitions from a guided wave into an unbounded wave, every point of its wavefront serves as a source of spherical secondary wavelets. The aperture may then be represented as a distribution of isotropic radiators. At a distant point  $Q$ , the combination of all the waves arriving from all of these radiators constitutes the total wave that would be observed by



**Figure 9-20** A horn antenna with aperture field distribution  $E_a(x_a, y_a)$ .

a receiver placed at that point.

The radiation process described for the horn antenna is equally applicable to any aperture upon which an electromagnetic wave is incident. For example, if a light source is used to illuminate an opening in an opaque screen through a collimating lens, as shown in [Fig. 9-21\(a\)](#), the opening becomes a source of secondary spherical wavelets, much like the aperture of the horn antenna. In the case of the parabolic reflector shown in [Fig. 9-21\(b\)](#), it can be described in terms of an imaginary aperture representing the electric-field distribution across a



**Figure 9-21** Radiation by apertures: (a) an opening in an opaque screen illuminated by a light source through a collimating lens and (b) a parabolic dish reflector illuminated by a small horn antenna.

练习9-10: 抛物面碟形天线的有效面积大约等于其物理孔径。如果它的方向性在 10 GHz 时为 30 dB, 那么它的有效面积是多少? 如果频率提高到30GHz, 它新的指向性会是什么?

答案:  $A_e = 0.07 \text{ m}^2$ ,  $D = 39.44 \text{ dB}$ . (参见<sup>EM</sup>.)

### 9-7 大孔径天线的辐射

对于线天线, 辐射源是由沿线的电流分布组成的无穷小电流元件, 空间中给定点的总辐射场等于所有元件辐射场的总和或积分。类似的情况适用于孔径天线, 只不过现在的辐射源是孔径上的电场分布。考虑图 9-20 所示的喇叭天线。它通过同轴传输线连接到信号源, 该线的外导体连接到喇叭的金属主体, 而内导体则通过小孔部分突出到喇叭的喉部末端。突出的导体充当单极天线, 产生向喇叭孔径向外辐射的波。到达孔径的波的电场可能随着喇叭孔径上的  $x_a$  和  $y_a$  的函数而变化, 称为电场孔径分布或照明,  $E_a(x_a, y_a)$ 。在喇叭内部, 波传播由喇叭的几何形状引导, 但当波从导波转变为无界波时, 其波前的每个点都充当球形次级小波的源。然后孔径可以被表示为各向同性辐射器的分布。在远处的点  $Q$ , 来自所有这些辐射器的所有波的组合构成了通过以下方式观测到的总波:

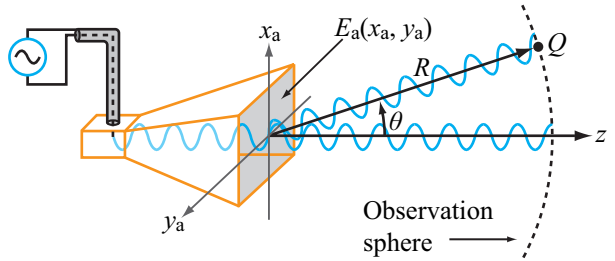


图 9-20 孔径场分布为  $E_a(x_a, y_a)$  的喇叭天线。

放置在该点的接收器。

针对喇叭天线描述的辐射过程同样适用于电磁波入射的任何孔径。例如, 如果使用光源通过准直透镜照亮不透明屏幕上的开口, 如图9-21 (a) 所示, 则该开口就成为二次球面小波的源, 就像喇叭天线的孔径一样。在图 9-21(b) 所示的抛物面反射器的情况下, 它可以用代表电场分布的假想孔径来描述。

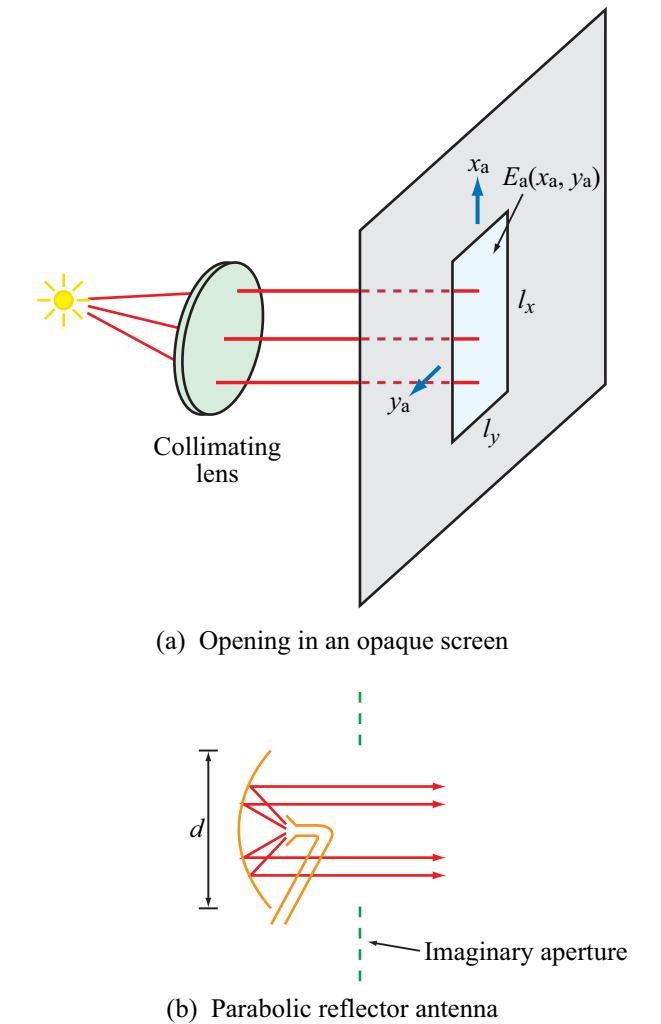


图 9-21 孔径辐射: (a) 不透明屏幕上的开口, 通过准直透镜由光源照明; (b) 由小喇叭天线照明的抛物面碟形反射器。

plane in front of the reflector.

Two types of mathematical formulations are available for computing the electromagnetic fields of waves radiated by apertures. The first is a **scalar formulation** based on Kirchhoff's work, and the second is a **vector formulation** based on Maxwell's equations. In this section, we limit our presentation to the scalar diffraction technique in part because of its inherent simplicity and also because it is applicable across a wide range of practical applications.

► The key requirement for the validity of the scalar formulation is that the antenna aperture be at least several wavelengths long along each of its principal dimensions. ◀

A distinctive feature of such an antenna is its high directivity and correspondingly narrow beam, which makes it attractive for radar and free-space microwave communication systems. The frequency range commonly used for such applications is the 1 to 30 GHz microwave band. Because the corresponding wavelength range is 30 to 1 cm, respectively, it is quite

practical to construct and use antennas (in this frequency range) with aperture dimensions that are many wavelengths in size.

The  $x_a$ - $y_a$  plane in **Fig. 9-22** (denoted plane  $A$ ) contains an aperture with an electric field distribution  $E_a(x_a, y_a)$ . For the sake of convenience, the opening has been chosen to be rectangular in shape, with dimensions  $l_x$  along  $x_a$  and  $l_y$  along  $y_a$ , even though the formulation we are about to discuss is general enough to accommodate any two-dimensional aperture distribution, including those associated with circular and elliptical apertures. At a distance  $z$  from the aperture plane  $A$  in **Fig. 9-22**, we have an observation plane  $O$  with axes  $(x, y)$ . The two planes have parallel axes and are separated by a distance  $z$ . Moreover,  $z$  is sufficiently large that any point  $Q$  in the observation plane is in the far-field region of the aperture. To satisfy the far-field condition, it is necessary that

$$R \geq 2d^2/\lambda, \quad (\text{far-field range}) \quad (9.73)$$

where  $d$  is the longest linear dimension of the radiating aperture.

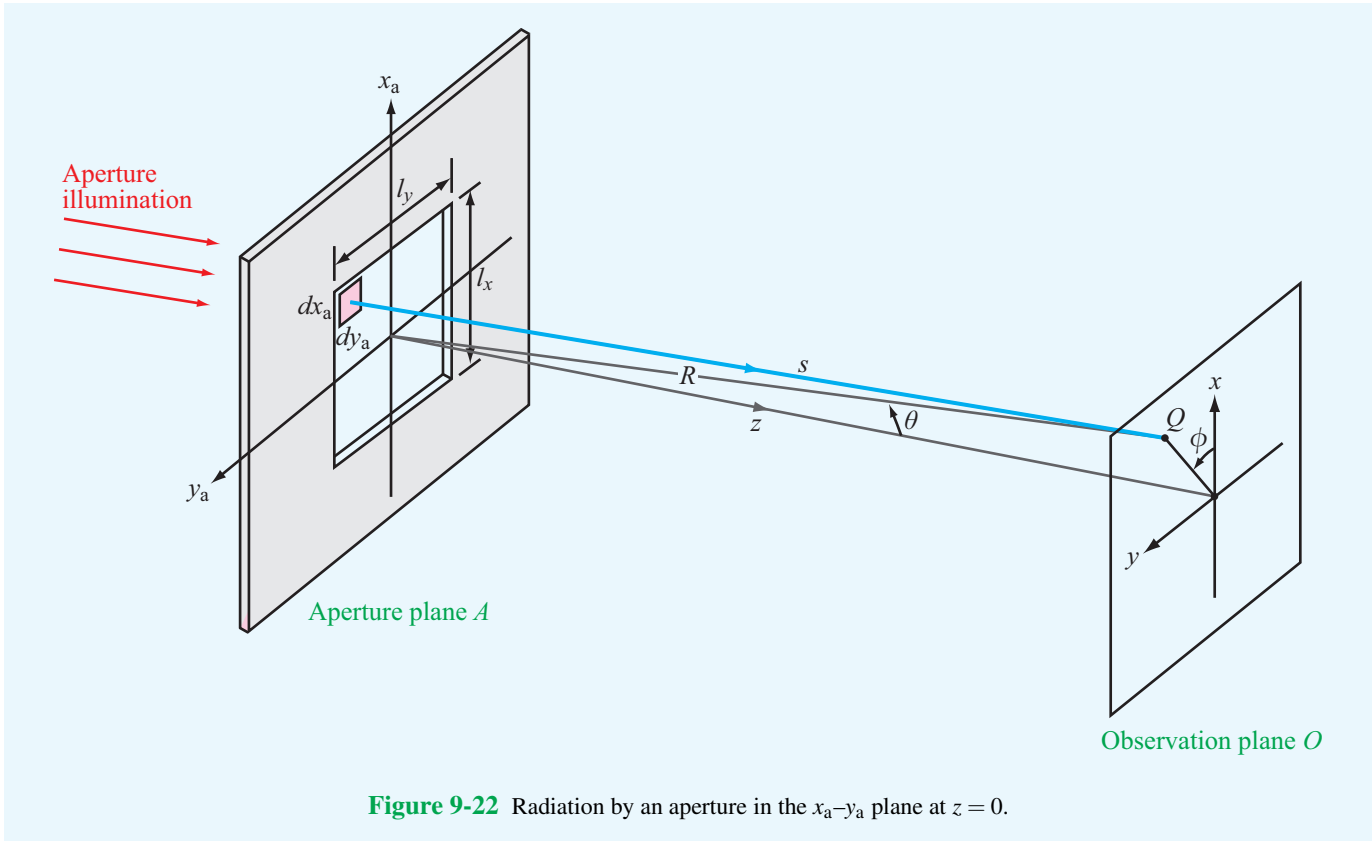


Figure 9-22 Radiation by an aperture in the  $x_a$ - $y_a$  plane at  $z = 0$ .

平面位于反射镜前面。

有两种类型的数学公式可用于计算孔径辐射波的电磁场。第一个是基于基尔霍夫工作的标量公式，第二个是基于麦克斯韦方程组的矢量公式。在本节中，我们将介绍限于标量衍射技术，部分原因是其固有的简单性，也因为适用于广泛的实际应用。

► 标量公式有效性的关键要求是天线孔径沿其每个主要维度至少有几个波长长。 ◀

这种天线的一个显著特点是其高方向性和相应的窄波束，这使其对雷达和自由空间微波通信系统具有吸引力。此类应用通常使用的频率范围是 1 至 30 GHz 微波频段。因为对应的波长范围分别是 30 到 1 cm，所以相当

构造和使用孔径尺寸为多个波长的天线（在此频率范围内）是可行的。

图 9-22 中的  $x_a$ - $y_a$  平面（表示为平面  $A$ ）包含电场分布为  $E_a(x_a, y_a)$  的孔径。为了方便起见，开口形状选择为矩形，尺寸为沿  $x_a$  的  $l_x$  和沿  $y_a$  的  $l_y$ ，尽管我们将要讨论的公式足够通用，足以适应任何二维孔径分布，包括与圆形和椭圆形孔径相关的分布。在图 9-22 中，距孔径平面  $A$  的距离为  $z$  处，我们有一个轴为  $(x, y)$  的观察平面  $O$ 。这两个平面具有平行轴并且相距距离  $z$ 。此外， $z$  足够大，使得观察平面中的任何点  $Q$  都位于孔径的远场区域。为了满足远场条件，需要

$$R \geq 2d^2/\lambda, \quad (\text{far-field range}) \quad (9.73)$$

其中  $d$  是辐射孔径的最长线性尺寸。

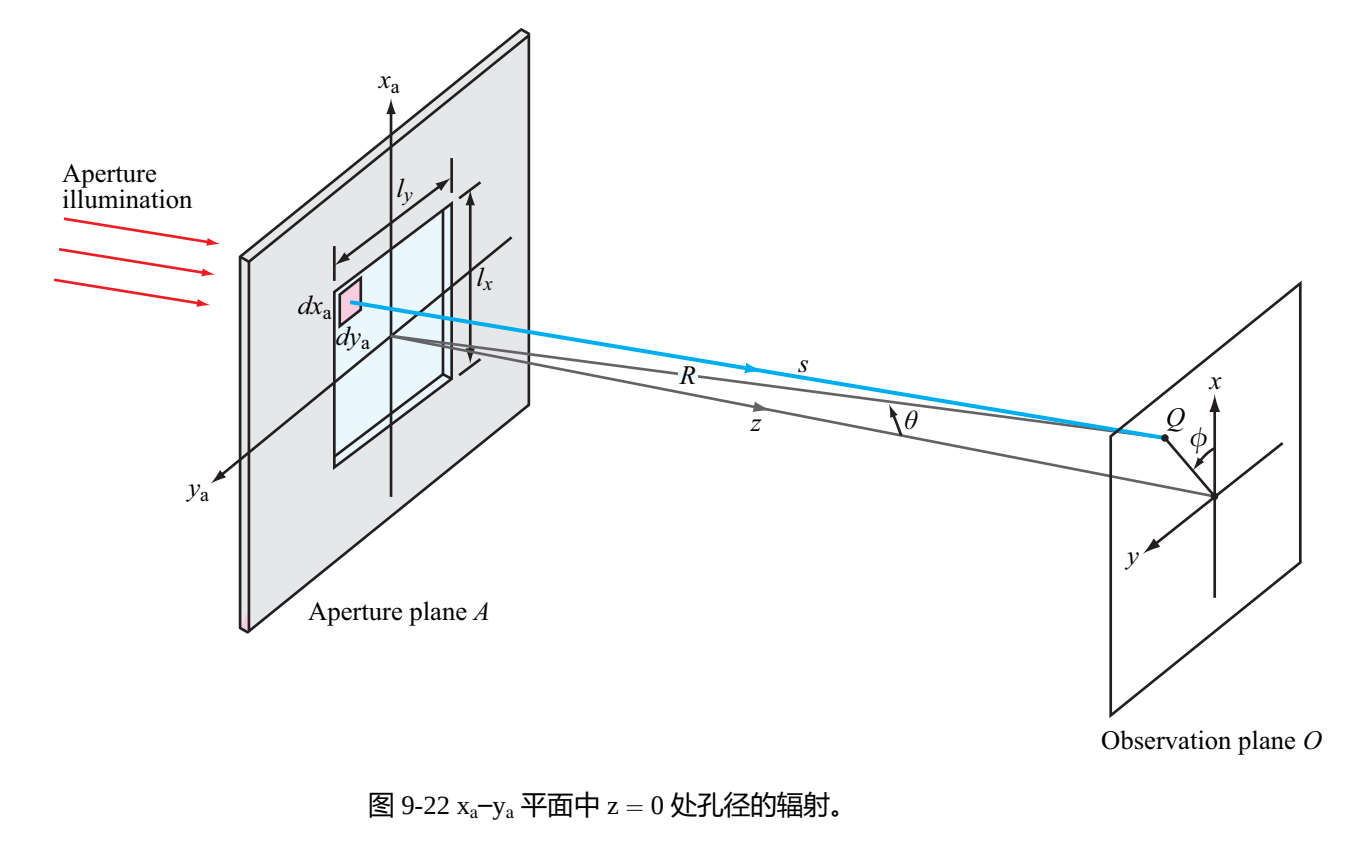


图 9-22  $x_a$ - $y_a$  平面中  $z = 0$  处孔径的辐射。

The position of observation point  $Q$  is specified by the range  $R$  between the center of the aperture and point  $Q$  and by the angles  $\theta$  and  $\phi$  (**Fig. 9-22**), which jointly define the direction of the observation point relative to the coordinate system of the aperture. In our treatment of the dipole antenna, we oriented the dipole along the  $z$  axis, and we called  $\theta$  the zenith angle. In the present context, the  $z$  axis is orthogonal to the plane containing the antenna aperture. Also,  $\theta$  usually is called the **elevation angle**. The electric field phasor of the wave incident upon point  $Q$  is denoted  $\tilde{E}(R, \theta, \phi)$ . Kirchhoff’s scalar diffraction theory provides the following relationship between the radiated field  $\tilde{E}(R, \theta, \phi)$  and the aperture illumination  $\tilde{E}_a(x_a, y_a)$ :

$$\tilde{E}(R, \theta, \phi) = \frac{j}{\lambda} \left( \frac{e^{-jkR}}{R} \right) \tilde{h}(\theta, \phi), \quad (9.74)$$

where

$$\tilde{h}(\theta, \phi) = \iint_{-\infty}^{\infty} \tilde{E}_a(x_a, y_a) \cdot \exp[jk \sin \theta (x_a \cos \phi + y_a \sin \phi)] dx_a dy_a. \quad (9.75)$$

We shall refer to  $\tilde{h}(\theta, \phi)$  as the **form factor** of  $\tilde{E}(R, \theta, \phi)$ . Its integral is written with infinite limits with the understanding that  $\tilde{E}_a(x_a, y_a)$  is identically zero outside the aperture. The spherical propagation factor ( $e^{-jkR}/R$ ) accounts for wave propagation between the center of the aperture and the observation point, and  $\tilde{h}(\theta, \phi)$  represents an integration of the exciting field  $\tilde{E}_a(x_a, y_a)$  over the extent of the aperture, taking into account [through the exponential function in Eq. (9.75)] the approximate deviation in distance between  $R$  and  $s$ , where  $s$  is the distance to any point  $(x_a, y_a)$  in the aperture plane (see **Fig. 9-22**).

► In Kirchhoff’s scalar formulation, the polarization direction of the radiated field  $\tilde{E}(R, \theta, \phi)$  is the same as that of the aperture field  $\tilde{E}_a(x_a, y_a)$ . ◀

Also, the power density of the radiated wave is given by

$$S(R, \theta, \phi) = \frac{|\tilde{E}(R, \theta, \phi)|^2}{2\eta_0} = \frac{|\tilde{h}(\theta, \phi)|^2}{2\eta_0\lambda^2R^2}. \quad (9.76)$$

### 9-8 Rectangular Aperture with Uniform Aperture Distribution

To illustrate the scalar diffraction technique, consider a rectangular aperture of height  $l_x$  and width  $l_y$ , both at least a few

wavelengths long. The aperture is excited by a **uniform field distribution** (i.e., constant value) given by

$$\tilde{E}_a(x_a, y_a) = \begin{cases} E_0 & \text{for } -l_x/2 \leq x_a \leq l_x/2 \\ & \text{and } -l_y/2 \leq y_a \leq l_y/2, \\ 0 & \text{otherwise.} \end{cases} \quad (9.77)$$

To keep the mathematics simple, let us confine our examination to the radiation pattern at a fixed range  $R$  in the  $x$ – $z$  plane, which corresponds to  $\phi = 0$ . In this case, Eq. (9.75) simplifies to

$$\tilde{h}(\theta) = \int_{y_a=-l_y/2}^{l_y/2} \int_{x_a=-l_x/2}^{l_x/2} E_0 \exp[jkx_a \sin \theta] dx_a dy_a. \quad (9.78)$$

In preparation for performing the integration in Eq. (9.78), we introduce the intermediate variable  $u$  defined as

$$u = k \sin \theta = \frac{2\pi \sin \theta}{\lambda}. \quad (9.79)$$

Hence,

$$\begin{aligned} \tilde{h}(\theta) &= E_0 \int_{-l_x/2}^{l_x/2} e^{ju x_a} dx_a \cdot \int_{-l_y/2}^{l_y/2} dy_a \\ &= E_0 \left[ \frac{e^{ju l_x/2} - e^{-ju l_x/2}}{ju} \right] \cdot l_y \\ &= \frac{2E_0 l_y}{u} \left[ \frac{e^{ju l_x/2} - e^{-ju l_x/2}}{2j} \right] = \frac{2E_0 l_y}{u} \sin(ul_x/2). \end{aligned} \quad (9.80)$$

Upon replacing  $u$  with its defining expression, we have

$$\begin{aligned} \tilde{h}(\theta) &= \frac{2E_0 l_y}{\left( \frac{2\pi}{\lambda} \sin \theta \right)} \sin(\pi l_x \sin \theta / \lambda) \\ &= E_0 l_x l_y \frac{\sin(\pi l_x \sin \theta / \lambda)}{\pi l_x \sin \theta / \lambda} \\ &= E_0 A_p \operatorname{sinc}(\pi l_x \sin \theta / \lambda), \end{aligned} \quad (9.81)$$

where  $A_p = l_x l_y$  is the physical area of the aperture. Also, we used the standard definition of the sinc function, which for any argument  $t$  is defined as

$$\operatorname{sinc} t = \frac{\sin t}{t}. \quad (9.82)$$

Using Eq. (9.76), we obtain the following expression for the power density at the observation point:

$$S(R, \theta) = S_0 \operatorname{sinc}^2(\pi l_x \sin \theta / \lambda) \quad (x\text{--}z \text{ plane}), \quad (9.83)$$

where  $S_0 = E_0^2 A_p^2 / (2\eta_0 \lambda^2 R^2)$ .

观察点 Q 的位置由孔径中心和点 Q 之间的范围 R 以及角度  $\theta$  和  $\phi$  (指定。9-22), 共同定义了观察点相对于孔径坐标系的方向。在我们处理偶极子天线时, 我们将偶极子沿  $z$  轴定向, 并将  $\theta$  称为天顶角。在本上下文中,  $z$  轴与包含天线孔径的平面正交。另外,  $\theta$  通常称为仰角。入射到 Q 点的波的电场相量表示为  $\tilde{E}(R, \theta, \phi)$ 。基尔霍夫的标量衍射理论提供了辐射场  $\tilde{E}(R, \theta, \phi)$  与孔径照明  $\tilde{E}_a(x_a, y_a)$  之间的关系:

$$\tilde{E}(R, \theta, \phi) = \frac{j}{\lambda} \left( \frac{e^{-jkR}}{R} \right) \tilde{h}(\theta, \phi), \quad (9.74)$$

在哪里

$$\tilde{h}(\theta, \phi) = \iint_{-\infty}^{\infty} \tilde{E}_a(x_a, y_a) \cdot \exp[jk \sin \theta (x_a \cos \phi + y_a \sin \phi)] dx_a dy_a. \quad (9.75)$$

我们将  $\tilde{h}(\theta, \phi)$  称为  $\tilde{E}(R, \theta, \phi)$  的形状因子。它的积分被写成无限极限, 且  $\tilde{E}_a(x_a, y_a)$  在孔径外同样为零。球面传播因子 ( $e^{-jkR}/R$ ) 解释了孔径中心和观察点之间的波传播, 而  $\tilde{h}(\theta, \phi)$  表示激发场  $\tilde{E}_a(x_a, y_a)$  在孔径范围内的积分, 并考虑[通过方程中的指数函数]。 (9.75)]R 和  $s$  之间距离的近似偏差, 其中  $s$  是到孔径平面中任意点  $(x_a, y_a)$  的距离 (见图 9-22) 。

► 在基尔霍夫标量公式中, 辐射场  $\tilde{E}(R, \theta, \phi)$  的偏振方向与孔径场  $\tilde{E}_a(x_a, y_a)$  的偏振方向相同。 ◀

此外, 辐射波的功率密度由下式给出

$$S(R, \theta, \phi) = \frac{|\tilde{E}(R, \theta, \phi)|^2}{2\eta_0} = \frac{|\tilde{h}(\theta, \phi)|^2}{2\eta_0\lambda^2R^2}. \quad (9.76)$$

### 9-8 孔径分布均匀的矩形孔径

为了说明标量衍射技术, 考虑一个高度为  $l_x$  和宽度为  $l_y$  的矩形孔径, 两者都至少是几个

波长长。孔径由均匀场分布 (即恒定值) 激发, 由下式给出

$$\tilde{E}_a(x_a, y_a) = \begin{cases} E_0 & \text{for } -l_x/2 \leq x_a \leq l_x/2 \\ & \text{and } -l_y/2 \leq y_a \leq l_y/2, \\ 0 & \text{otherwise.} \end{cases} \quad (9.77)$$

为了保持数学简单, 让我们将我们的检查限制在  $x$ - $z$  平面中固定范围 R 的辐射方向图上, 该范围对应于  $\phi = 0$ 。 (9.75) 简化为

$$\tilde{h}(\theta) = \int_{y_a=-l_y/2}^{l_y/2} \int_{x_a=-l_x/2}^{l_x/2} E_0 \exp[jkx_a \sin \theta] dx_a dy_a. \quad (9.78)$$

准备执行方程式中的积分。 (9.78), 我们引入中间变量  $u$  定义为

$$u = k \sin \theta = \frac{2\pi \sin \theta}{\lambda}. \quad (9.79)$$

因此,

$$\begin{aligned} \tilde{h}(\theta) &= E_0 \int_{-l_x/2}^{l_x/2} e^{ju x_a} dx_a \cdot \int_{-l_y/2}^{l_y/2} dy_a \\ &= E_0 \left[ \frac{e^{ju l_x/2} - e^{-ju l_x/2}}{ju} \right] \cdot l_y \\ &= \frac{2E_0 l_y}{u} \left[ \frac{e^{ju l_x/2} - e^{-ju l_x/2}}{2j} \right] = \frac{2E_0 l_y}{u} \sin(ul_x/2). \end{aligned} \quad (9.80)$$

将  $u$  替换为其定义表达式后, 我们有

$$\begin{aligned} \tilde{h}(\theta) &= \frac{2E_0 l_y}{\left( \frac{2\pi}{\lambda} \sin \theta \right)} \sin(\pi l_x \sin \theta / \lambda) \\ &= E_0 l_x l_y \frac{\sin(\pi l_x \sin \theta / \lambda)}{\pi l_x \sin \theta / \lambda} \\ &= E_0 A_p \operatorname{sinc}(\pi l_x \sin \theta / \lambda), \end{aligned} \quad (9.81)$$

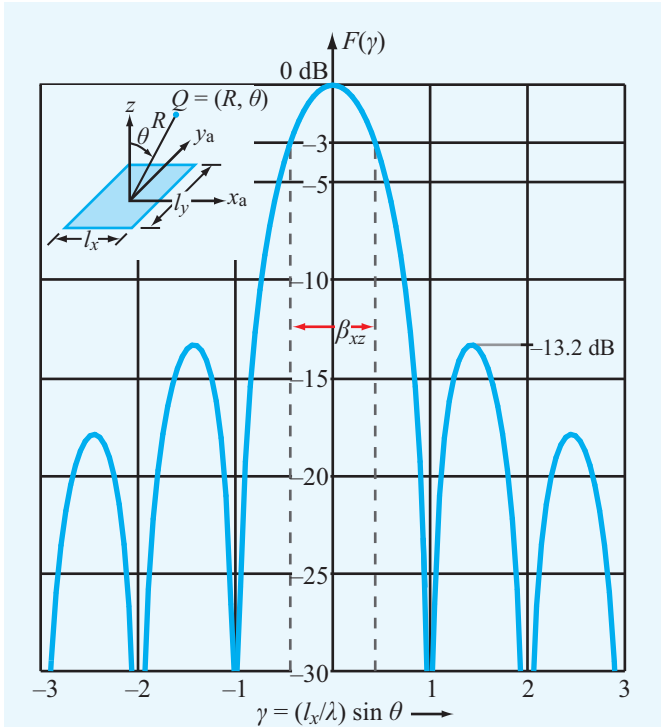
其中  $A_p = l_x l_y$  是孔径的物理面积。此外, 我们使用了  $\operatorname{sinc}$  函数的标准定义, 对于任何参数  $t$  都定义为

$$\operatorname{sinc} t = \frac{\sin t}{t}. \quad (9.82)$$

使用方程式根据方程 (9.76) , 我们得到观测点处功率密度的表达式如下:

$$S(R, \theta) = S_0 \operatorname{sinc}^2(\pi l_x \sin \theta / \lambda) \quad (x\text{--}z \text{ plane}), \quad (9.83)$$

其中  $S_0 = E_0^2 A_p^2 / (2\eta_0 \lambda^2 R^2)$ 。



**Figure 9-23** Normalized radiation pattern of a uniformly illuminated rectangular aperture in the  $x$ - $z$  plane ( $\phi = 0$ ).

► The sinc function is maximum when its argument is zero;  $\text{sinc}(0) = 1$ . ◀

This occurs when  $\theta = 0$ . Hence, at a fixed range  $R$ ,  $S_{\max} = S(\theta = 0) = S_0$ . The normalized radiation intensity is then given by

$$F(\theta) = \frac{S(R, \theta)}{S_{\max}} = \text{sinc}^2(\pi l_x \sin \theta / \lambda) = \text{sinc}^2(\pi \gamma) \quad (x\text{-}z \text{ plane}). \quad (9.84)$$

**Figure 9-23** shows  $F(\theta)$  plotted (on a decibel scale) as a function of the intermediate variable  $\gamma = (l_x/\lambda) \sin \theta$ . The pattern exhibits nulls at nonzero integer values of  $\gamma$ .

### 9-8.1 Beamwidth

The normalized radiation intensity  $F(\theta)$  is symmetrical in the  $x$ - $z$  plane, and its maximum is along the boresight direction ( $\theta = 0$ , in this case). Its half-power beamwidth  $\beta_{xz} = \theta_2 - \theta_1$ , where  $\theta_1$  and  $\theta_2$  are the values of  $\theta$  at which  $F(\theta, 0) = 0.5$  (or

$-3$  dB on a decibel scale), as shown in **Fig. 9-23**. Since the pattern is symmetrical with respect to  $\theta = 0$ ,  $\theta_1 = -\theta_2$ , and  $\beta_{xz} = 2\theta_2$ . The angle  $\theta_2$  can be obtained from a solution of

$$F(\theta_2) = \text{sinc}^2(\pi l_x \sin \theta / \lambda) = 0.5. \quad (9.85)$$

From tabulated values of the sinc function, it is found that Eq. (9.85) yields the result

$$\frac{\pi l_x}{\lambda} \sin \theta_2 = 1.39, \quad (9.86)$$

or

$$\sin \theta_2 = 0.44 \frac{\lambda}{l_x}. \quad (9.87)$$

Because  $\lambda/l_x \ll 1$  (a fundamental condition of scalar diffraction theory is that the aperture dimensions be much larger than the wavelength  $\lambda$ ),  $\theta_2$  is a small angle, in which case we can use the approximation  $\sin \theta_2 \approx \theta_2$ . Hence,

$$\beta_{xz} = 2\theta_2 \approx 2 \sin \theta_2 = 0.88 \frac{\lambda}{l_x} \quad (\text{rad}). \quad (9.88a)$$

A similar solution for the  $y$ - $z$  plane ( $\phi = \pi/2$ ) gives

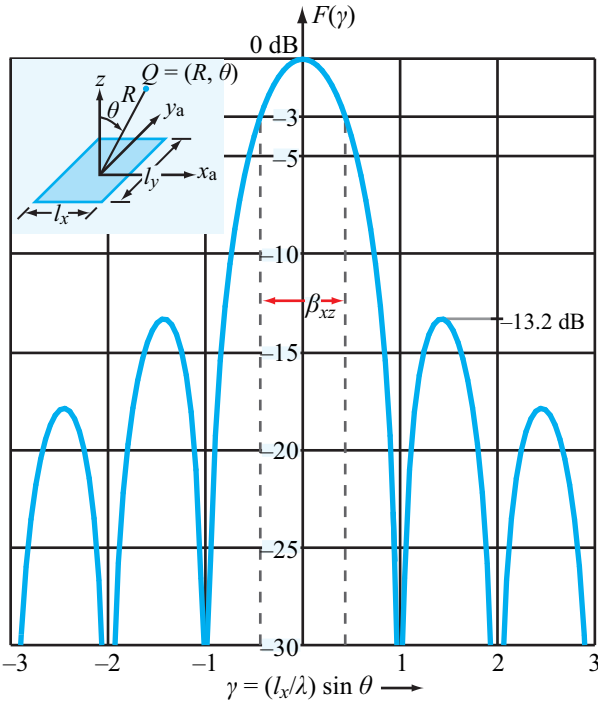
$$\beta_{yz} = 0.88 \frac{\lambda}{l_y} \quad (\text{rad}). \quad (9.88b)$$

► The uniform aperture distribution ( $\tilde{E}_a = E_0$  across the aperture) gives a far-field pattern with the narrowest possible beamwidth. ◀

The first sidelobe level is 13.2 dB below the peak value (see **Fig. 9-23**), which is equivalent to 4.8% of the peak value. If the intended application calls for a pattern with a lower sidelobe level (to avoid interference with signals from sources along directions outside the main beam of the antenna pattern), this can be accomplished by using a **tapered aperture distribution**—one that is a maximum at the center of the aperture and decreases toward the edges.

► A tapered distribution provides a pattern with lower side lobes, but the main lobe becomes wider. ◀

The steeper the taper, the lower are the side lobes and the wider is the main lobe. In general, the beamwidth in a given plane,



**图 9-23**  $x$ - $z$  平面 ( $\phi = 0$ ) 中均匀照明的矩形孔径的归一化辐射方向图。

► 当参数为零时，sinc 函数最大；正弦(0) = 1. ◀

这发生在  $\theta = 0$  时。因此，在固定范围  $R$ ， $S_{\max} = S(\theta = 0) = S_0$  处。归一化辐射强度由下式给出

$$F(\theta) = \frac{S(R, \theta)}{S_{\max}} = \text{sinc}^2(\pi l_x \sin \theta / \lambda) = \text{sinc}^2(\pi \gamma) \quad (x\text{-}z \text{ plane}). \quad (9.84)$$

**图 9-23** 显示了  $F(\theta)$  (以分贝刻度) 作为中间变量  $\gamma = (l_x/\lambda) \sin \theta$  的函数。该模式在  $\gamma$  的非零整数值处呈现空值。

### 9-8.1 波束宽度

归一化辐射强度  $F(\theta)$  在  $x$ - $z$  平面上对称，其最大值沿视轴方向 (在本例中为  $\theta = 0$ )。其半功率波束宽度  $\beta_{xz} = \theta_2 - \theta_1$ ，其中  $\theta_1$  和  $\theta_2$  是  $\theta$  的值，其中  $F(\theta, 0) = 0.5$  (或

$-3$  dB (分贝级)，如图 9-23 所示。由于该图案相对于  $\theta = 0$ 、 $\theta_1 = -\theta_2$  和  $\beta_{xz} = 2\theta_2$  对称。角度  $\theta_2$  可以从以下解中获得

$$F(\theta_2) = \text{sinc}^2(\pi l_x \sin \theta / \lambda) = 0.5. \quad (9.85)$$

从 sinc 函数的表格值可以发现，方程：(9.85) 得出结果

$$\frac{\pi l_x}{\lambda} \sin \theta_2 = 1.39, \quad (9.86)$$

或者

$$\sin \theta_2 = 0.44 \frac{\lambda}{l_x}. \quad (9.87)$$

由于  $\lambda/l_x \ll 1$  (标量衍射理论的基本条件是孔径尺寸远大于波长  $\lambda$ )， $\theta_2$  是一个小角度，在这种情况下我们可以使用近似  $\sin \theta_2 \approx \theta_2$ 。因此，

$$\beta_{xz} = 2\theta_2 \approx 2 \sin \theta_2 = 0.88 \frac{\lambda}{l_x} \quad (\text{rad}). \quad (9.88a)$$

$y$ - $z$  平面 ( $\phi = \pi/2$ ) 的类似解给出

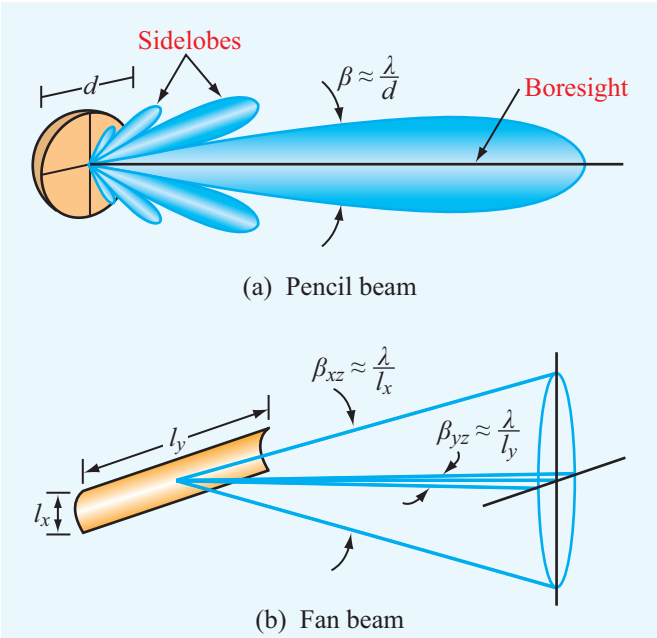
$$\beta_{yz} = 0.88 \frac{\lambda}{l_y} \quad (\text{rad}). \quad (9.88b)$$

► 均匀的孔径分布 (孔径上的  $\tilde{E}_a = E_0$ ) 提供了具有尽可能窄的波束宽度的远场方向图。◀

第一旁瓣电平比峰值低 13.2 dB (见图 9-23)，相当于峰值的 4.8%。如果预期应用需要具有较低旁瓣电平的方向图 (以避免沿天线方向图主波束之外的方向对来自源的信号产生干扰)，则可以通过使用锥形孔径分布来实现，即孔径中心处最大值并向边缘减小。

► 锥形分布提供了具有较低旁瓣的模式，但主瓣变得更宽。◀

锥度越陡，旁瓣越低，主瓣越宽。一般来说，给定平面上的波束宽度，



**Figure 9-24** Radiation patterns of (a) a circular reflector and (b) a cylindrical reflector (side lobes not shown).

say the  $x$ - $z$  plane, is given by

$$\beta_{xz} = k_x \frac{\lambda}{l_x}, \quad (9.89)$$

where  $k_x$  is a constant related to the steepness of the taper. For a uniform distribution with no taper,  $k_x = 0.88$ , and for a highly tapered distribution,  $k_x \approx 2$ . In the typical case,  $k_x \approx 1$ .

To illustrate the relationship between the antenna dimensions and the corresponding beam shape, we show in **Fig. 9-24** the radiation patterns of a circular reflector and a cylindrical reflector. The circular reflector has a circularly symmetric pattern, whereas the pattern of the cylindrical reflector has a narrow beam in the azimuth plane corresponding to its long dimension and a wide beam in the elevation plane corresponding to its narrow dimension. For a circularly symmetric antenna pattern, the beamwidth  $\beta$  is related to the diameter  $d$  by the approximate relation  $\beta \approx \lambda/d$ .

### 9-8.2 Directivity and Effective Area

In Section 9-2.3, we derived an approximate expression [Eq. (9.26)] for the antenna directivity  $D$  in terms of the half-power beamwidths  $\beta_{xz}$  and  $\beta_{yz}$  for antennas characterized by a

single major lobe whose boresight is along the  $z$  direction:

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \quad (9.90)$$

If we use the approximate relations  $\beta_{xz} \approx \lambda/l_x$  and  $\beta_{yz} \approx \lambda/l_y$ , we obtain

$$D \approx \frac{4\pi l_x l_y}{\lambda^2} = \frac{4\pi A_p}{\lambda^2}. \quad (9.91)$$

For any antenna, its directivity is related to its effective area  $A_e$  by Eq. (9.64):

$$D = \frac{4\pi A_e}{\lambda^2}. \quad (9.92)$$

▶ For aperture antennas, their effective apertures are approximately equal to their physical apertures; that is,  $A_e \approx A_p$ . ◀

**Exercise 9-11:** Verify that Eq. (9.86) is a solution of Eq. (9.85) by calculating  $\text{sinc}^2 t$  for  $t = 1.39$ .

**Exercise 9-12:** With its boresight direction along  $z$ , a square aperture was observed to have half-power beamwidths of  $3^\circ$  in both the  $x$ - $z$  and  $y$ - $z$  planes. Determine its directivity in decibels.

**Answer:**  $D = 4,583.66 = 36.61$  dB. (See [EM](#).)

**Exercise 9-13:** What condition must be satisfied in order to use scalar diffraction to compute the field radiated by an aperture antenna? Can we use it to compute the directional pattern of the eye's pupil ( $d \approx 0.2$  cm) in the visible part of the spectrum ( $\lambda = 0.35$  to  $0.7$   $\mu\text{m}$ )? What would the beamwidth of the eye's directional pattern be at  $\lambda = 0.5$   $\mu\text{m}$ ?

**Answer:**  $\beta \approx \lambda/d = 2.5 \times 10^{-4}$  rad  $= 0.86'$  (arc minute, with  $60' = 1^\circ$ ). (See [EM](#).)

## 9-9 Antenna Arrays

AM broadcast services operate in the 535 to 1605 kHz band. The antennas they use are vertical dipoles mounted along tall towers. The antennas range in height from  $\lambda/6$  to  $5\lambda/8$ , depending on the operating characteristics desired and other considerations. Their physical heights vary from 46 m (150 ft) to 274 m (900 ft); the wavelength at 1 MHz, approximately in the middle of the AM band, is 300 m. Because the field radiated by a single dipole is uniform in the horizontal plane

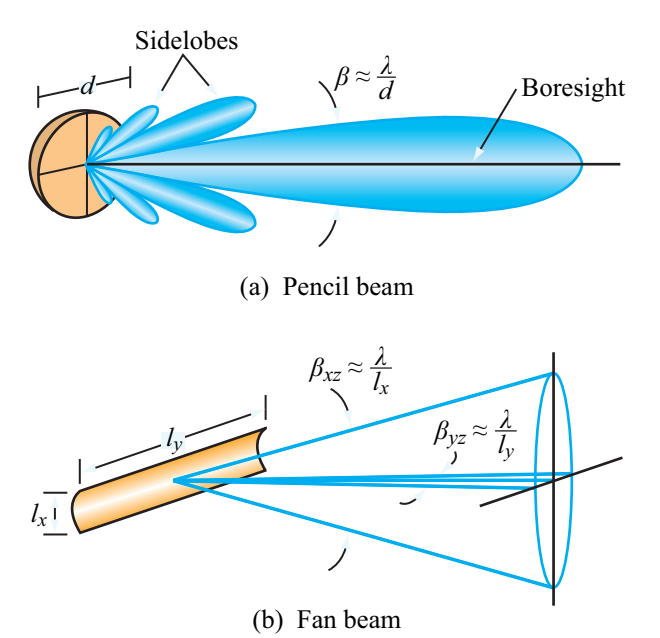


图 9-24 (a) 圆形反射器和 (b) 圆柱形反射器的辐射方向图（未显示旁瓣）。

说  $x$ - $z$  平面，由下式给出

$$\beta_{xz} = k_x \frac{\lambda}{l_x}, \quad (9.89)$$

其中  $k_x$  是与锥度陡度相关的常数。对于无锥度的均匀分布,  $k_x = 0.88$ ; 对于高锥度分布,  $k_x \approx 2$ 。在典型情况下,  $k_x \approx 1$ 。

为了说明天线尺寸和相应波束形状之间的关系，我们在图 9-24 中显示了圆形反射器和圆柱形反射器的辐射方向图。圆形反射器具有圆形对称图案，而圆柱形反射器的图案在与其长尺寸对应的方位平面中具有窄光束，在与其窄尺寸对应的仰角平面中具有宽光束。对于圆形对称天线方向图，波束宽度  $\beta$  与直径  $d$  通过近似关系  $\beta \approx \lambda/d$  相关。

### 9-8.2 方向性和有效面积

在9-2.3节中，我们推导出了一个近似表达式[Eq. (9.26)]对于天线方向性  $D$ ，以半功率波束宽度  $\beta_{xz}$  和  $\beta_{yz}$  表示，天线特征为

视轴沿  $z$  方向的主瓣：

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}. \quad (9.90)$$

如果我们使用近似关系  $\beta_{xz} \approx \lambda/l_x$  和  $\beta_{yz} \approx \lambda/l_y$ ，我们得到

$$D \approx \frac{4\pi l_x l_y}{\lambda^2} = \frac{4\pi A_p}{\lambda^2}. \quad (9.91)$$

对于任何天线，其方向性与其有效面积 $A_e$ 有关，如式(1)所示。 (9.64)：

$$D = \frac{4\pi A_e}{\lambda^2}. \quad (9.92)$$

▶ 对于孔径天线，其有效孔径约等于其物理孔径；即  $A_e \approx A_p$ 。 ◀

练习 9-11：验证方程(9.86) 是方程 (9.86) 的解。 (9.85) 通过计算  $t = 1.39$  的  $\text{sinc}^2 t$ 。

练习 9-12：当视轴方向沿  $z$  方向时，观察到方形孔径在  $x$ - $z$  和  $y$ - $z$  平面上的半功率波束宽度均为  $3^\circ$ 。确定其方向性（以分贝为单位）。

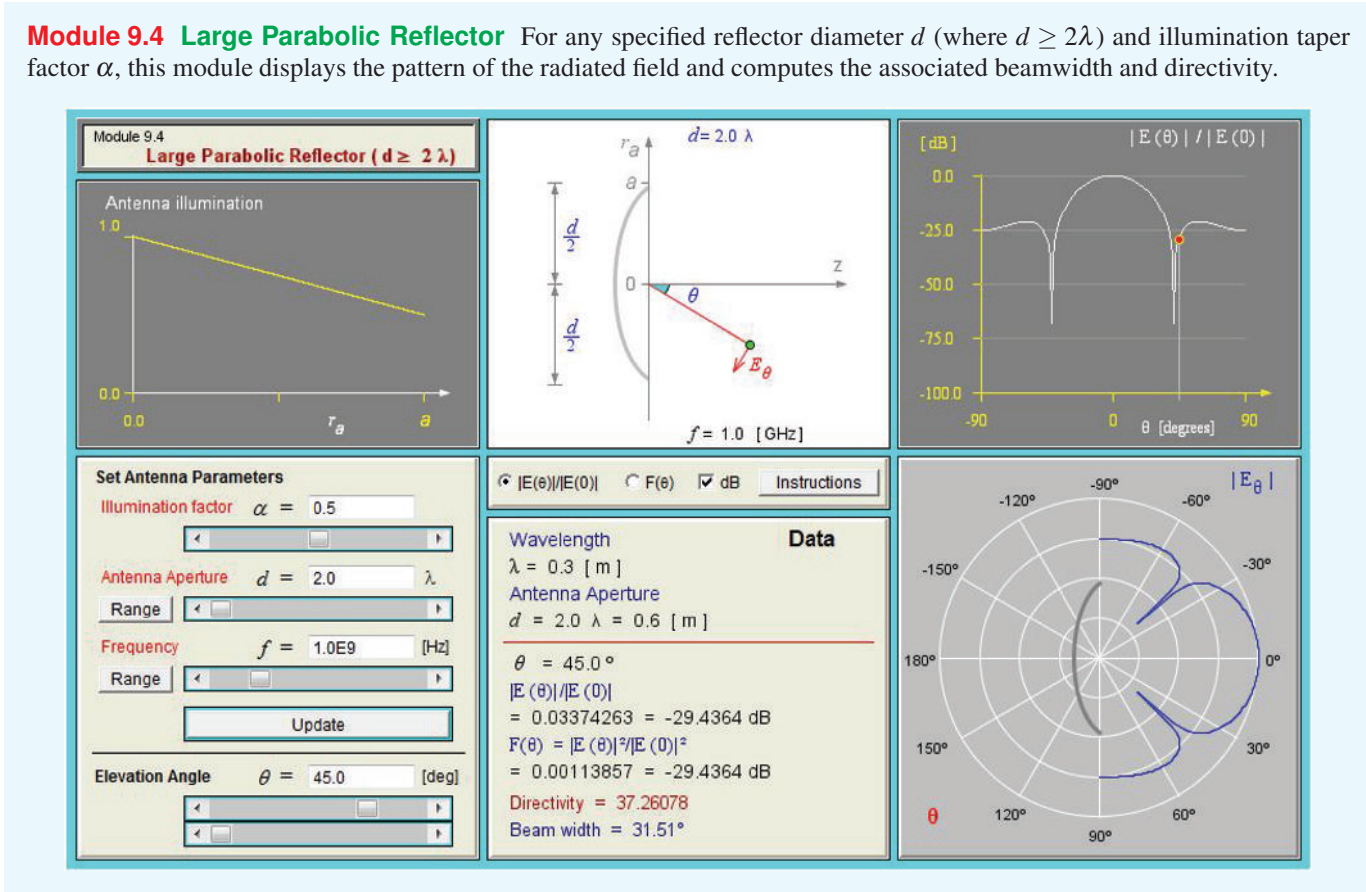
**Answer:**  $D = 4,583.66 = 36.61$  dB. (See [EM](#).)

练习9-13：为了使用标量衍射来计算孔径天线的辐射场，必须满足什么条件？我们可以用它来计算光谱可见部分 ( $\lambda = 0.35$  至  $0.7$   $\mu\text{m}$ ) 中眼睛瞳孔 ( $d \approx 0.2$  cm) 的方向图吗？  $\lambda = 0.5$   $\mu\text{m}$  处眼睛方向图的波束宽度是多少？

答案：  $\beta \approx \lambda/d = 2.5 \times 10^{-4}$  rad  $= 0.86'$  (角分,  $60' = 1^\circ$ )。 (参见<sup>EM</sup>。) [EM](#)

## 9-9 天线阵列

AM 广播服务在 535 至 1605 kHz 频段运行。他们使用的天线是沿着高塔安装的垂直偶极子。天线的高度范围从  $\lambda/6$  到  $5\lambda/8$ ，具体取决于所需的操作特性和其他考虑因素。它们的物理高度从 46 m (150 英尺) 到 274 m (900 英尺) 不等； 1 MHz 处的波长 (大约位于 AM 波段的中) 为 300 m。因为单个偶极子辐射的场在水平面内是均匀的



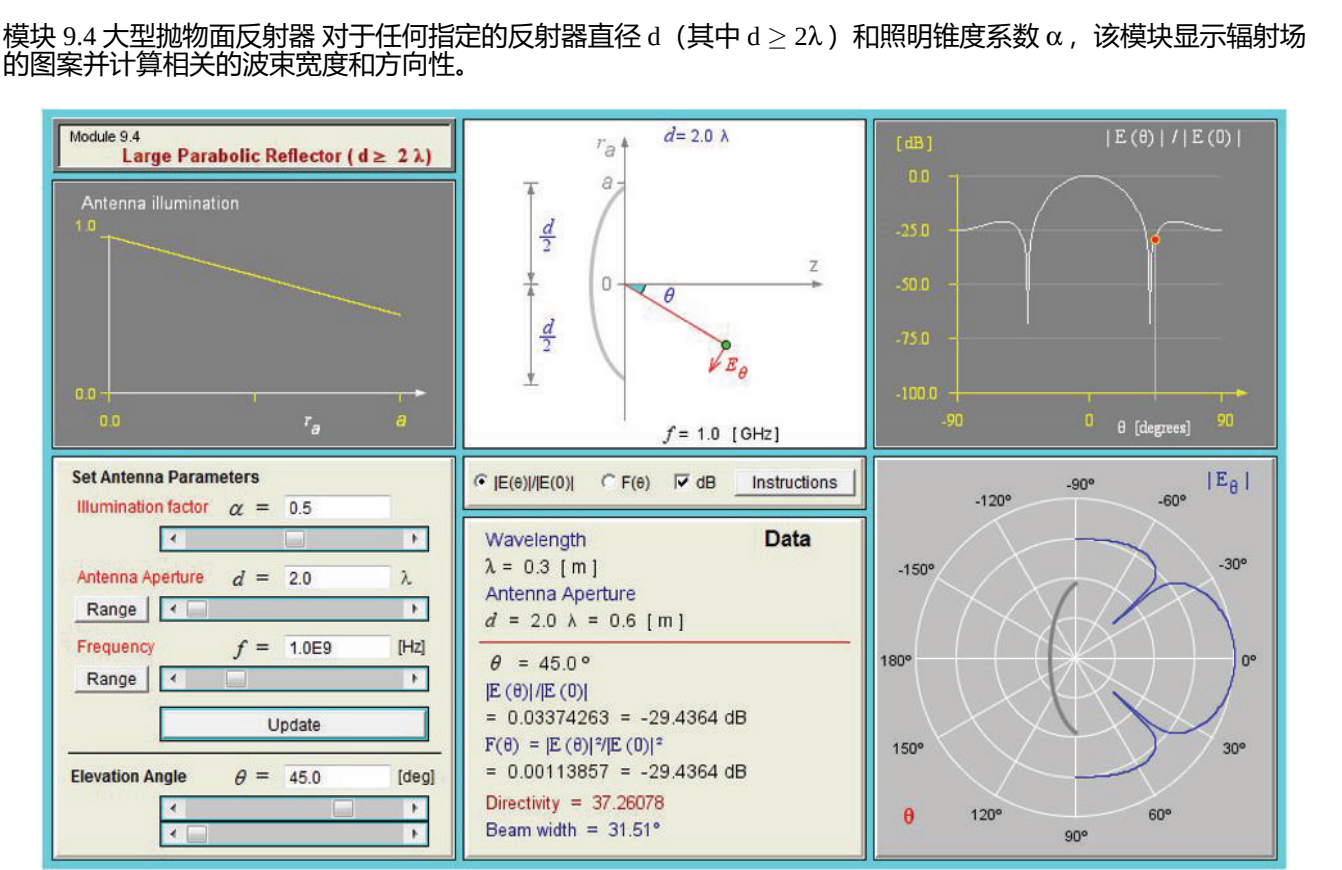
(as discussed in Sections 9-1 and 9-3), it is not possible to direct the horizontal pattern along specific directions of interest, unless two or more antenna towers are used simultaneously. Directions of interest may include cities serviced by the AM station, and directions to avoid may include areas serviced by another station operating at the same frequency (thereby avoiding interference effects). When two or more antennas are used together, the combination is called an **antenna array**.

The AM broadcast antenna array is only one example of the many antenna arrays used in communication systems and radar applications. Antenna arrays provide the antenna designer the flexibility to obtain high directivity, narrow beams, low side lobes, steerable beams, and shaped antenna patterns starting from very simple antenna elements. **Figure 9-25** shows a very large radar system consisting of a transmitter array composed of 5,184 individual dipole antenna elements and a receiver array composed of 4,660 elements. The radar system, part of the Space Surveillance Network operated by the U.S. Air Force, operates at 442 MHz and transmits a combined peak power of 30 MW!

Although an array need not consist of similar radiating elements, most arrays actually use identical elements, such as dipoles, slots, horn antennas, or parabolic dishes. The antenna elements comprising an array may be arranged in various configurations, but the most common are the linear one-dimensional configuration—wherein the elements are arranged along a straight line—and the two-dimensional lattice configuration in which the elements sit on a planar grid. The desired shape of the far-field radiation pattern of the array can be synthesized by controlling the relative amplitudes of the array elements' excitations.

► Also, through the use of electronically controlled solid-state phase shifters, the beam direction of the antenna array can be steered electronically by controlling the relative phases of the array elements. ◀

This flexibility of the array antenna has led to numerous applications, including **electronic steering** and **multiple-beam generation**.



(如第 9-1 和 9-3 节中讨论的)，除非同时使用两个或更多天线塔，否则不可能沿着特定的感兴趣方向引导水平方向图。感兴趣的方向可能包括 AM 电台服务的城市，而要避开方向可能包括以相同频率运行的另一个电台服务的区域（从而避免干扰影响）。当两个或多个天线一起使用时，这种组合称为天线阵列。

AM 广播天线阵列只是通信系统和雷达应用中使用的众多天线阵列之一。天线阵列为天线设计者提供了从非常简单的天线元件开始获得高方向性、窄波束、低旁瓣、可操纵波束和成形天线方向图的灵活性。图9-25显示了一个非常大的雷达系统，其中包括由5,184个独立偶极天线元件组成的发射器阵列和由4,660个元件组成的接收器阵列。该雷达系统是美国空军运营的太空监视网络的一部分，工作频率为 442 MHz，发射的综合峰值功率为 30 MW!

尽管阵列不需要由类似的辐射元件组成，但大多数阵列实际上使用相同的元件，例如偶极子、缝隙、喇叭天线或抛物面天线。组成阵列的天线元件可以以各种配置排列，但最常见的是线性一维配置（其中元件沿直线排列）和二维晶格配置（其中元件位于平面网格上）。可以通过控制阵列元件激励的相对幅度来合成阵列远场辐射图的所需形状。

► 此外，通过使用电子控制的固态移相器，可以通过控制阵列元件的相对相位来电子地控制天线阵列的波束方向。 ◀

阵列天线的这种灵活性带来了众多应用，包括电子转向和多波束生成。



**Figure 9-25** The AN/FPS-85 Phased Array Radar Facility in the Florida panhandle is near the city of Freeport. A several-mile no-fly zone surrounds the radar installation as a safety concern for electroexplosive devices, such as ejection seats and munitions that are carried on military aircraft.

The purpose of this and the next two sections is to introduce the reader to the basic principles of array theory and design techniques used in shaping the antenna pattern and steering the main lobe. This presentation is confined to the one-dimensional linear array with equal spacing between adjacent elements.

A linear array of  $N$  identical radiators is arranged along the  $z$  axis, as shown in **Fig. 9-26**. The radiators are fed by a common oscillator through a branching network. In each branch, an attenuator (or amplifier) and phase shifter are inserted in series to control the amplitude and phase of the signal feeding the antenna element in that branch.

In the far-field region of any radiating element, the **element electric-field intensity**  $\tilde{E}_e(R, \theta, \phi)$  may be expressed as a product of two functions: the spherical propagation factor  $e^{-jkR}/R$ , which accounts for the dependence on the range  $R$ , and  $\tilde{f}_e(\theta, \phi)$ , which accounts for the directional dependence of the element's electric field. Thus, for an isolated element, the radiated field is

$$\tilde{E}_e(R, \theta, \phi) = \frac{e^{-jkR}}{R} \tilde{f}_e(\theta, \phi), \tag{9.93}$$

and the corresponding power density  $S_e$  is

$$S_e(R, \theta, \phi) = \frac{1}{2\eta_0} |\tilde{E}_e(R, \theta, \phi)|^2 = \frac{1}{2\eta_0 R^2} |\tilde{f}_e(\theta, \phi)|^2. \tag{9.94}$$

Hence, for the array shown in **Fig. 9-26(b)**, the far-zone field

due to element  $i$  at range  $R_i$  from observation point  $Q$  is

$$\tilde{E}_i(R_i, \theta, \phi) = A_i \frac{e^{-jkR_i}}{R_i} \tilde{f}_e(\theta, \phi), \tag{9.95}$$

where  $A_i = a_i e^{j\psi_i}$  is a **complex feeding coefficient** representing the amplitude  $a_i$  and phase  $\psi_i$  of the excitation giving rise to  $\tilde{E}_i$ , which is relative to a reference excitation. In practice, the excitation of one of the elements is used as reference. Note that  $R_i$  and  $A_i$  may be different for different elements in the array, but  $\tilde{f}_e(\theta, \phi)$  is the same for all of them because they are all identical; hence, they exhibit identical directional patterns.

The total field at the observation point  $Q(R_0, \theta, \phi)$  is the sum of the fields due to the  $N$  elements:

$$\tilde{E}(R_0, \theta, \phi) = \sum_{i=0}^{N-1} \tilde{E}_i(R_i, \theta, \phi) = \left[ \sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right] \tilde{f}_e(\theta, \phi), \tag{9.96}$$

where  $R_0$  denotes the range of  $Q$  from the center of the coordinate system, which is chosen to be at the location of the zeroth element. To satisfy the far-field condition given by Eq. (9.73) for an array of length  $l = (N - 1)d$ , where  $d$  is the interelement spacing, the range  $R_0$  should be sufficiently large to satisfy

$$R_0 \geq \frac{2l^2}{\lambda} = \frac{2(N - 1)^2 d^2}{\lambda}. \tag{9.97}$$



图 9-25 佛罗里达州狭长地带的 AN/FPS-85 相控阵雷达设施靠近自由港市。雷达装置周围有一个数英里的禁飞区，这是军用飞机上携带的弹射座椅和弹药等电爆装置的安全问题。

本节和接下来两节的目的是向读者介绍阵列理论的基本原理以及用于塑造天线方向图和控制主瓣的设计技术。本演示仅限于相邻元素之间间距相等的一维线性阵列。

N个相同的辐射器沿z轴排列成线性阵列，如图9-26所示。辐射器由公共振荡器通过分支网络供电。在每个分支中，串联插入衰减器（或放大器）和移相器，以控制馈送到该分支中的天线元件的信号幅度和相位。

在任何辐射元件的远场区域，元件电场强度  $\tilde{E}_e(R, \theta, \phi)$  可以表示为两个函数的乘积：球面传播因子  $e^{-jkR}/R$ （解释对范围  $R$  的依赖性）和  $\tilde{f}_e(\theta, \phi)$ （解释元件电场的方向依赖性）。因此，对于孤立元件，辐射场为

$$\tilde{E}_e(R, \theta, \phi) = \frac{e^{-jkR}}{R} \tilde{f}_e(\theta, \phi), \tag{9.93}$$

相应的功率密度 $S_e$ 为

$$S_e(R, \theta, \phi) = \frac{1}{2\eta_0} |\tilde{E}_e(R, \theta, \phi)|^2 = \frac{1}{2\eta_0 R^2} |\tilde{f}_e(\theta, \phi)|^2. \tag{9.94}$$

因此，对于图9-26(b)所示的阵列，远区域

由于距观察点  $Q$  的范围  $R_i$  内的元素  $i$  为

$$\tilde{E}_i(R_i, \theta, \phi) = A_i \frac{e^{-jkR_i}}{R_i} \tilde{f}_e(\theta, \phi), \tag{9.95}$$

其中  $A_i = a_i e^{j\psi_i}$  是一个复数馈送系数，表示产生  $\tilde{E}_i$  的激励的幅度  $a_i$  和相位  $\psi_i$ ，相对于参考激励。在实践中，以其中一种元素的激励作为参考。请注意，对于数组中的不同元素， $R_i$  和  $A_i$  可能不同，但  $\tilde{f}_e(\theta, \phi)$  对于所有元素来说都是相同的，因为它们都是相同的；因此，它们表现出相同的方向模式。

观察点  $Q(R_0, \theta, \phi)$  处的总场是  $N$  个元素产生的场之和：

$$\tilde{E}(R_0, \theta, \phi) = \sum_{i=0}^{N-1} \tilde{E}_i(R_i, \theta, \phi) = \left[ \sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right] \tilde{f}_e(\theta, \phi), \tag{9.96}$$

其中 $R_0$ 表示Q距坐标系中心的范围，该坐标系选择在第零个元素的位置。满足方程给出的远场条件。(9.73)对于长度为  $l = (N - 1)d$  的数组，其中  $d$  是元素间间距，范围  $R_0$  应足够大以满足

$$R_0 \geq \frac{2l^2}{\lambda} = \frac{2(N - 1)^2 d^2}{\lambda}. \tag{9.97}$$

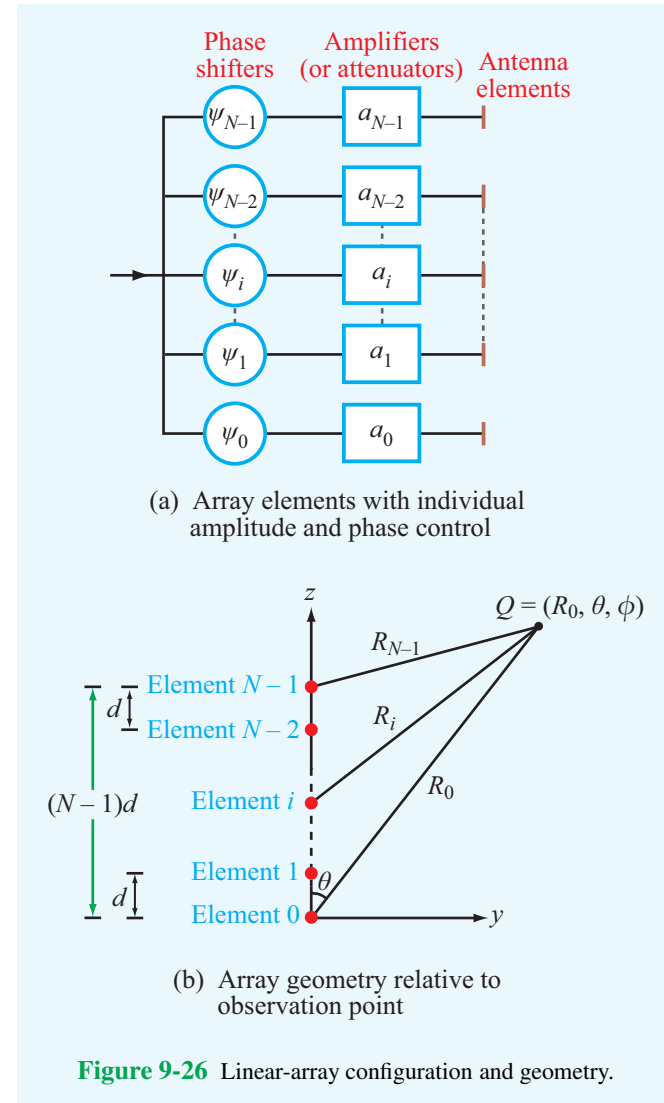
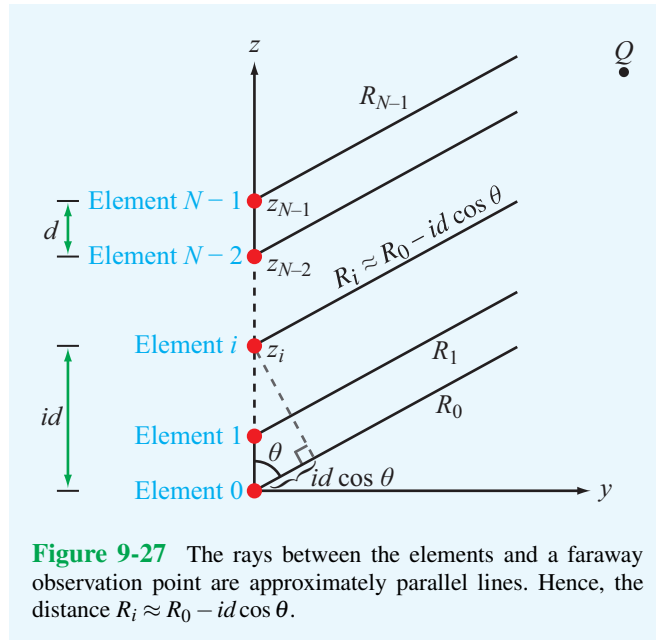


Figure 9-26 Linear-array configuration and geometry.

This condition allows us to ignore differences in the distances from  $Q$  to the individual elements as far as the magnitudes of the radiated fields are concerned. Thus, we can set  $R_i = R_0$  in the denominator in Eq. (9.96) for all  $i$ . With regard to the phase part of the propagation factor, we can use the parallel-ray approximation given by

$$R_i \approx R_0 - z_i \cos \theta = R_0 - id \cos \theta, \quad (9.98)$$

where  $z_i = id$  is the distance between the  $i$ th element and the zeroth element (Fig. 9-27). Employing these two approxima-

Figure 9-27 The rays between the elements and a faraway observation point are approximately parallel lines. Hence, the distance  $R_i \approx R_0 - id \cos \theta$ .

tions in Eq. (9.96) leads to

$$\tilde{E}(R_0, \theta, \phi) = \tilde{f}_e(\theta, \phi) \left( \frac{e^{-jkR_0}}{R_0} \right) \left[ \sum_{i=0}^{N-1} A_i e^{j ikd \cos \theta} \right], \quad (9.99)$$

and the corresponding array-antenna power density is given by

$$\begin{aligned} S(R_0, \theta, \phi) &= \frac{1}{2\eta_0} |\tilde{E}(R_0, \theta, \phi)|^2 \\ &= \frac{1}{2\eta_0 R_0^2} |\tilde{f}_e(\theta, \phi)|^2 \left| \sum_{i=0}^{N-1} A_i e^{j ikd \cos \theta} \right|^2 \\ &= S_e(R_0, \theta, \phi) \left| \sum_{i=0}^{N-1} A_i e^{j ikd \cos \theta} \right|^2, \quad (9.100) \end{aligned}$$

where use was made of Eq. (9.94). This expression is a product of two factors. The first factor,  $S_e(R_0, \theta, \phi)$ , is the power density of the energy radiated by an individual element, and the second, called the **array factor**, is a function of the positions of the individual elements and their feeding coefficients but not a function of the specific type of radiators used.

► The array factor represents the far-field radiation intensity of the  $N$  elements—had the elements been isotropic radiators. ◀

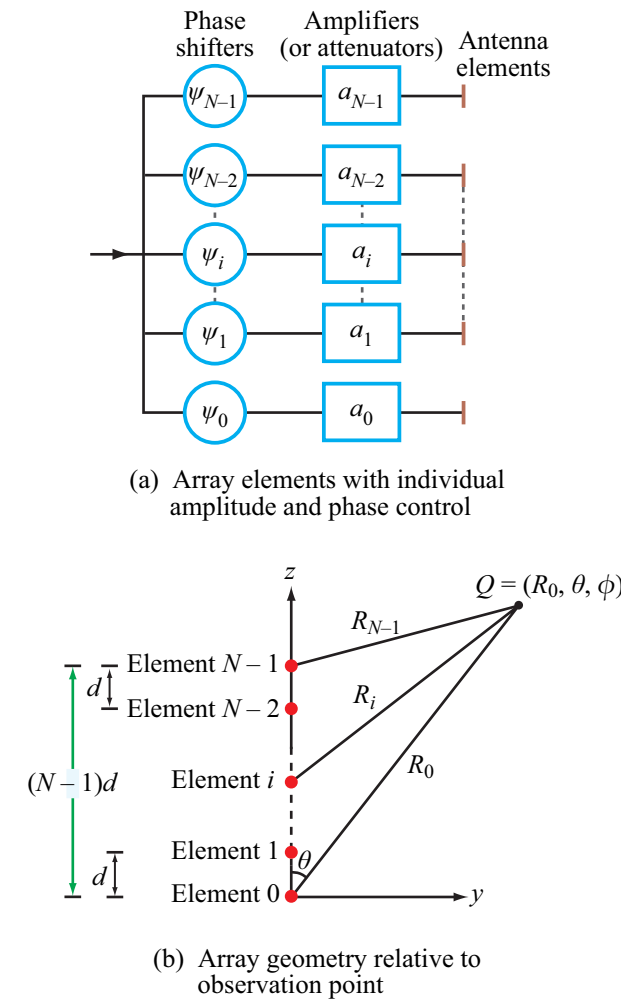
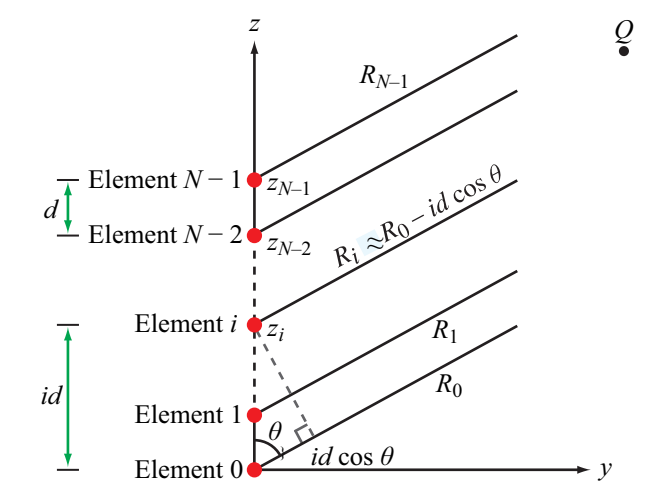


图 9-26 线性阵列配置和几何结构。

就辐射场的大小而言，这种条件允许我们忽略  $Q$  到各个元件的距离差异。因此，我们可以将  $R_i = R_0$  设置为式 (1) 中的分母。(9.96) 对于所有  $i$ 。对于传播因子的相位部分，我们可以使用下式给出的平行射线近似：

$$R_i \approx R_0 - z_i \cos \theta = R_0 - id \cos \theta, \quad (9.98)$$

其中  $z_i = id$  是第  $i$  个元素和第  $0$  个元素之间的距离 (图 9-27)。采用这两个近似

图9-27 单元与远处观察点之间的光线近似为平行线。因此，距离  $R_i \approx R_0 - id \cos \theta$ 。

方程式中的蒸发散(9.96) 导致

$$\tilde{E}(R_0, \theta, \phi) = \tilde{f}_e(\theta, \phi) \left( \frac{e^{-jkR_0}}{R_0} \right) \left[ \sum_{i=0}^{N-1} A_i e^{j ikd \cos \theta} \right], \quad (9.99)$$

相应的阵列天线功率密度由下式给出

$$\begin{aligned} S(R_0, \theta, \phi) &= \frac{1}{2\eta_0} |\tilde{E}(R_0, \theta, \phi)|^2 \\ &= \frac{1}{2\eta_0 R_0^2} |\tilde{f}_e(\theta, \phi)|^2 \left| \sum_{i=0}^{N-1} A_i e^{j ikd \cos \theta} \right|^2 \\ &= S_e(R_0, \theta, \phi) \left| \sum_{i=0}^{N-1} A_i e^{j ikd \cos \theta} \right|^2, \quad (9.100) \end{aligned}$$

其中使用了方程式。(9.94)。该表达式是两个因素的产物。第一个因子  $S_e(R_0, \theta, \phi)$  是单个元件辐射的能量功率密度，第二个因子称为阵列因子，是单个元件的位置及其馈电系数的函数，但不是所使用的特定类型辐射器的函数。

► 阵列因子代表  $N$  个元件的远场辐射强度 (假设这些元件是各向同性辐射器)。◀

Denoting the array factor by

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{j i k d \cos \theta} \right|^2,$$

(9.101)

the power density of the antenna array is then written as

$$S(R_0, \theta, \phi) = S_e(R_0, \theta, \phi) F_a(\theta).$$

(9.102)

This equation demonstrates the **pattern multiplication principle**. It allows us to find the far-field power density of the antenna array by first computing the far-field power pattern with the array elements replaced with isotropic radiators, which yields the array factor  $F_a(\theta)$ , and then multiplying the result by  $S_e(R_0, \theta, \phi)$ , which is the power density for a single element (and is the same for all elements).

The feeding coefficient  $A_i$  is, in general, a complex amplitude consisting of an amplitude factor  $a_i$  and a phase factor  $\psi_i$ :

$$A_i = a_i e^{j \psi_i}.$$

(9.103)

Insertion of Eq. (9.103) into Eq. (9.101) leads to

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{j \psi_i} e^{j i k d \cos \theta} \right|^2.$$

(9.104)

The array factor is governed by two input functions: the **array amplitude distribution** given by the  $a_i$ 's and the **array phase distribution** given by the  $\psi_i$ 's.

► The amplitude distribution serves to control the shape of the array radiation pattern, while the phase distribution can be used to steer its direction. ◀

Example 9-5: Array of Two Vertical Dipoles

An AM radio station uses two vertically oriented half-wave dipoles separated by a distance of  $\lambda/2$ , as shown in **Fig. 9-28(a)**. The vector from the location of the first dipole to the location of the second dipole points toward the east. The two dipoles are fed with equal-amplitude excitations, and the dipole farther east is excited with a phase shift of  $-\pi/2$  relative to the other one. Find and plot the antenna pattern of the antenna array in the horizontal plane.

**Solution:** The array factor given by Eq. (9.104) was derived for radiators arranged along the  $z$  axis. To keep the coordinate system the same, we choose the easterly direction to be the  $z$  axis, as shown in **Fig. 9-28(b)**, and we place the first dipole at  $z = -\lambda/4$  and the second at  $z = \lambda/4$ . A dipole radiates uniformly in the plane perpendicular to its axis, which in this case is the horizontal plane. Hence,  $S_e = S_0$  for all angles  $\theta$  in **Fig. 9-28(b)**, where  $S_0$  is the maximum value of the power density radiated by each dipole individually. Consequently, the power density radiated by the two-dipole array is

$$S(R, \theta) = S_0 F_a(\theta).$$

For two elements separated by  $d = \lambda/2$  and excited with equal amplitudes ( $a_0 = a_1 = 1$ ) and with phase angles  $\psi_0 = 0$  and

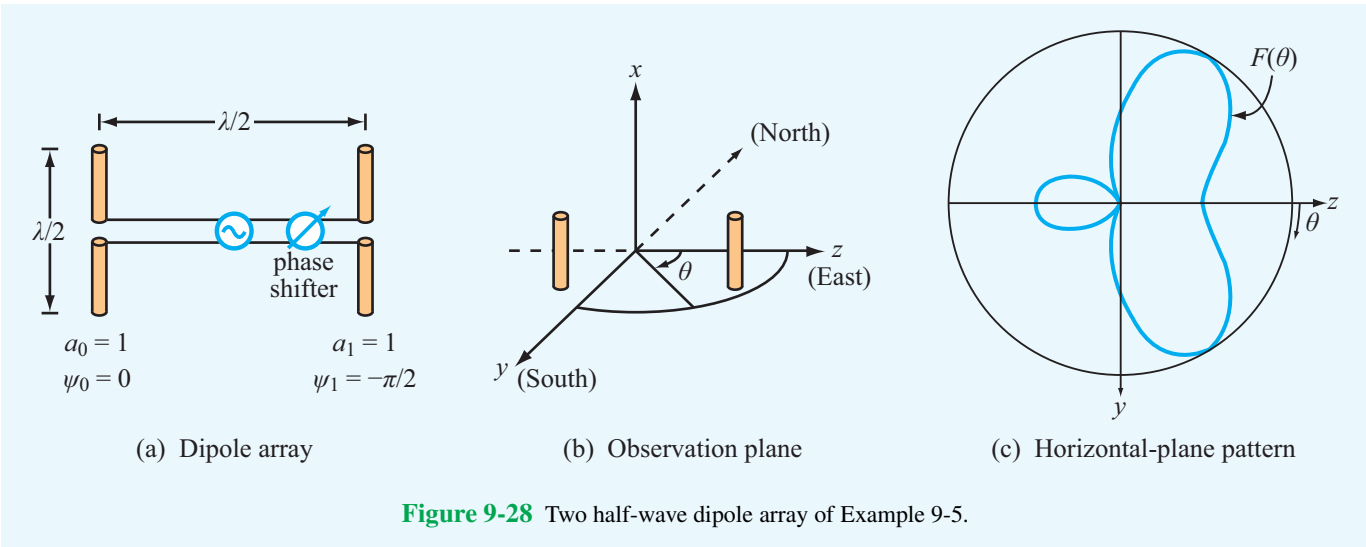


Figure 9-28 Two half-wave dipole array of Example 9-5.

将数组因子表示为

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{j i k d \cos \theta} \right|^2,$$

(9.101)

天线阵列的功率密度可写为

$$S(R_0, \theta, \phi) = S_e(R_0, \theta, \phi) F_a(\theta).$$

(9.102)

该方程演示了模式乘法原理。它允许我们首先计算用各向同性辐射器替换阵列元件的远场功率方向图，从而得出阵列因子  $F_a(\theta)$ ，然后将结果乘以  $S_e(R_0, \theta, \phi)$ ，这就是单个元件的功率密度（并且对于所有元件都相同），从而找到天线阵列的远场功率密度。

一般来说，馈电系数  $A_i$  是一个复振幅，由振幅因子  $a_i$  和相位因子  $\psi_i$  组成：

$$A_i = a_i e^{j \psi_i}.$$

(9.103)

插入方程。 (9.103) 代入方程。 (9.101) 导致

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{j \psi_i} e^{j i k d \cos \theta} \right|^2.$$

(9.104)

阵列因子由两个输入函数控制：由  $a_i$  给出的阵列幅度分布和由  $\psi_i$  给出的阵列相位分布。

► 幅度分布用于控制阵列辐射方向图的形状，而相位分布可用于控制其方向。 ◀

例 9-5：两个垂直偶极子的阵列

AM广播电台使用两个垂直定向的半波偶极子，其间隔距离为  $\lambda/2$ ，如图9-28(a)所示。从第一个偶极子位置到第二个偶极子位置的矢量指向东方。两个偶极子受到等幅激励，更东的偶极子受到相对于另一个偶极子  $-\pi/2$  相移的激励。查找并绘制水平面上天线阵列的天线方向图。

解：由方程给出的阵列因子。 (9.104) 是针对沿  $z$  轴排列的散热器得出的。为了保持坐标系相同，我们选择东向作为  $z$  轴，如图9-28(b)所示，并将第一个偶极子放置在  $z = -\lambda/4$  处，第二个偶极子放置在  $z = \lambda/4$  处。偶极子在垂直于其轴的平面（在本例中为水平面）内均匀辐射。因此，图9-28(b)中所有角度  $\theta$  的  $S_e = S_0$ ，其中  $S_0$  是每个偶极子单独辐射的功率密度的最大值。因此，双偶极子阵列辐射的功率密度为

$$S(R, \theta) = S_0 F_a(\theta).$$

对于由  $d = \lambda/2$  分隔并以相等幅度 ( $a_0 = a_1 = 1$ ) 和相角  $\psi_0 = 0$  激励的两个元素，

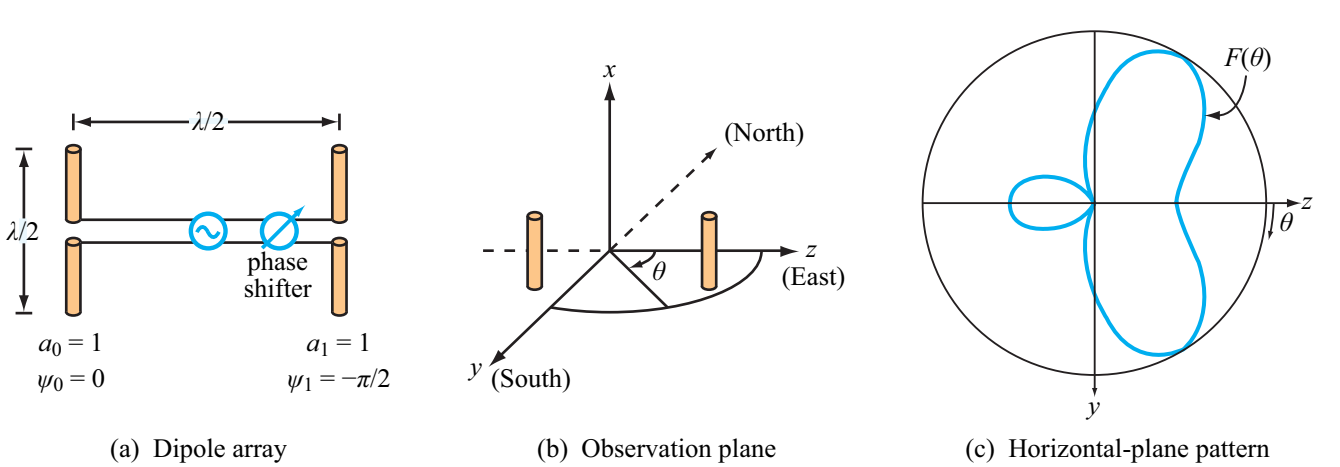
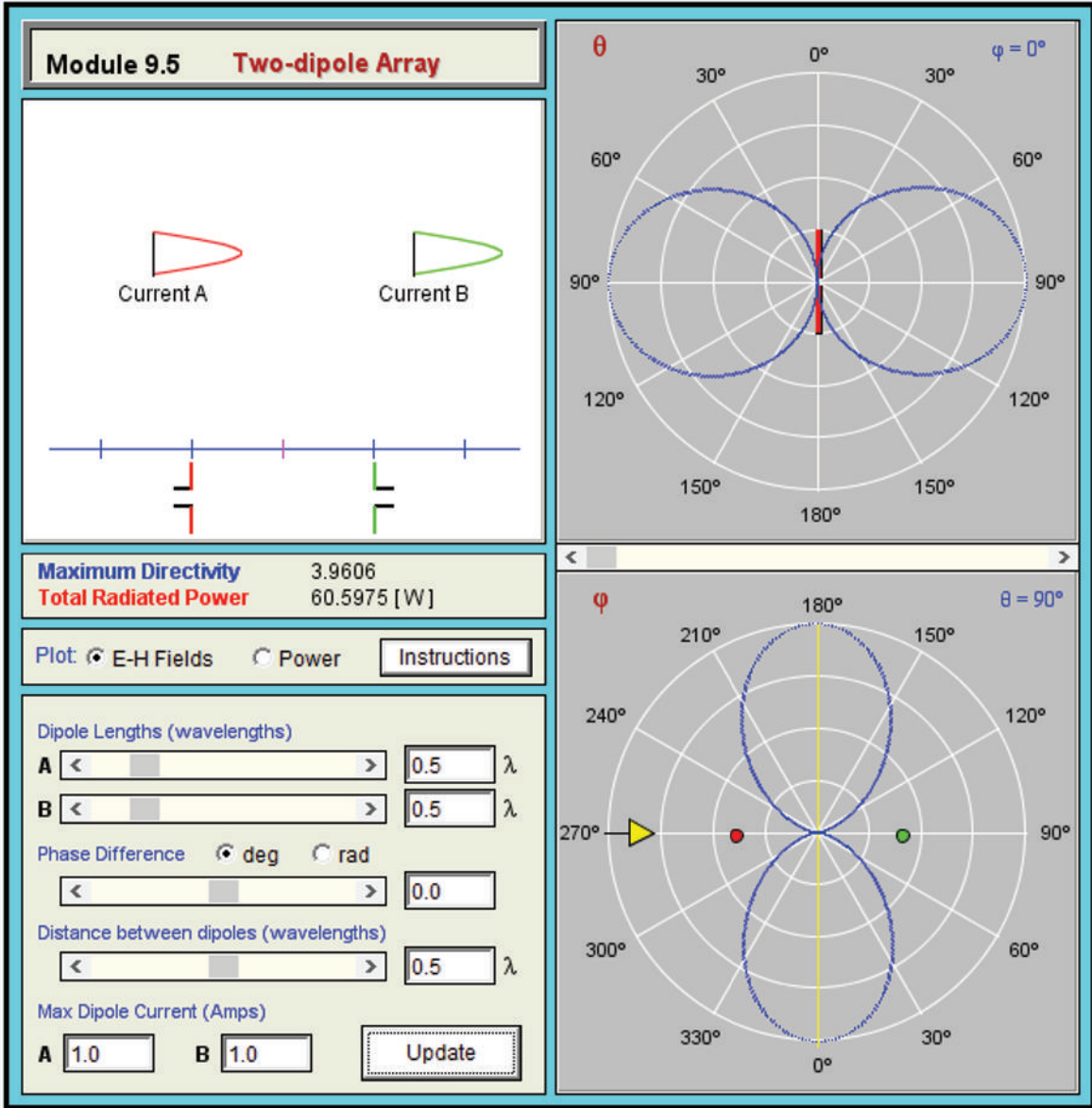
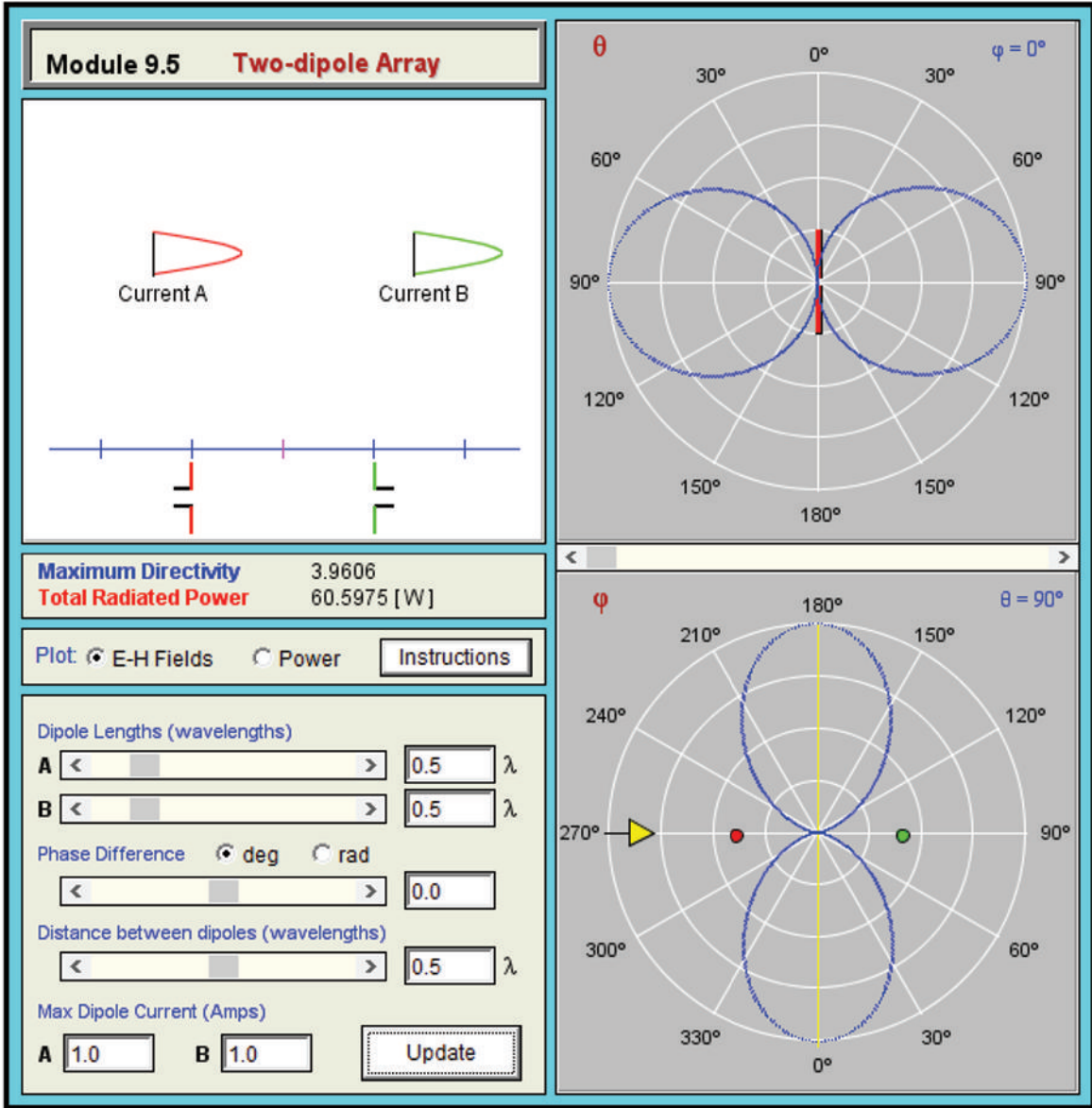


图9-28 例9-5的两个半波偶极子阵列。

**Module 9.5 Two-Dipole Array** Given two vertical dipoles, the user can specify their individual lengths and current maxima, as well as the distance between them and the phase difference between their current excitations. The module generates plots of the field and power patterns in the far zone and calculates the maximum directivity and total radiated power.



模块 9.5 双偶极子阵列 给定两个垂直偶极子，用户可以指定它们各自的长度和电流最大值，以及它们之间的距离和电流激励之间的相位差。该模块生成远区的场和功率图，并计算最大方向性和总辐射功率。



$\psi_1 = -\pi/2$ , Eq. (9.104) becomes

$$F_a(\theta) = \left| \sum_{i=0}^1 a_i e^{j\psi_i} e^{j i k d \cos \theta} \right|^2 = \left| 1 + e^{-j\pi/2} e^{j(2\pi/\lambda)(\lambda/2) \cos \theta} \right|^2 \\ = \left| 1 + e^{j(\pi \cos \theta - \pi/2)} \right|^2.$$

A function of the form  $|1 + e^{jx}|^2$  can be evaluated by factoring out  $e^{jx/2}$  from both terms:

$$|1 + e^{jx}|^2 = |e^{jx/2}(e^{-jx/2} + e^{jx/2})|^2 \\ = |e^{jx/2}|^2 |e^{-jx/2} + e^{jx/2}|^2 \\ = |e^{jx/2}|^2 \left| 2 \frac{[e^{-jx/2} + e^{jx/2}]}{2} \right|^2.$$

The absolute value of  $e^{jx/2}$  is 1, and we recognize the function inside the square bracket as  $\cos(x/2)$ . Hence,

$$|1 + e^{jx}|^2 = 4 \cos^2 \left( \frac{x}{2} \right).$$

Applying this result to the expression for  $F_a(\theta)$ , we have

$$F_a(\theta) = 4 \cos^2 \left( \frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right).$$

The power density radiated by the array is then

$$S(R, \theta) = S_0 F_a(\theta) = 4S_0 \cos^2 \left( \frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right).$$

This function has a maximum value  $S_{\max} = 4S_0$ , and it occurs when the argument of the cosine function is equal to zero. Thus,

$$\frac{\pi}{2} \cos \theta - \frac{\pi}{4} = 0,$$

which leads to the solution  $\theta = 60^\circ$ . Upon normalizing  $S(R, \theta)$  by its maximum value, we obtain the normalized radiation intensity given by

$$F(\theta) = \frac{S(R, \theta)}{S_{\max}} = \cos^2 \left( \frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right).$$

The pattern of  $F(\theta)$  is shown in **Fig. 9-28(c)**.

Module 9.6 Detailed Analysis of Two-Dipole Array

This module extends the display and computational capabilities of Module 9.5 by offering plots for individual components of **E** and **H** at any range from the antenna, including the near field.

Example 9-6: Pattern Synthesis

In Example 9-5, we were given the array parameters  $a_0$ ,  $a_1$ ,  $\psi_0$ ,  $\psi_1$ , and  $d$ , and we were then asked to determine the pattern of the two-element dipole array. We now consider the reverse process; given specifications on the desired pattern, we specify the array parameters to meet those specifications.

Given two vertical dipoles, as depicted in **Fig. 9-28(b)**, specify the array parameters so that the array exhibits maximum radiation toward the east and no radiation toward the north or south.

**Solution:** From Example 9-5, we established that, because each dipole radiates equally along all directions in the y–z plane, the radiation pattern of the two-dipole array in that plane is governed solely by the array factor  $F_a(\theta)$ . The shape of the pattern of the array factor depends on three parameters: the amplitude ratio  $a_1/a_0$ , the phase difference  $\psi_1 - \psi_0$ , and the spacing  $d$  (**Fig. 9-29(a)**). For convenience, we choose  $a_0 = 1$  and  $\psi_0 = 0$ . Accordingly, Eq. (9.101) becomes

$$F_a(\theta) = \left| \sum_{i=0}^1 a_i e^{j\psi_i} e^{j i k d \cos \theta} \right|^2 = |1 + a_1 e^{j\psi_1} e^{j(2\pi d/\lambda) \cos \theta}|^2.$$

Next, we consider the specification that  $F_a$  be equal to zero when  $\theta = 90^\circ$  (north and south directions in **Fig. 9-29(a)**). For any observation point on the y axis, the ranges  $R_0$  and  $R_1$  shown in **Fig. 9-29(a)** are equal, which means that the propagation phases associated with the time travel of the waves radiated by the two dipoles to that point are identical. Hence, to satisfy the stated condition, we need to choose  $a_1 = a_0$  and  $\psi_1 = \pm\pi$ . With these choices, the signals radiated by the two dipoles have equal amplitudes and opposite phases—thereby interfering destructively. This conclusion can be ascertained by evaluating the array factor at  $\theta = 90^\circ$  with  $a_0 = a_1 = 1$  and  $\psi_1 = \pm\pi$ :

$$F_a(\theta = 90^\circ) = |1 + 1e^{\pm j\pi}|^2 = |1 - 1| = 0.$$

The two values of  $\psi_1$ , namely  $\pi$  and  $-\pi$ , lead to the same solution for the value of the spacing  $d$  to meet the specification that the array radiation pattern is maximum toward the east, corresponding to  $\theta = 0^\circ$ . Let us choose  $\psi_1 = -\pi$  and examine the array factor at  $\theta = 0^\circ$ :

$$F_a(\theta = 0) = |1 + e^{-j\pi} e^{j2\pi d/\lambda}|^2 = |1 + e^{j(-\pi+2\pi d/\lambda)}|^2.$$

For  $F_a(\theta = 0)$  to be a maximum, we require the phase angle of the second term to be zero or a multiple of  $2\pi$ . That is,

$$-\pi + \frac{2\pi d}{\lambda} = 2n\pi,$$

$\psi_1 = -\pi/2$ , 等式。 (9.104) 变为

$$F_a(\theta) = \left| \sum_{i=0}^1 a_i e^{j\psi_i} e^{j i k d \cos \theta} \right|^2 = \left| 1 + e^{-j\pi/2} e^{j(2\pi/\lambda)(\lambda/2) \cos \theta} \right|^2 \\ = \left| 1 + e^{j(\pi \cos \theta - \pi/2)} \right|^2.$$

$|1 + e^{jx}|^2$  形式的函数可以通过从两项中分解出  $e^{jx/2}$  来计算:

$$|1 + e^{jx}|^2 = |e^{jx/2}(e^{-jx/2} + e^{jx/2})|^2 \\ = |e^{jx/2}|^2 |e^{-jx/2} + e^{jx/2}|^2 \\ = |e^{jx/2}|^2 \left| 2 \frac{[e^{-jx/2} + e^{jx/2}]}{2} \right|^2.$$

$e^{jx/2}$  的绝对值为 1, 我们将方括号内的函数识别为  $\cos(x/2)$ 。因此,

$$|1 + e^{jx}|^2 = 4 \cos^2 \left( \frac{x}{2} \right).$$

将此结果应用于  $F_a(\theta)$  的表达式, 我们有

$$F_a(\theta) = 4 \cos^2 \left( \frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right).$$

则阵列辐射的功率密度为

$$S(R, \theta) = S_0 F_a(\theta) = 4S_0 \cos^2 \left( \frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right).$$

该函数具有最大值  $S_{\max} = 4S_0$ , 并且当余弦函数的参数等于 0 时出现。因此,

$$\frac{\pi}{2} \cos \theta - \frac{\pi}{4} = 0,$$

这导致解  $\theta = 60^\circ$ 。通过最大值对  $S(R, \theta)$  进行归一化, 我们得到归一化辐射强度:

$$F(\theta) = \frac{S(R, \theta)}{S_{\max}} = \cos^2 \left( \frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right).$$

$F(\theta)$  的模式如图9-28(c)所示。

模块 9.6 二偶极子阵列的详细分析 该模块通过提供距天线 线任意范围 (包括近场) 的 **E** 和 **H** 各个分量的绘图, 扩展了模块 9.5 的显示和计算能力。

例 9-6: 模式合成

在示例 9-5 中, 我们获得了数组参数  $a_0$ 、 $a_1$ 、 $\psi_0$ 、 $\psi_1$  和  $d$ , 然后要求我们确定二元偶极子数组的模式。我们现在考虑相反的过程; 给定所需模式的规格后, 我们指定数组参数以满足这些规格。

给定两个垂直偶极子, 如图 9-28(b) 所示, 指定阵列参数, 使阵列向东呈现最大辐射, 而向北或向南没有辐射。

解决方案: 从示例 9-5 中, 我们确定, 由于每个偶极子沿 y-z 平面中的所有方向均匀辐射, 因此该平面中双偶极子阵列的辐射方向图仅由阵列因子  $F_a(\theta)$  控制。阵列因子的图案形状取决于三个参数: 幅度比 $a_1/a_0$ 、相位差 $\psi_1 - \psi_0$ 和间距 $d$  (图9-29 (a) )。为了方便起见, 我们选择  $a_0 = 1$  和  $\psi_0 = 0$ 。 (9.101) 变为

$$F_a(\theta) = \left| \sum_{i=0}^1 a_i e^{j\psi_i} e^{j i k d \cos \theta} \right|^2 = |1 + a_1 e^{j\psi_1} e^{j(2\pi d/\lambda) \cos \theta}|^2.$$

接下来, 我们考虑图 9-29(a) 中  $\theta = 90^\circ$  (南北方向时  $F_a$  等于 0 的规定。对于 y 轴上的任意观测点, 图 9-29 (a) 所示的范围  $R_0$  和  $R_1$  相等, 这意味着两个偶极子辐射到该点的波的时间传播相关的传播相位是相同的。因此, 为了满足规定的条件, 我们需要选择  $a_1 = a_0$  和  $\psi_1 = \pm\pi$ 。通过这些选择, 两个偶极子辐射的信号具有相等的幅度和相反的相位, 从而产生破坏性干扰。这个结论可以通过使用  $a_0 = a_1 = 1$  和  $\psi_1 = \pm\pi$  评估  $\theta = 90^\circ$  处的数组因子来确定:

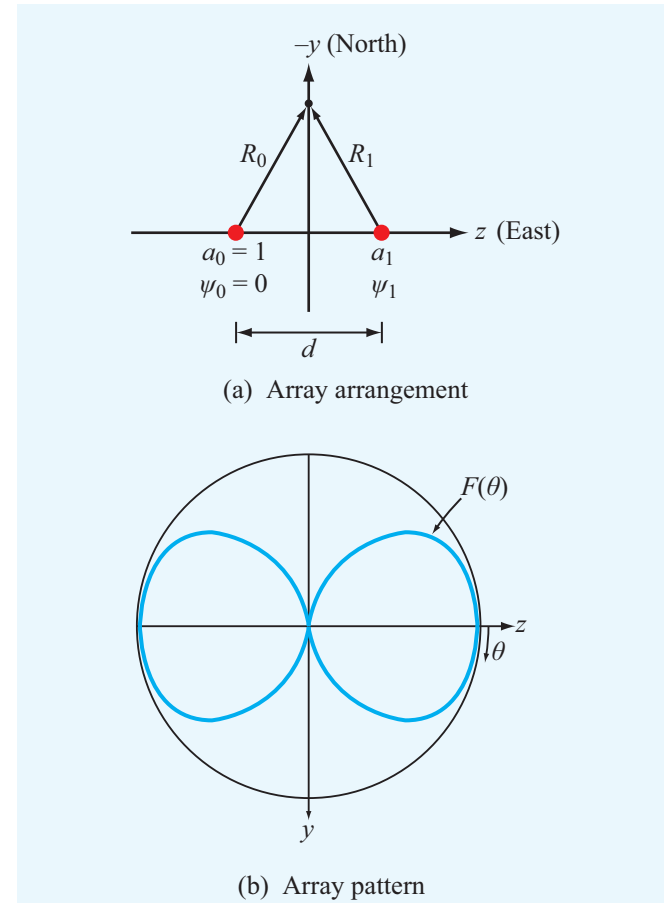
$$F_a(\theta = 90^\circ) = |1 + 1e^{\pm j\pi}|^2 = |1 - 1| = 0.$$

$\psi_1$  的两个值, 即  $\pi$  和  $-\pi$ , 对于间距  $d$  的值得出相同的解, 以满足阵列辐射方向图向东最大的规定, 对应于  $\theta = 0^\circ$ 。让我们选择  $\psi_1 = -\pi$  并检查  $\theta = 0^\circ$  处的数组因子:

$$F_a(\theta = 0) = |1 + e^{-j\pi} e^{j2\pi d/\lambda}|^2 = |1 + e^{j(-\pi+2\pi d/\lambda)}|^2.$$

为了使  $F_a(\theta = 0)$  成为最大值, 我们要求第二项的相位角为零或  $2\pi$  的倍数。那是,

$$-\pi + \frac{2\pi d}{\lambda} = 2n\pi,$$



**Figure 9-29** (a) Two vertical dipoles separated by a distance  $d$  along the  $z$  axis; (b) normalized array pattern in the  $y$ - $z$  plane for  $a_0 = a_1 = 1$ ,  $\psi_1 = \psi_0 = -\pi$ , and  $d = \lambda/2$ .

or

$$d = (2n + 1) \frac{\lambda}{2}, \quad n = 0, 1, 2, \dots$$

In summary, the two-dipole array will meet the given specifications if  $a_0 = a_1$ ,  $\psi_1 - \psi_0 = -\pi$ , and  $d = (2n + 1)\lambda/2$ .

For  $d = \lambda/2$ , the array factor is

$$\begin{aligned} F_a(\theta) &= |1 + e^{-j\pi} e^{j\pi \cos \theta}|^2 \\ &= |1 - e^{j\pi \cos \theta}|^2 \\ &= \left| 2je^{-j(\pi/2) \cos \theta} \left[ \frac{e^{j(\pi/2) \cos \theta} - e^{-j(\pi/2) \cos \theta}}{2j} \right] \right|^2 \\ &= 4 \sin^2 \left( \frac{\pi}{2} \cos \theta \right). \end{aligned}$$

The array factor has a maximum value of 4, which is the maximum level attainable from a two-element array with unit amplitudes. The directions along which  $F_a(\theta)$  is a maximum are those corresponding to  $\theta = 0$  (east) and  $\theta = 180^\circ$  (west), as shown in **Fig. 9-29(b)**.

**Exercise 9-14:** Derive an expression for the array factor of a two-element array excited in phase with  $a_0 = 1$  and  $a_1 = 3$ . The elements are positioned along the  $z$  axis and are separated by  $\lambda/2$ .

**Answer:**  $F_a(\theta) = [10 + 6 \cos(\pi \cos \theta)]$ . (See **EM**.)

**Exercise 9-15:** An equally spaced  $N$ -element array arranged along the  $z$  axis is fed with equal amplitudes and phases; that is,  $A_i = 1$  for  $i = 0, 1, \dots, (N - 1)$ . What is the magnitude of the array factor in the broadside direction?

**Answer:**  $F_a(\theta = 90^\circ) = N^2$ . (See **EM**.)

## 9-10 N-Element Array with Uniform Phase Distribution

We now consider an array of  $N$  elements with equal spacing  $d$  and equal-phase excitations; that is,  $\psi_i = \psi_0$  for  $i = 1, 2, \dots, (N - 1)$ . Such an array of in-phase elements is sometimes referred to as a **broadside array** because the main beam of the radiation pattern of its array factor is always in the direction broadside to the array axis. From Eq. (9.104), the array factor is given by

$$\begin{aligned} F_a(\theta) &= \left| e^{j\psi_0} \sum_{i=0}^{N-1} a_i e^{j i k d \cos \theta} \right|^2 \\ &= |e^{j\psi_0}|^2 \left| \sum_{i=0}^{N-1} a_i e^{j i k d \cos \theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{j i k d \cos \theta} \right|^2. \end{aligned} \quad (9.105)$$

The phase difference between the fields radiated by adjacent elements is

$$\gamma = k d \cos \theta = \frac{2\pi d}{\lambda} \cos \theta. \quad (9.106)$$

In terms of  $\gamma$ , Eq. (9.105) takes the compact form

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{j i \gamma} \right|^2. \quad (\text{uniform phase}) \quad (9.107)$$

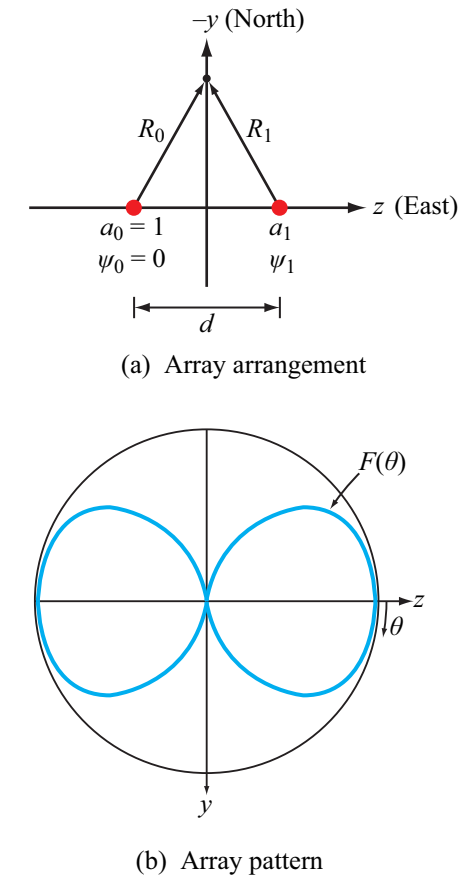


图9-29(a)沿z轴间隔距离d的两个垂直偶极子; (b)  $a_0 = a_1 = 1$ 、 $\psi_1 = \psi_0 = -\pi$  和  $d = \lambda/2$  在  $y$ - $z$  平面中的归一化阵列模式。

或者

$$d = (2n + 1) \frac{\lambda}{2}, \quad n = 0, 1, 2, \dots$$

总之, 如果  $a_0 = a_1$ 、 $\psi_1 - \psi_0 = -\pi$  和  $d = (2n + 1)\lambda/2$ , 则二偶极子阵列将满足给定的规格。

对于  $d = \lambda/2$ , 数组因子为

$$\begin{aligned} F_a(\theta) &= |1 + e^{-j\pi} e^{j\pi \cos \theta}|^2 \\ &= |1 - e^{j\pi \cos \theta}|^2 \\ &= \left| 2je^{-j(\pi/2) \cos \theta} \left[ \frac{e^{j(\pi/2) \cos \theta} - e^{-j(\pi/2) \cos \theta}}{2j} \right] \right|^2 \\ &= 4 \sin^2 \left( \frac{\pi}{2} \cos \theta \right). \end{aligned}$$

阵列因子的最大值为 4, 这是具有单位幅度的二元阵列可达到的最大水平。  $F_a(\theta)$  最大的方向是  $\theta = 0$  (东) 和  $\theta = 180^\circ$  (西) 对应的方向, 如图9-29 (b) 所示。

练习 9-14: 推导出与  $a_0 = 1$  和  $a_1 = 3$  同相激励的二元阵列的阵列因子表达式。元素沿  $z$  轴定位, 并以  $\lambda/2$  分隔。

**Answer:**  $F_a(\theta) = [10 + 6 \cos(\pi \cos \theta)]$ . (See **EM**.)

练习 9-15: 沿  $z$  轴排列的等距  $N$  元阵列具有相等的幅度和相位; 即,  $A_i = 1$  对于  $i = 0, 1, \dots, (N - 1)$ 。侧面方向上的阵列因子的大小是多少?

**Answer:**  $F_a(\theta = 90^\circ) = N^2$ . (See **EM**.)

## 9-10 具有均匀相位分布的 N 元阵列

现在我们考虑一个由  $N$  个元素组成的阵列, 其具有等间距  $d$  和等相位激励; 即,  $\psi_i = \psi_0$  代表  $i = 1, 2, \dots, (N - 1)$ 。这种同相元件的阵列有时被称为侧边阵列, 因为其阵列因子的辐射方向图的主波束总是在阵列轴的侧面方向上。从方程。 (9.104), 阵列因子由下式给出

$$\begin{aligned} F_a(\theta) &= \left| e^{j\psi_0} \sum_{i=0}^{N-1} a_i e^{j i k d \cos \theta} \right|^2 \\ &= |e^{j\psi_0}|^2 \left| \sum_{i=0}^{N-1} a_i e^{j i k d \cos \theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{j i k d \cos \theta} \right|^2. \end{aligned} \quad (9.105)$$

相邻元件辐射的场之间的相位差为

$$\gamma = k d \cos \theta = \frac{2\pi d}{\lambda} \cos \theta. \quad (9.106)$$

就  $\gamma$  而言, 等式: (9.105) 采取紧凑形式

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{j i \gamma} \right|^2. \quad (\text{uniform phase}) \quad (9.107)$$

For a uniform amplitude distribution with  $a_i = 1$  for  $i = 0, 1, \dots, (N - 1)$ , Eq. (9.107) becomes

$$F_a(\gamma) = |1 + e^{j\gamma} + e^{j2\gamma} + \dots + e^{j(N-1)\gamma}|^2. \tag{9.108}$$

This geometric series can be rewritten in a more compact form by applying the following recipe. First, we define

$$F_a(\gamma) = |f_a(\gamma)|^2, \tag{9.109}$$

with

$$f_a(\gamma) = [1 + e^{j\gamma} + e^{j2\gamma} + \dots + e^{j(N-1)\gamma}]. \tag{9.110}$$

Next, we multiply  $f_a(\gamma)$  by  $e^{j\gamma}$  to obtain

$$f_a(\gamma) e^{j\gamma} = (e^{j\gamma} + e^{j2\gamma} + \dots + e^{jN\gamma}). \tag{9.111}$$

Subtracting Eq. (9.111) from Eq. (9.110) gives

$$f_a(\gamma) (1 - e^{j\gamma}) = 1 - e^{jN\gamma}, \tag{9.112}$$

which in turn gives

$$\begin{aligned} f_a(\gamma) &= \frac{1 - e^{jN\gamma}}{1 - e^{j\gamma}} = \frac{e^{jN\gamma/2}}{e^{j\gamma/2}} \frac{(e^{-jN\gamma/2} - e^{jN\gamma/2})}{(e^{-j\gamma/2} - e^{j\gamma/2})} \\ &= e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)}. \end{aligned} \tag{9.113}$$

After multiplying  $f_a(\gamma)$  by its complex conjugate, we obtain the result:

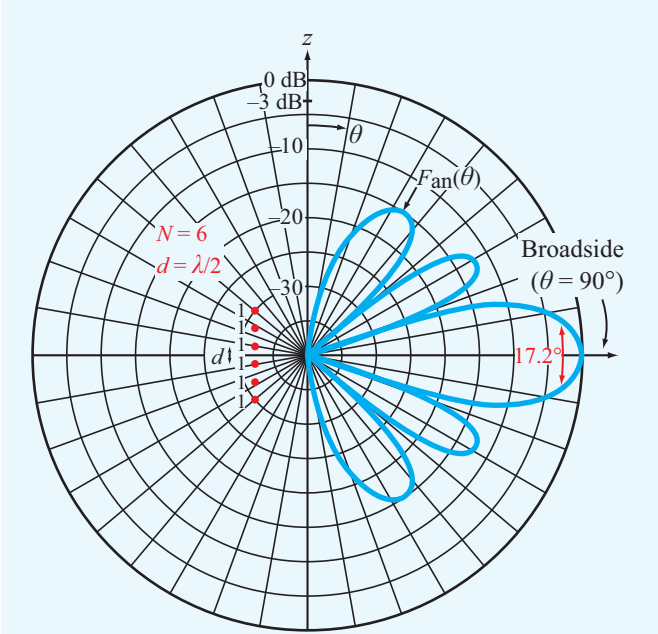
$$F_a(\gamma) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}. \tag{9.114}$$

(uniform amplitude and phase)

From Eq. (9.108),  $F_a(\gamma)$  is maximum when all terms are 1, which occurs when  $\gamma = 0$  (or equivalently,  $\theta = \pi/2$ ). Moreover,  $F_a(0) = N^2$ . Hence, the normalized array factor is given by

$$\begin{aligned} F_{an}(\gamma) &= \frac{F_a(\gamma)}{F_{a,\max}} \\ &= \frac{\sin^2(N\gamma/2)}{N^2 \sin^2(\gamma/2)} = \frac{\sin^2\left(\frac{N\pi d}{\lambda} \cos \theta\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda} \cos \theta\right)}. \end{aligned} \tag{9.115}$$

A polar plot of  $F_{an}(\theta)$  is shown in **Fig. 9-30** for  $N = 6$  and  $d = \lambda/2$ . The reader is reminded that this is a plot of the



**Figure 9-30** Normalized array pattern of a uniformly excited six-element array with interelement spacing  $d = \lambda/2$ .

radiation pattern of the array factor alone; the pattern for the antenna array is equal to the product of this pattern and that of a single element, as discussed earlier in connection with the pattern multiplication principle.

**Example 9-7: Multiple-Beam Array**

Obtain an expression for the array factor of a two-element array with equal excitation and a separation  $d = 7\lambda/2$ , and then plot the array pattern.

**Solution:** The array factor of a two-element array ( $N = 2$ ) with equal excitation ( $a_0 = a_1 = 1$ ) is given by

$$\begin{aligned} F_a(\gamma) &= \left| \sum_{i=0}^1 a_i e^{ji\gamma} \right|^2 = |1 + e^{j\gamma}|^2, \\ &= |e^{j\gamma/2}(e^{-j\gamma/2} + e^{j\gamma/2})|^2 \\ &= |e^{j\gamma/2}|^2 |e^{-j\gamma/2} + e^{j\gamma/2}|^2 = 4 \cos^2(\gamma/2), \end{aligned}$$

where  $\gamma = (2\pi d/\lambda) \cos \theta$ . The normalized array pattern shown in **Fig. 9-31** consists of seven beams; all have the same peak value but not the same angular width. The number of beams in the angular range between  $\theta = 0$  and  $\theta = \pi$  is equal to the

对于均匀振幅分布， $a_i = 1$  对于  $i = 0, 1, \dots, (N - 1)$ ，方程：(9.107) 变为

$$F_a(\gamma) = |1 + e^{j\gamma} + e^{j2\gamma} + \dots + e^{j(N-1)\gamma}|^2. \tag{9.108}$$

通过应用以下配方，可以将这个几何级数重写为更紧凑的形式。首先，我们定义

$$F_a(\gamma) = |f_a(\gamma)|^2, \tag{9.109}$$

和

$$f_a(\gamma) = [1 + e^{j\gamma} + e^{j2\gamma} + \dots + e^{j(N-1)\gamma}]. \tag{9.110}$$

接下来，我们将  $f_a(\gamma)$  乘以  $e^{j\gamma}$  得到

$$f_a(\gamma) e^{j\gamma} = (e^{j\gamma} + e^{j2\gamma} + \dots + e^{jN\gamma}). \tag{9.111}$$

减法(9.111) 来自方程式(9.110) 给出

$$f_a(\gamma) (1 - e^{j\gamma}) = 1 - e^{jN\gamma}, \tag{9.112}$$

这又给出了

$$\begin{aligned} f_a(\gamma) &= \frac{1 - e^{jN\gamma}}{1 - e^{j\gamma}} = \frac{e^{jN\gamma/2}}{e^{j\gamma/2}} \frac{(e^{-jN\gamma/2} - e^{jN\gamma/2})}{(e^{-j\gamma/2} - e^{j\gamma/2})} \\ &= e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)}. \end{aligned} \tag{9.113}$$

将  $f_a(\gamma)$  与其复共轭相乘后，我们得到结果：

$$F_a(\gamma) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}. \tag{9.114}$$

(均匀的幅度和相位)

从方程。(9.108)，当所有项均为 1 时， $F_a(\gamma)$  最大，这发生在  $\gamma = 0$  (或等效地， $\theta = \pi/2$ ) 时。此外， $F_a(0) = N^2$ 。因此，归一化数组因子由下式给出

$$\begin{aligned} F_{an}(\gamma) &= \frac{F_a(\gamma)}{F_{a,\max}} \\ &= \frac{\sin^2(N\gamma/2)}{N^2 \sin^2(\gamma/2)} = \frac{\sin^2\left(\frac{N\pi d}{\lambda} \cos \theta\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda} \cos \theta\right)}. \end{aligned} \tag{9.115}$$

N = 6 和  $d = \lambda/2$  的  $F_{an}(\theta)$  极坐标图如图 9-30 所示。提醒读者，这是一个情節

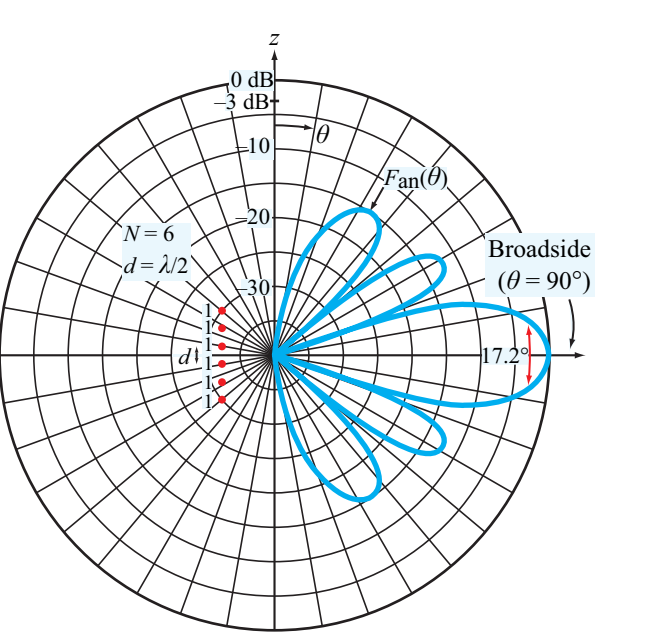


图 9-30 元素间间距为  $d = \lambda/2$  的均匀激发六元素阵列的归一化阵列图案。

单独的阵列因子的辐射方向图；天线阵列的方向图等于该方向图与单个元件的方向图的乘积，如前面结合方向图乘法原理所讨论的。

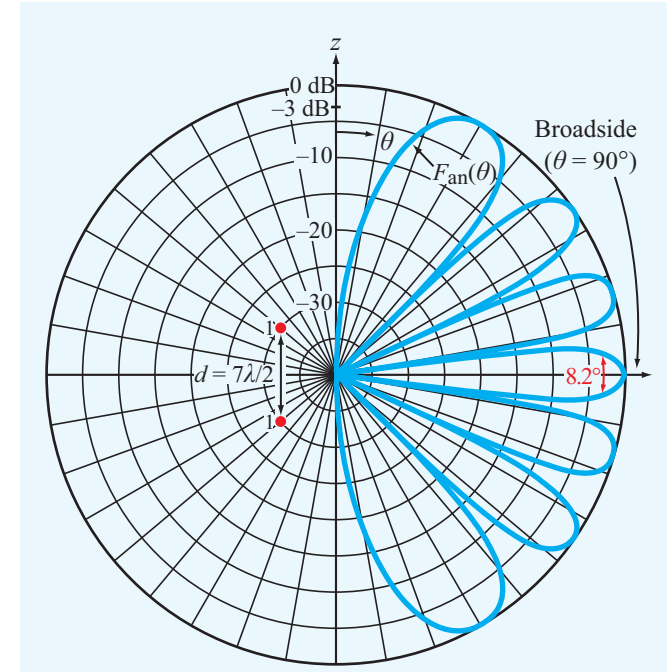
示例 9-7：多光束阵列

获取具有相等激励和间隔  $d = 7\lambda/2$  的二元阵列的阵列因子的表达式，然后绘制阵列模式。

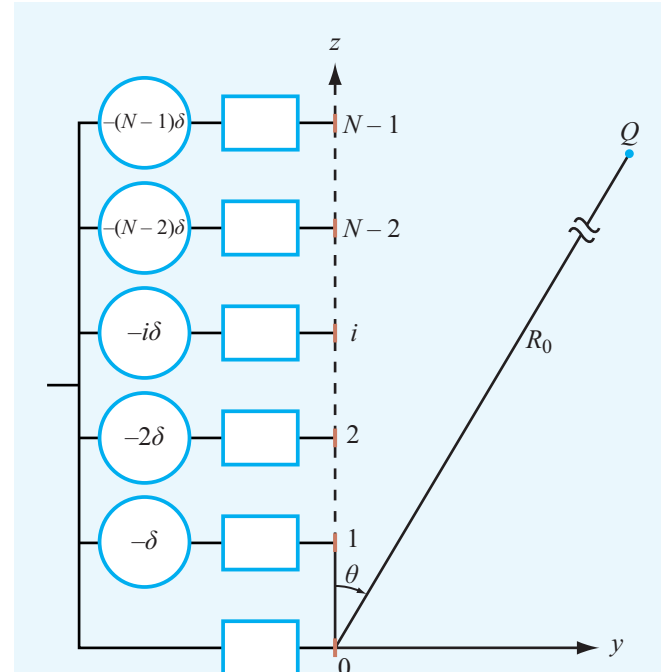
解：具有相等激励 ( $a_0 = a_1 = 1$ ) 的二元阵列 ( $N = 2$ ) 的阵列因子由下式给出

$$\begin{aligned} F_a(\gamma) &= \left| \sum_{i=0}^1 a_i e^{ji\gamma} \right|^2 = |1 + e^{j\gamma}|^2, \\ &= |e^{j\gamma/2}(e^{-j\gamma/2} + e^{j\gamma/2})|^2 \\ &= |e^{j\gamma/2}|^2 |e^{-j\gamma/2} + e^{j\gamma/2}|^2 = 4 \cos^2(\gamma/2), \end{aligned}$$

其中  $\gamma = (2\pi d/\lambda) \cos \theta$ 。图9-31所示的归一化阵列图案由七个波束组成；都具有相同的峰值，但角宽度不同。 $\theta = 0$  和  $\theta = \pi$  之间的角度范围内的光束数量等于



**Figure 9-31** Normalized array pattern of a two-element array with spacing  $d = 7\lambda/2$ .



**Figure 9-32** The application of linear phase.

separation between the array elements,  $d$ , measured in units of  $\lambda/2$ .

## 9-11 Electronic Scanning of Arrays

The discussion in the preceding section was concerned with uniform-phase arrays, in which the phases of the feeding coefficients,  $\psi_0$  to  $\psi_{N-1}$ , are all equal. In this section, we examine the use of phase delay between adjacent elements as a tool to **electronically steer** the direction of the array-antenna beam from broadside at  $\theta = 90^\circ$  to any desired angle  $\theta_0$ . In addition to eliminating the need to mechanically steer an antenna to change its beam's direction, electronic steering allows beam scanning at very fast rates.

► Electronic steering is achieved by applying a **linear phase distribution** across the array:  $\psi_0 = 0$ ,  $\psi_1 = -\delta$ ,  $\psi_2 = -2\delta$ , etc. ◀

As shown in **Fig. 9-32**, the phase of the  $i$ th element relative to that of the zeroth element is

$$\psi_i = -i\delta, \quad (9.116)$$

where  $\delta$  is the **incremental phase delay** between adjacent elements. Use of Eq. (9.116) in Eq. (9.104) leads to

$$\begin{aligned} F_a(\theta) &= \left| \sum_{i=0}^{N-1} a_i e^{-ji\delta} e^{j i k d \cos \theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{j i (k d \cos \theta - \delta)} \right|^2 = \left| \sum_{i=0}^{N-1} a_i e^{j i \gamma'} \right|^2 = F_a(\gamma'), \end{aligned} \quad (9.117)$$

where we introduced a new variable given by

$$\gamma' = k d \cos \theta - \delta. \quad (9.118)$$

For reasons that become clear later, we define the phase shift  $\delta$  in terms of an angle  $\theta_0$ , which we call the **scan angle**, as follows:

$$\delta = k d \cos \theta_0. \quad (9.119)$$

Hence,  $\gamma'$  becomes

$$\gamma' = k d (\cos \theta - \cos \theta_0). \quad (9.120)$$

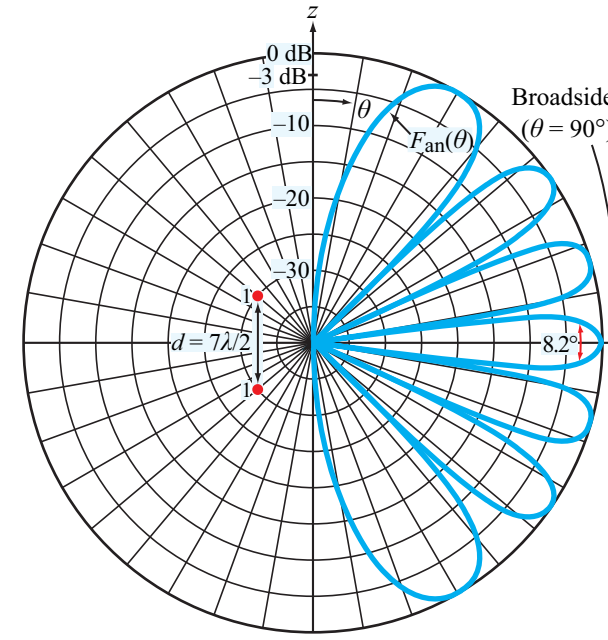


图 9-31 间距为  $d = 7\lambda/2$  的二元素数组的归一化数组模式。

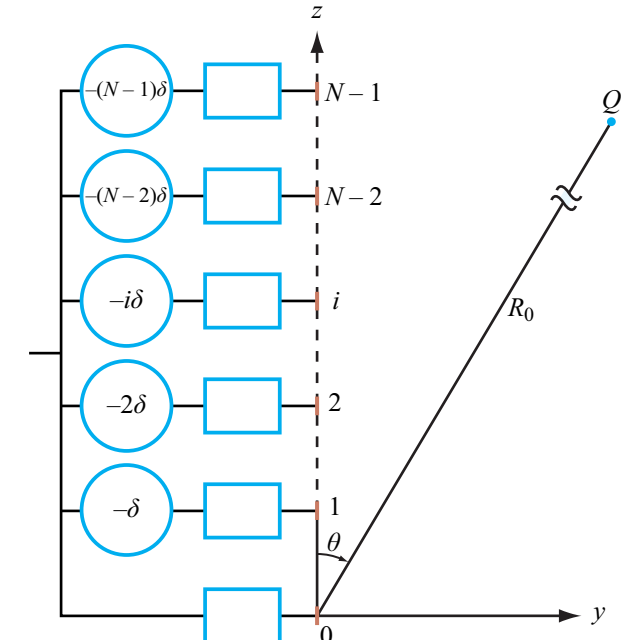


图9-32 线性相位的应用。

数组元素之间的间隔  $d$ ，以  $\lambda/2$  为单位测量。

## 9-11 阵列的电子扫描

上一节中的讨论涉及均匀相位阵列，其中馈电系数  $\psi_0$  至  $\psi_{N-1}$  的相位均相等。在本节中，我们将研究如何使用相邻元件之间的相位延迟作为一种工具，以电子方式将阵列天线波束的方向从  $\theta = 90^\circ$  的侧面转向任何所需的角  $\theta_0$ 。除了无需机械操纵天线来改变波束方向外，电子操纵还可以以非常快的速率进行波束扫描。

► 电子转向是通过在阵列上应用线性相位分布来实现的： $\psi_0 = 0$ 、 $\psi_1 = -\delta$ 、 $\psi_2 = -2\delta$  等。◀

如图9-32所示，第 $i$ 个单元相对于第0个单元的相位为

$$\psi_i = -i\delta, \quad (9.116)$$

其中  $\delta$  是相邻元件之间的增量相位延迟。使用等式 (9.116) 式(9.104) 导致

$$\begin{aligned} F_a(\theta) &= \left| \sum_{i=0}^{N-1} a_i e^{-ji\delta} e^{j i k d \cos \theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{j i (k d \cos \theta - \delta)} \right|^2 = \left| \sum_{i=0}^{N-1} a_i e^{j i \gamma'} \right|^2 = F_a(\gamma'), \end{aligned} \quad (9.117)$$

我们引入了一个新变量，由下式给出

$$\gamma' = k d \cos \theta - \delta. \quad (9.118)$$

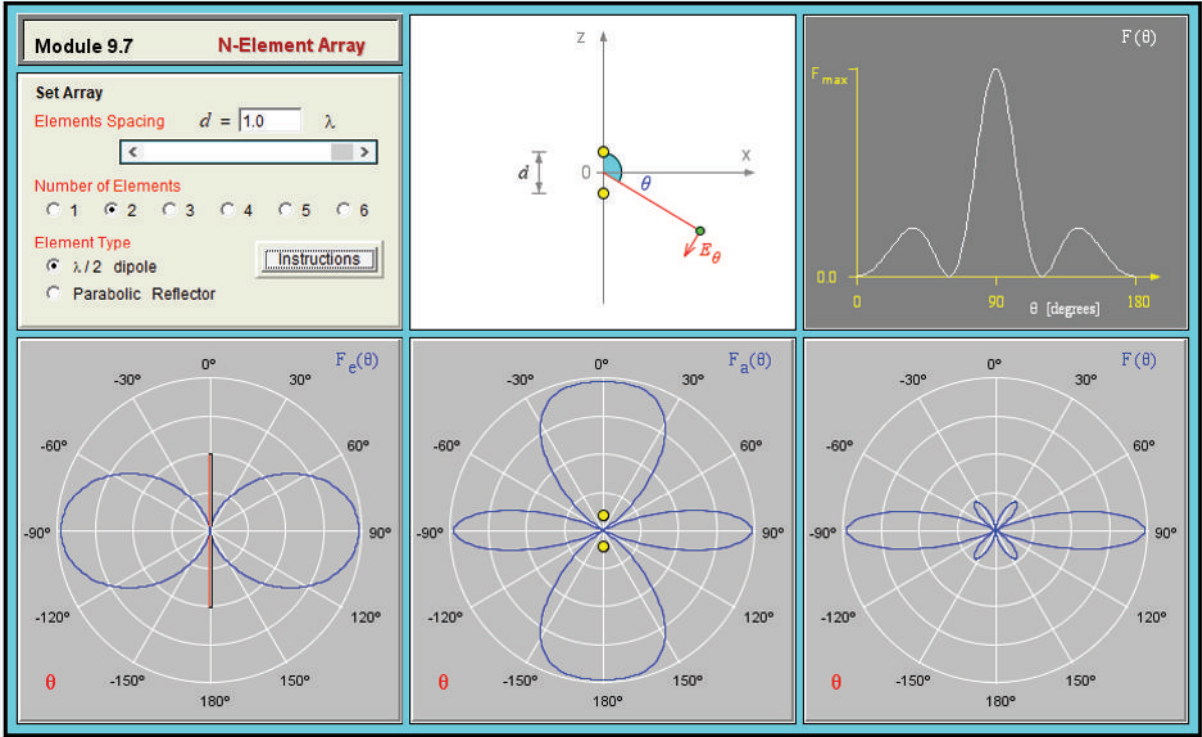
出于稍后会清楚的原因，我们用角  $\theta_0$  定义相移  $\delta$ ，我们将其称为扫描角，如下所示：

$$\delta = k d \cos \theta_0. \quad (9.119)$$

因此， $\gamma'$  变为

$$\gamma' = k d (\cos \theta - \cos \theta_0). \quad (9.120)$$

**Module 9.7 N-Element Array** This module displays the far-field patterns of an array of  $N$  identical, equally spaced antennas, with  $N$  being a selectable integer between 1 and 6. Two types of antennas can be simulated:  $\lambda/2$ -dipoles and parabolic reflectors. This module provides visual examples of the pattern multiplication principle.



The array factor given by Eq. (9.117) has the same functional form as the array factor developed earlier for the uniform-phase array [see Eq. (9.107)], except that  $\gamma$  is replaced with  $\gamma'$ . Hence,

► Regardless of the amplitude distribution across an array, its array factor  $F_a(\gamma')$ , when excited by a linear-phase distribution, can be obtained from  $F_a(\gamma)$ , the expression developed for the array assuming a uniform-phase distribution, by replacing  $\gamma$  with  $\gamma'$ . ◀

If the amplitude distribution is symmetrical with respect to the array center, the array factor  $F_a(\gamma')$  is maximum when its argument  $\gamma' = 0$ . When the phase is uniform ( $\delta = 0$ ), this condition corresponds to the direction  $\theta = 90^\circ$ , which is why the uniform-phase arrangement is called a broadside array. According to Eq. (9.120), in a linearly phased array,  $\gamma' = 0$  when  $\theta = \theta_0$ . Thus, by applying linear phase across the array, the array pattern is shifted along the  $\cos \theta$  axis by an amount

$\cos \theta_0$ , and the direction of maximum radiation is **steered** from the broadside direction ( $\theta = 90^\circ$ ) to the direction  $\theta = \theta_0$ . To steer the beam all the way to the **end-fire direction** ( $\theta = 0$ ), the incremental phase shift  $\delta$  should be equal to  $kd$  radians.

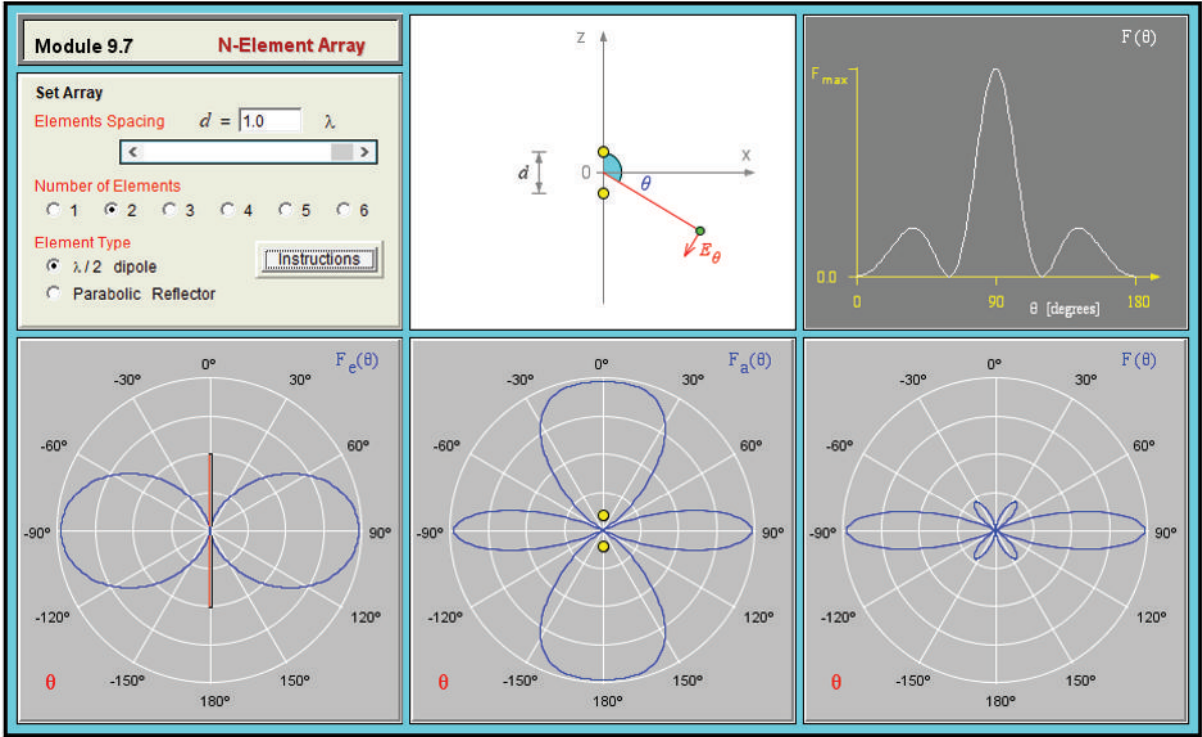
9-11.1 Uniform-Amplitude Excitation

To illustrate the process with an example, consider the case of the  $N$ -element array excited by a uniform-amplitude distribution. Its normalized array factor is given by Eq. (9.115). Upon replacing  $\gamma$  with  $\gamma'$ , we have

$$F_{an}(\gamma') = \frac{\sin^2(N\gamma'/2)}{N^2 \sin^2(\gamma'/2)} \quad (9.121)$$

with  $\gamma'$  as defined by Eq. (9.120). For an array with  $N = 10$  and  $d = \lambda/2$ , plots of the main lobe of  $F_{an}(\theta)$  are shown in Fig. 9-33 for  $\theta_0 = 0^\circ, 45^\circ$ , and  $90^\circ$ . We note that the half-power beamwidth increases as the array beam is steered from broadside to end-fire directions.

模块 9.7 N 元阵列 该模块显示 N 个相同、等距天线阵列的远场方向图，N 是 1 到 6 之间的可选整数。可以模拟两种类型的天线： $\lambda/2$  偶极子和抛物面反射器。该模块提供了模式乘法原理的直观示例。



方程给出的阵列因子。(9.117) 与之前为均匀相位阵列开发的阵列因子具有相同的函数形式[见方程1]。(9.107)]，只不过  $\gamma$  被替换为  $\gamma'$ 。因此，

► 无论阵列上的幅度分布如何，其阵列因子  $F_a(\gamma')$ ，当由线性相位分布激励时，可以从  $F_a(\gamma)$  获得，为假设均匀相位分布的阵列开发的表达式，通过替换  $\gamma$  和  $\gamma'$ 。◀

如果振幅分布相对于阵列中心对称，则当其自变量  $\gamma' = 0$  时，阵列因子  $F_a(\gamma')$  最大。当相位均匀 ( $\delta = 0$ ) 时，该条件对应于方向  $\theta = 90^\circ$ ，这就是为什么均匀相位排列被称为宽边阵列的原因。根据方程。(9.120)，在线性相控阵中，当  $\theta = \theta_0$  时， $\gamma' = 0$ 。因此，通过在阵列上应用线性相位，阵列图案沿  $\cos \theta$  轴移动一定量

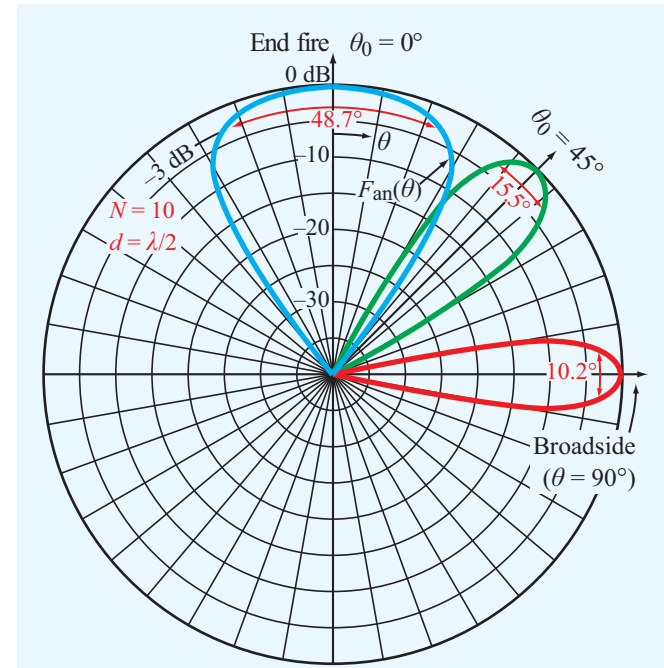
$\cos \theta_0$ ，最大辐射方向从舷侧方向 ( $\theta = 90^\circ$ ) 转向  $\theta = \theta_0$  方向。为了将光束一直转向端射方向 ( $\theta = 0$ )，增量相移  $\delta$  应等于  $kd$  弧度。

9-11.1 等幅激励

为了用一个例子来说明这个过程，考虑由均匀幅度分布激励的 N 元阵列的情况。其归一化阵列因子由等式给出。(9.115)。将  $\gamma$  替换为  $\gamma'$  后，我们有

$$F_{an}(\gamma') = \frac{\sin^2(N\gamma'/2)}{N^2 \sin^2(\gamma'/2)} \quad (9.121)$$

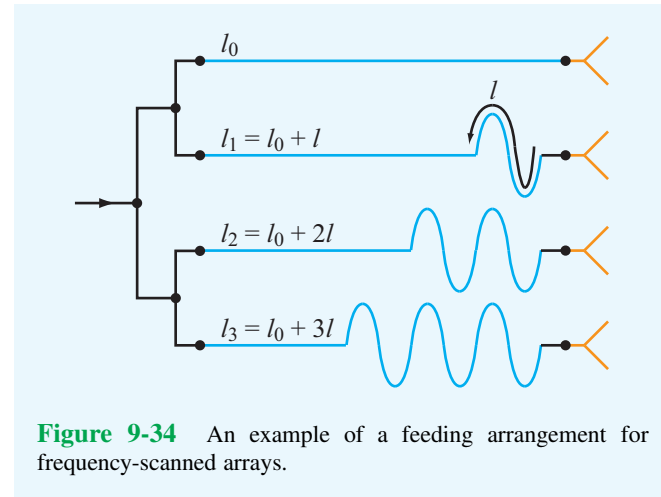
与  $\gamma'$  如方程所定义。(9.120)。对于  $N = 10$  和  $d = \lambda/2$  的阵列，对于  $\theta_0 = 0^\circ, 45^\circ$  和  $90^\circ$  的  $F_{an}(\theta)$  主瓣图如图 9-33 所示。我们注意到，随着阵列波束从侧面转向端射方向，半功率波束宽度增加。



**Figure 9-33** Normalized array pattern of a 10-element array with  $\lambda/2$  spacing between adjacent elements. All elements are excited with equal amplitude. Through the application of linear phase across the array, the main beam can be steered from the broadside direction ( $\theta_0 = 90^\circ$ ) to any scan angle  $\theta_0$ . Equiphasic excitation corresponds to  $\theta_0 = 90^\circ$ .

### 9-11.2 Array Feeding

According to the foregoing discussion, to steer the antenna beam to an angle  $\theta_0$ , two conditions must be met: (1) the phase distribution must be linear across the array and (2) the magnitude of the incremental phase delay  $\delta$  must satisfy Eq. (9.119). The combination of these two conditions provides the necessary tilting of the beam from  $\theta = 90^\circ$  (broadside) to  $\theta = \theta_0$ . This can be accomplished by controlling the excitation of each radiating element individually through the use of electronically controlled phase shifters. Alternatively, a technique known as **frequency scanning** can be used to provide control of the phases of all the elements simultaneously. **Figure 9-34** shows an example of a simple feeding arrangement employed in frequency-scanning arrays. A common feed point is connected to the radiating elements through transmission lines of varying lengths. Relative to the zeroth element, the path between the common feed point and a radiating element is longer by  $l$  for the first element, by  $2l$  for the second, and by  $3l$  for the third.



**Figure 9-34** An example of a feeding arrangement for frequency-scanned arrays.

Thus, the path length for the  $i$ th element is

$$l_i = il + l_0, \quad (9.122)$$

where  $l_0$  is the path length of the zeroth element. Waves of frequency  $f$  propagating through a transmission line of length  $l_i$  are characterized by a phase factor  $e^{-j\beta l_i}$ , where  $\beta = 2\pi f/u_p$  is the phase constant of the line and  $u_p$  is its phase velocity. Hence, the **incremental** phase delay of the  $i$ th element relative to the phase of the zeroth element is

$$\psi_i(f) = -\beta(l_i - l_0) = -\frac{2\pi}{u_p} f(l_i - l_0) = -\frac{2\pi i}{u_p} fl. \quad (9.123)$$

Suppose that at a given reference frequency  $f_0$  we choose the incremental length  $l$  so that

$$l = \frac{n_0 u_p}{f_0}, \quad (9.124)$$

where  $n_0$  is a *specific* positive integer. In this case, the phase delay  $\psi_1(f_0)$  becomes

$$\psi_1(f_0) = -2\pi \left( \frac{f_0 l}{u_p} \right) = -2n_0\pi \quad (9.125)$$

Similarly,  $\psi_2(f_0) = -4n_0\pi$  and  $\psi_3(f_0) = -6n_0\pi$ . That is, at  $f_0$  all the elements have equal phase (within multiples of  $2\pi$ ) and the array radiates in the broadside direction. If  $f$  is changed to

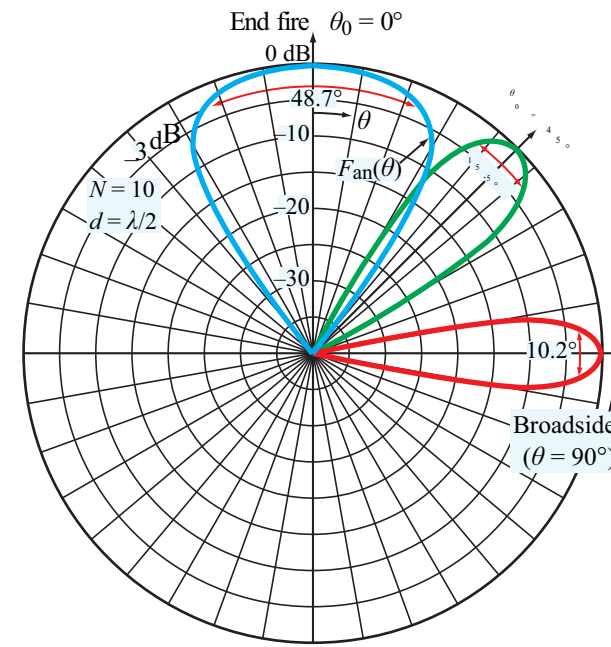


图 9-33 10 元素数组的标准化数组模式，相邻元素之间的间距为  $\lambda/2$ 。所有元素都以相等的幅度被激发。通过在阵列上应用线性相位，主波束可以从侧边方向 ( $\theta_0 = 90^\circ$ ) 转向任何扫描角度  $\theta_0$ 。等相激励对应于  $\theta_0 = 90^\circ$ 。

### 9-11.2 数组馈送

根据前面的讨论，为了将天线波束转向  $\theta_0$  角度，必须满足两个条件：(1) 整个阵列的相位分布必须是线性的，(2) 增量相位延迟  $\delta$  的幅度必须满足方程 1。 (9.119)。这两个条件的组合提供了光束从  $\theta = 90^\circ$  (侧边) 到  $\theta = \theta_0$  的必要倾斜。这可以通过使用电子控制移相器单独控制每个辐射元件的激励来实现。或者，可以使用称为频率扫描的技术来同时提供对所有元件的相位的控制。图 9-34 显示了频率扫描阵列中采用的简单馈电布置的示例。公共馈电点通过不同长度的传输线连接到辐射元件。相对于第 0 个元件，公共馈电点和辐射元件之间的路径第一个元件长  $l$ ，第二个元件长  $2l$ ，第三个元件长  $3l$ 。

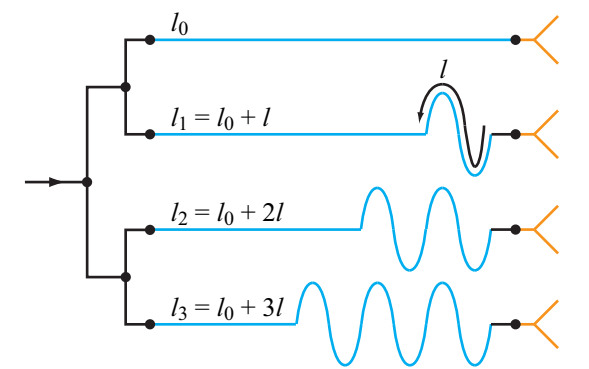


图 9-34 频率扫描阵列的馈电布置示例。

因此，第  $i$  个元素的路径长度为

$$l_i = il + l_0, \quad (9.122)$$

其中  $l_0$  是第零个元素的路径长度。通过长度为  $l_i$  的传输线传播的频率为  $f$  的波由相位因子  $e^{-j\beta l_i}$  表征，其中  $\beta = 2\pi f/u_p$  是传输线的相位常数， $u_p$  是其相速度。因此，第  $i$  个元件相对于第 0 个元件的相位延迟增量为

$$\psi_i(f) = -\beta(l_i - l_0) = -\frac{2\pi}{u_p} f(l_i - l_0) = -\frac{2\pi i}{u_p} fl. \quad (9.123)$$

假设在给定的参考频率  $f_0$  我们选择增量长度  $l$  使得

$$l = \frac{n_0 u_p}{f_0}, \quad (9.124)$$

其中  $n_0$  是一个特定的正整数。在这种情况下，相位延迟  $\psi_1(f_0)$  变为

$$\psi_1(f_0) = -2\pi \left( \frac{f_0 l}{u_p} \right) = -2n_0\pi \quad (9.125)$$

类似地， $\psi_2(f_0) = -4n_0\pi$  和  $\psi_3(f_0) = -6n_0\pi$ 。也就是说，在  $f_0$  处，所有单元都具有相等的相位（在  $2\pi$  的倍数内），并且阵列沿宽边方向辐射。如果  $f$  改为

$f_0 + \Delta f$ , the new phase shift of the first element relative to the zeroth element is

$$\begin{aligned}\psi_1(f_0 + \Delta f) &= -\frac{2\pi}{u_p}(f_0 + \Delta f)l \\ &= -\frac{2\pi f_0 l}{u_p} - \left(\frac{2\pi l}{u_p}\right)\Delta f \\ &= -2n_0\pi - 2n_0\pi\left(\frac{\Delta f}{f_0}\right) \\ &= -2n_0\pi - \delta,\end{aligned}\tag{9.126}$$

where use was made of Eq. (9.124) and  $\delta$  is defined as

$$\delta = 2n_0\pi\left(\frac{\Delta f}{f_0}\right).\tag{9.127}$$

Similarly,  $\psi_2(f_0 + \Delta f) = 2\psi_1$  and  $\psi_3(f_0 + \Delta f) = 3\psi_1$ . Ignoring the factor of  $2\pi$  and its multiples (since they exercise no influence on the relative phases of the radiated fields), we see that the incremental phase shifts are directly proportional to the fractional frequency deviation  $(\Delta f/f_0)$ . Thus, in an array with  $N$  elements, controlling  $\Delta f$  provides a direct control of  $\delta$ , which in turn controls the scan angle  $\theta_0$  according to Eq. (9.119). Equating Eq. (9.119) to Eq. (9.127) and then solving for  $\cos \theta_0$  leads to

$$\cos \theta_0 = \frac{2n_0\pi}{kd}\left(\frac{\Delta f}{f_0}\right)\tag{9.128}$$

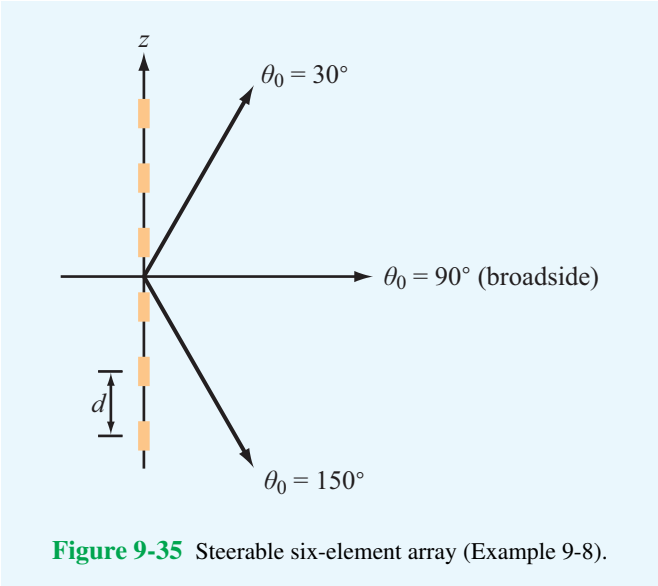
As  $f$  is changed from  $f_0$  to  $f_0 + \Delta f$ ,  $k = 2\pi/\lambda = 2\pi f/c$  also changes with frequency. However, if  $\Delta f/f_0$  is small, we may treat  $k$  as a constant equal to  $2\pi f_0/c$ ; the error in  $\cos \theta_0$  resulting from the use of this approximation in Eq. (9.128) is on the order of  $\Delta f/f_0$ .

**Example 9-8: Electronic Steering**

Design a steerable six-element array with the following specifications:

- All elements are excited with equal amplitudes.
- At  $f_0 = 10$  GHz, the array radiates in the broadside direction, and the interelement spacing  $d = \lambda_0/2$ , where  $\lambda_0 = c/f_0 = 3$  cm.
- The array pattern is to be electronically steerable in the elevation plane over the angular range extending between  $\theta_0 = 30^\circ$  and  $\theta_0 = 150^\circ$ .

- The antenna array is fed by a voltage-controlled oscillator whose frequency can be varied over the range from 9.5 to 10.5 GHz.
- The array uses a feeding arrangement of the type shown in **Fig. 9-34**, and the transmission lines have a phase velocity  $u_p = 0.8c$ .



**Figure 9-35** Steerable six-element array (Example 9-8).

**Solution:** The array is to be steerable from  $\theta_0 = 30^\circ$  to  $\theta_0 = 150^\circ$  (**Fig. 9-35**). For  $\theta_0 = 30^\circ$  and

$$kd = \left(\frac{2\pi}{\lambda_0}\right)\left(\frac{\lambda_0}{2}\right) = \pi,$$

Eq. (9.128) gives

$$0.87 = 2n_0\left(\frac{\Delta f}{f_0}\right).\tag{9.129}$$

We are given that  $f_0 = 10$  GHz and the oscillator frequency can be varied between  $(f_0 - 0.5$  GHz) and  $(f_0 + 0.5$  GHz). Thus,  $\Delta f_{\max} = 0.5$  GHz. To satisfy Eq. (9.129), we need to choose  $n_0$  so that  $\Delta f$  is as close as possible to, but not larger than,  $\Delta f_{\max}$ . Solving Eq. (9.129) for  $n_0$  with  $\Delta f = \Delta f_{\max}$  gives

$$n_0 = \frac{0.87}{2} \frac{f_0}{\Delta f_{\max}} = 8.7.$$

Since  $n_0$  is not an integer, we need to modify its value by rounding it upward to the next whole-integer value. Hence, we set  $n_0 = 9$ .

$f_0 + \Delta f$ , 第一个元素相对于第零个元素的新相移为

$$\begin{aligned}\psi_1(f_0 + \Delta f) &= -\frac{2\pi}{u_p}(f_0 + \Delta f)l \\ &= -\frac{2\pi f_0 l}{u_p} - \left(\frac{2\pi l}{u_p}\right)\Delta f \\ &= -2n_0\pi - 2n_0\pi\left(\frac{\Delta f}{f_0}\right) \\ &= -2n_0\pi - \delta,\end{aligned}\tag{9.126}$$

其中使用了方程式。(9.124) 和  $\delta$  定义为

$$\delta = 2n_0\pi\left(\frac{\Delta f}{f_0}\right).\tag{9.127}$$

相似地,  $\psi_2(f_0 + \Delta f) = 2\psi_1$  和  $\psi_3(f_0 + \Delta f) = 3\psi_1$ 。忽略因子  $2\pi$  及其倍数 (因为它们对辐射场的相对相位没有影响), 我们看到增量相移与分数频率偏差成正比  $(\Delta f/f_0)$ 。因此, 在具有  $N$  个元素的数组中, 控制  $\Delta f$  提供直接控制  $\delta$ , 进而控制扫描角度  $\theta_0$  根据方程式 (9.119)。等式(9.119) 至方程式(9.127) 然后求解  $\cos \theta_0$  导致

$$\cos \theta_0 = \frac{2n_0\pi}{kd}\left(\frac{\Delta f}{f_0}\right)\tag{9.128}$$

当  $f$  从  $f_0$  变为  $f_0 + \Delta f$  时,  $k = 2\pi/\lambda = 2\pi f/c$  也随频率变化。然而, 如果  $\Delta f/f_0$  很小, 我们可以将  $k$  视为等于  $2\pi f_0/c$  的常数;  $\cos \theta_0$  中的误差是由于在方程式中使用此近似值而产生的。(9.128) 的量级为  $\Delta f/f_0$ 。

**例 9-8: 电子转向**

设计一个具有以下规格的可操纵六元阵列:

- 所有元件均以相等的幅度激励。
- 在  $f_0 = 10$  GHz 处, 阵列沿侧面方向辐射, 元件间间距  $d = \lambda_0/2$ , 其中  $\lambda_0 = c/f_0 = 3$  cm。
- 阵列方向图应在  $\theta_0 = 30^\circ$  和  $\theta_0 = 150^\circ$  之间延伸的角度范围内, 在仰角平面中可电子操纵。

4. 天线阵列由压控振荡器供电, 其频率可在 9.5 至 10.5 GHz 范围内变化。

5. 该阵列使用图9-34所示类型的馈电布置, 并且传输线具有相速度  $u_p = 0.8c$ 。

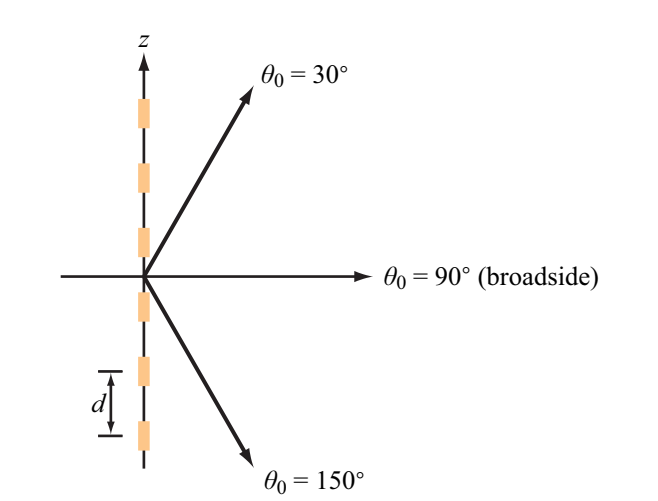


图 9-35 可操纵六元素阵列 (示例 9-8)。

解决方案: 阵列应可从  $\theta_0 = 30^\circ$  转向  $\theta_0 = 150^\circ$  (。 9-35)。对于  $\theta_0 = 30^\circ$  和

$$kd = \left(\frac{2\pi}{\lambda_0}\right)\left(\frac{\lambda_0}{2}\right) = \pi,$$

等式。(9.128) 给出

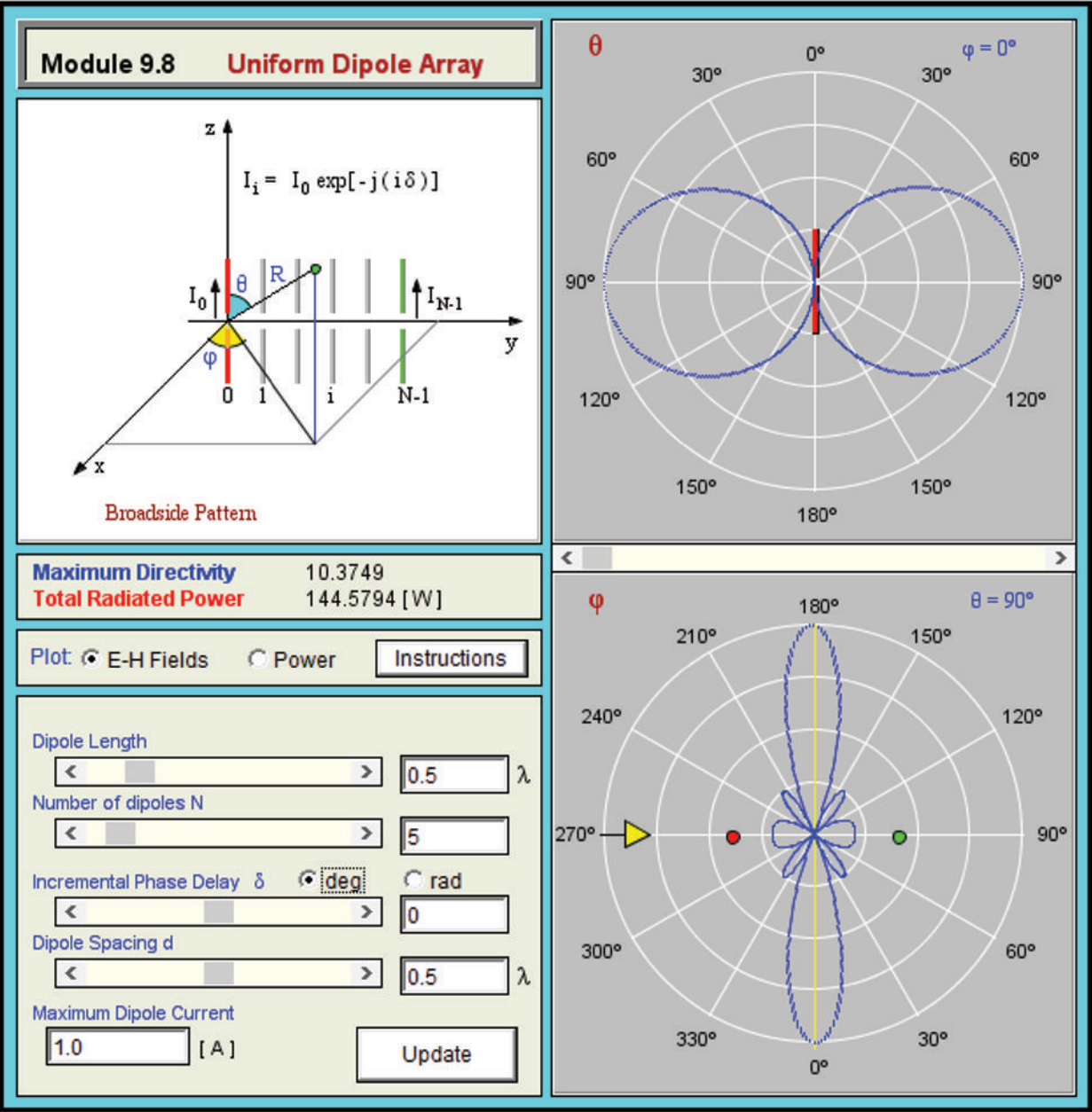
$$0.87 = 2n_0\left(\frac{\Delta f}{f_0}\right).\tag{9.129}$$

我们已知  $f_0 = 10$  GHz, 并且振荡器频率可以在  $(f_0 - 0.5$  GHz) 和  $(f_0 + 0.5$  GHz) 之间变化。因此,  $\Delta f_{\max} = 0.5$  GHz。满足方程。(9.129), 我们需要选择  $n_0$ , 以便  $\Delta f$  尽可能接近但不大于  $\Delta f_{\max}$ 。求解方程。(9.129) 对于  $n_0$  和  $\Delta f = \Delta f_{\max}$  给出

$$n_0 = \frac{0.87}{2} \frac{f_0}{\Delta f_{\max}} = 8.7.$$

由于  $n_0$  不是整数, 我们需要通过向上舍入到下一个整数值来修改它的值。因此, 我们将  $n_0 =$  设置为 9。

**Module 9.8 Uniform Dipole Array** For an array of up to 50 identical vertical dipoles of selectable length and current maximum that is excited with incremental phase delay  $\delta$  between adjacent elements, the module displays the elevation and azimuthal patterns of the array. By varying  $\delta$ , the array pattern can be steered in the horizontal plane.



模块 9.8 均匀偶极子阵列 对于最多 50 个具有可选长度和电流最大值的相同垂直偶极子阵列，该阵列由相邻元件之间的增量相位延迟  $\delta$  激励，该模块显示阵列的仰角和方位角图案。通过改变  $\delta$ ，可以在水平面上操纵阵列模式。

